TECHNICAL MANUAL

MAINTENANCE INSTRUCTIONS
WITH
ILLUSTRATED PARTS BREAKDOWN

INTERMEDIATE/DEPOT MAINTENANCE

THRUST CONTROL UNIT

282E261G4
282E261G5

PREPARED BY AFSC COMMODITY TEAM

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INTRODUCTION

1 PURPOSE AND SCOPE.

This manual provides intermediate and depot maintenance instructions for the Thrust Control Unit, PN 282E261G5, manufactured by the General Electric Company, Aircraft Control Systems Department.

2 ARRANGEMENT.

This manual is arranged into nine chapters as follows:

Chapter 1	Description
Chapter 2	Support Equipment And List Of Consumables
Chapter 3	Theory Of Operation
Chapter 4	Description Of System, Tie-In Of Equipment And Accessories
Chapter 5	Preparation For Maintenance
Chapter 6	Checkout And Troubleshooting
Chapter 7	Maintenance/Repair Instructions
Chapter 8	Overhaul Instructions
Chapter 9	Illustrated Parts Breakdown

3 ABBREVIATIONS.

All abbreviations used in this manual are in accordance with ASME Y14.38M for use on drawings, specifications, standards, and in technical documents, except as follows:

ADTS	Advanced Digital Test Station
AP	Attaching Parts
AR	As Required
A/T	Autothrottle
CCW	Counterclockwise
CW	Clockwise
CITS	Central Integrated Test System
CT	Control Transformer
DATSA	Depot Automatic Test Station for Avionics
ESD	Electrostatic Discharge
ESDS	Electrostatic Discharge Sensitive
ETCS	Engine Thrust Control System
IATE	Intermediate Automatic Test Equipment
LOAP	List of Applicable Publications
LRU	Line Replaceable Unit
NHA	Next Higher Assembly
PQ	Pilots Thrust Control Quadrant
PWM	Pulse Width Multiplier
SMR	Source, Maintenance, and Recoverability
SRU	Shop Replaceable Unit
TCL	Thrust Control Lever
TCU	Thrust Control Unit
VDATS	Versatile Diagnostic Automatic Test Station

4 SYMBOLS.

The following is an explanation of symbols used if applicable. All symbols used in this manual are in accordance with symbols per ASME Y14.38M.

-	Not illustrated
*	Requisition this marking in accordance with the requirements of DoDI 5330.3, DLA Document Services (DLA)
#	Detail parts are listed in a separate manual
F	Follows
$	Requisition this marking in accordance with the requirements of AFR 6-1
=	Alternate parts to the initial indexed part of the series

5 LIST OF RELATED PUBLICATIONS.

These publications contain information in support of this technical manual and are required to accomplish the prescribed maintenance:

List of Related Publications

Number	Title
TO 00-5-1	AF Technical Order System
TO 00-25-195	Source, Maintenance, and Recoverability Coding of Air Force Weapons, Systems, and Equipment
TO 00-25-234	General Shop Practice Requirements for Repair, Maintenance, Test of Electronic Equipment
TO 1-1-2	Corrosion Control, Prevention and Treatment for Aerospace Equipment

List of Related Publications - Continued

Number	Title
TO 1B-1B-2-76GS-00-1	General System - Engine Controls
TO 2JA8-28-8-1	Test Procedure Manual, Thrust Control Unit
TO 2JA8-28-8-2	Test Procedure Manual, Depot Maintenance using DATSA, Thrust Control Unit Shop Replaceable Units SRU’s
TO 2JA8-28-8-3	Test Procedure Manual, Module, Power Supply PS1
TO 2JA8-28-8-4	Test Procedure Manual Synchro Error Receiver 188C1254G2, -G1
TO 2JA8-28-8-6	Test Procedure Manual, Intermediate Level, Thrust Control Unit, Part Number 282E261G5 (ADTS)
TO 2JA8-28-8-7	Test Procedure Manual, Depot Maintenance using VDATS, Thrust Control Unit, Shop Replaceable Units (SRUs)

6 NUCLEAR HARDNESS.

CAUTION

The HCI HCI symbol emphasizes the uniqueness of hardness features and establishes special requirements limiting changes and substitutions, and that specific parts listed must be used to ensure hardness is not degraded.

If equipment covered has nuclear survivability requirements, all Hardness Critical Items (HCI) shall be marked. When the part is identified as HCI, the symbol HCI shall immediately precede the description of the item. When approved by the acquiring activity, the symbol **HCI** may be used in lieu of the boxed HCI symbol.

7 ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) PARTS.

CAUTION

The ESDS symbol requires that all ESDS parts be handled in accordance with the ESDS device handling procedures in TO 00-25-234.

If equipment covered contains ESDS parts, the parts shall be marked with the ESDS symbol immediately preceding the description of the item. When approved by the acquiring activity, the symbol **ESD** may be used in lieu of the ESDS symbol.

8 IMPROVEMENT REPORTS.

Recommendations for improvements to prescribed requirements and procedures will be submitted by a Recommended Change (RC) or AFTO Form 22, Technical Order System Publication Improvement Report, in accordance with TO 00-5-1.

SAFETY SUMMARY

1 GENERAL SAFETY INSTRUCTIONS.

This safety summary includes general safety precautions and instructions that must be understood and applied during operation and maintenance to ensure personnel safety and protection of equipment. Prior to performing any task, the WARNINGs, CAUTIONs, and NOTEs included in that task shall be reviewed and understood.

2 WARNINGS, CAUTIONS, AND NOTES.

WARNINGs and CAUTIONs are used in this manual to highlight operating or maintenance procedures, practices, conditions, or statements considered essential to protection of personnel (WARNING) or equipment (CAUTION). WARNINGs and CAUTIONs immediately precede the step or procedure to which they apply. WARNINGs and CAUTIONs consist of four parts: a heading (WARNING, CAUTION, or Icon), a statement of the hazard, minimum precautions, and possible result if disregarded. NOTEs are used in this manual to highlight operating or maintenance procedures, practices, conditions, or statements that are not essential to protection of personnel or equipment. NOTEs may precede or follow the step or procedure, depending upon the information to be highlighted. The headings used and their definitions are as follows.

WARNING

Highlights an essential operating or maintenance procedure, practice, condition, statement, etc., which if not strictly observed, could result in injury to, or death of, personnel or long-term health hazards.

CAUTION

Highlights an essential operating or maintenance procedure, practice, condition, statement, etc., which if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

NOTE

Highlights an essential operating or maintenance procedure, condition, or statement.

3 HAZARDOUS MATERIAL WARNINGS.

Consult the Safety Data Sheets (SDS) (Occupational Safety and Health Administration (OSHA) Form 20 or equivalent) for specific information on hazards, effects, and protective equipment requirements. If you do not have an SDS for the material involved, contact your supervisor, or the base Safety or Bioenvironmental Engineering Offices.

4 SAFETY PRECAUTIONS.

The following are general safety precautions that are not related to any specific procedure and therefore do not appear elsewhere in this publication. These are precautions that personnel must understand and apply during the various phases of equipment operation and maintenance.

CAUTION

- EXERCISE CAUTION AROUND LIVE CIRCUITS. Operations personnel must observe safety regulations at all times. Do not replace components or make adjustments with power applied. Under certain conditions, dangerous potentials could still exist after power has been removed.
- DO NOT SERVICE ALONE. Under no circumstances should any persons perform maintenance on the equipment except in the presence of someone who is capable of rendering aid.
- RESUSCITATION. Personnel working with or near high voltages shall be familiar with modern methods of resuscitation. Information and training sources may be obtained from the Director of the Base Medical Services.
- DO NOT WEAR JEWELRY. Remove all rings, watches, and exposed pendant jewelry before system power is turned on. Serious injury may result from failure to observe this and other standard safety precautions.
- LIFTING/HANDLING. Improper handling or lifting can result in injury.
- CLEANERS, CHEMICALS, PRIMERS, OR PAINTS. Cleaner and Isopropyl Alcohol are toxic and skin irritants. When mixing and using ensure adequate ventilation, avoid prolonged breathing of vapor or mist, and avoid prolonged or repeated skin contact to prevent injury, physical disorder or death.

- PERSONAL PROTECTIVE EQUIPMENT (PPE). To prevent personal injury, adequate eye protection is to be worn when spraying connectors with solvent, using compressed air, or soldering.

- COMPRESSED AIR. Use of compressed air can create an environment of propelled particles. Do not direct air stream toward self or other personnel. Compressed air must be reduced to 30 pounds per square inch (psi) or less.

CHAPTER 1
DESCRIPTION

1.1 MAJOR CHARACTERISTICS.

The Thrust Control Unit (TCU) consists of an electrical equipment chassis containing three circuit cards (A2, A3, and A4), a plug-in component module (A1), and a plug-in Power supply module (PS1). (Figure 1-1). The module and chassis connectors are keyed to preclude insertion of a module into the wrong connector. The TCU has three external connectors: input power connector A1J1, input/output signal interface connector J2, and test connector J3. J3 is covered with a protective cap attached to a chain on the TCU. For aircraft mounting, the TCU has two guide holes (opposite the connector side) to accommodate two 1-1/8 inch long pins. The connector side of the LRU has a tab at the bottom for aircraft attachment.

1.2 GENERAL INFORMATION.

The purpose of the TCU is to control engine speed from idle to maximum speed and back to idle. The TCU converts electrical input signals and computes an engine speed error signal. This signal is applied to an electromechanical actuator which drives the engine fuel valve. The actuator provides a position feedback signal to the Pilots Thrust Control Quadrant (PQ). When the actuator feedback position signal equals the quadrant demand position signal, the position error voltage will drop to zero. The TCU also converts 230 VAC, 400 Hz aircraft power to +13 and -13 VDC, 28 VDC, 26 VAC and 150 VAC. Another function provided by the TCU is to supply several test outputs.

1.3 FACTUAL DATA.

Factual data of the TCU are given in Table 1-1.

Table 1-1. Factual Data

Description	Data
Dimensions:	
Width	3-9/16 inches
Height	7-9/16 inches
Depth	8-9/16 inches (10-9/16 inches with handle)
Weight	5.8 pounds (approximate)
Power Requirements	230 VAC, 400 Hz, 1-phase

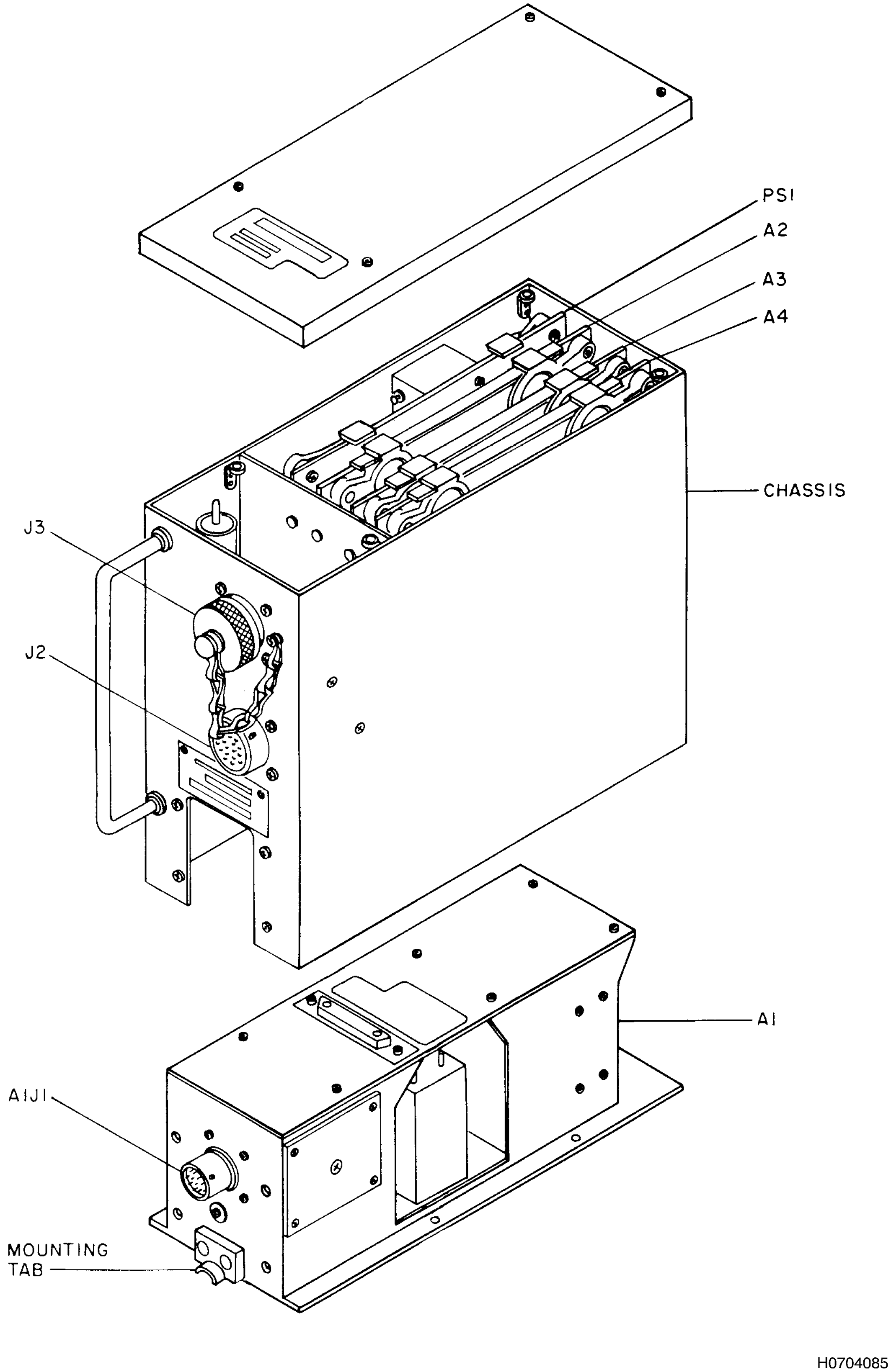

Figure 1-1. Thrust Control Unit

CHAPTER 2
SUPPORT EQUIPMENT AND LIST OF CONSUMABLES

2.1 GENERAL.

This chapter lists all the support equipment and consumables required for maintenance of the Thrust Control Unit (TCU).

2.2 SUPPORT EQUIPMENT.

Table 2-1 lsts the support equipment, with their part numbers and type designations, required for testing and maintaining the TCU. Common tools, such as screwdrivers, pliers, and sidecutters are not listed. These items are authorized support equipment required to perform the prescribed maintenance procedures. Use of authorized alternate parts is approved unless the restriction “NO SUBSTITUTIONS AUTHORIZED” appears in the “PART NUMBER” column. When “NO SUBSTITUTIONS AUTHORIZED” is specified, only the specified item shall be used. (Table 2-1).

2.3 CONSUMABLES.

Table 2-2 provides a list of consumables for cleaning and repairing the TCU. Consumable materials are expendable items and supplies consumed during component maintenance. All consumables not identified in Illustrated Parts Breakdown of Chapter 9 are listed in the List of Consumables, Table 2-2. Examples are: sealants, lubricants, cleaning solvents, paints, etc. Use of authorized alternate parts is approved unless the restriction “NO SUBSTITUTIONS AUTHORIZED” appears in the “PART NUMBER” column. When “NO SUBSTITUTIONS AUTHORIZED” is specified, only the specified item shall be used. (Table 2-2). In the text of this manual, consumables will be referenced by their index number.

2.4 LOCAL MANUFACTURED ITEMS.

There are no local manufactured items required.

Table 2-1. Support Equipment List

Part Number	Nomenclature	Type Designation	CAGE	Application
DMC621	Electrical Connector Maintenance Kit		11851	Repair
1432M	Decade Resistor Box		24655	Load Resistance
2465	Oscilloscope		80009	Peak to Peak Voltage Checks
275023/HT-900B	Heat Gun		06090	Repair
8012A	Multimeter, Digital		89536	Continuity Checks

Table 2-2. List of Consumables

Index No.	Nomenclature	Part Number	Specification	CAGE
1	Adhesive	RTV108	MIL-A-46106, Type I	01139
2	Alcohol, Isopropyl	--	TT-I-735	81348
3	Bag, ESD plastic	--	PPP-B-26	81348
4	Brush, fine camel hair	--	H-B-00377	
5	Brush, stiff bristle	--	MIL-B-43871	81349
6	Cap, dust, ESD	--	MS90376	96906
7	Cloth, crocus, fine grade	--	P-C-458	81348
8	Cloth, lint-free	--	CCC-C-46	81348
9	Coating, chemical conversion (Alodine)	--	MIL-C-5541	81349
10	Compound, cleaning, Trans-1,2- Dichloroethylene based solvent	--	--	--

Table 2-2. List of Consumables - Continued

Index No.	Nomenclature	Part Number	Specification	CAGE
11	Coating, Conformal	--	MIL-I-46058, Type UR	81349
12	Contacts		MS27491-22D	96906
13	Lacquer, lusterless gray, Color No. 36231	--	MIL-P-22808	81349
			FED STD-595	06542
14	Primer	SS4004	--	01139
15	Primer, zinc chromate	--	TT-P-1757	81348
16	Sleeving, heat shrinkable	--	MIL-I-23053	81349
17	Sleeving, insulation	--	MIL-I-22129	81349
18	Solder	--	QQ-S-571, WRMAP-2	81348
19	Solvent	Conap S-8	--	16245
20	Solvent	Girder No. 3A	--	13932
21	Tape, lacing, braided nylon	--	MIL-T-43435, Type I, Size 5, Finish B	81349
22	Wire, electrical	--	MIL-W-22759	81349

CHAPTER 3
THEORY OF OPERATION

3.1 GENERAL.

This chapter contains theory of operation of the Thrust Control Unit (TCU).

3.2 THEORY OF OPERATION.

The 26 VAC excitation to the rotor of the actuator control transmitter (CX) is transformed into 3-wire actuator shaft position information. (FO-1). It is transmitted to the pilot's thrust control lever (TCL) control transformer (CT). Any difference between the actuator shaft angle and the angle commanded by the TCL results in an error signal being generated in the rotor of the CT. The error signal is fed to the demodulator in the TCU. The rectified error signal is filtered to remove the 800 Hz fundamental frequency and any harmonics. It is amplified and applied to inputs of three level detectors. If the error voltage is positive and larger than the voltage corresponding to 2 degrees, the ±2.0 degree level detector will switch the brake driver off. The +0.2 degree level detector will switch the CW motor drive on. This condition releases the normal mode brake. It causes the motor to drive at full voltage and torque in a CW direction. This will cause the output shaft to drive the actuator CX rotor in a direction to reduce the error voltage. As the error voltage is reduced below a level corresponding to +2.0 degrees, the ±2.0 degree level detector causes the normal mode brake to be pulsed on and off. The rate and duty cycle of the pulser causes the shaft position error to be reduced from +2.0 degrees to less than +0.2 degrees with a minimum of overshoot. When the error has been reduced to +0.2 degrees, the +0.2 degree level detector switches off the motor drive (K1 off). It also removes excitation from the normal mode brake. With both the motor and brake de-energized, the output shaft is held in the commanded position until a new command is generated by the TCL. Operation for a command of the opposite polarity is directly analogous. The brake will stop the motor when the motor is excited at full voltage. The actuator idle switch, S1, disconnects the CCW motor drive when the TCL is moved to idle. This prevents inadvertent engine shutdown. Normal mode control is from idle to maximum augmentation and back to idle. TCU signal interfaces are shown in Figure 3-1.

3.2.1 Power Supply Module (PS1). Power supply module (PS1) receives 230 VAC, 400 Hz input voltage from component module (A1) (Figure 3-2). The stepdown transformer converts the 230 VAC to center tapped 31.0 VAC (15.5 VAC per half) and 26.0 VAC. The 15.5 VAC is full-wave rectified and choke-input filtered to produce plus and minus 13 VDC. The ±13 VDC is applied to the A2, A3, and A4 circuit card assemblies (CCAs). The +13 VDC is also applied to the pilots thrust control quadrant to excite the TCL potentiometers. The 26 VAC is applied to CCAs A2 and A4 and is used as a demodulation voltage. The 26 VAC is also used as the thrust control actuator synchro excitation voltage.

3.2.1.1 The ±13 VDC and 26 VAC supply voltages are applied to test connector J3 and are available as buffered test points.

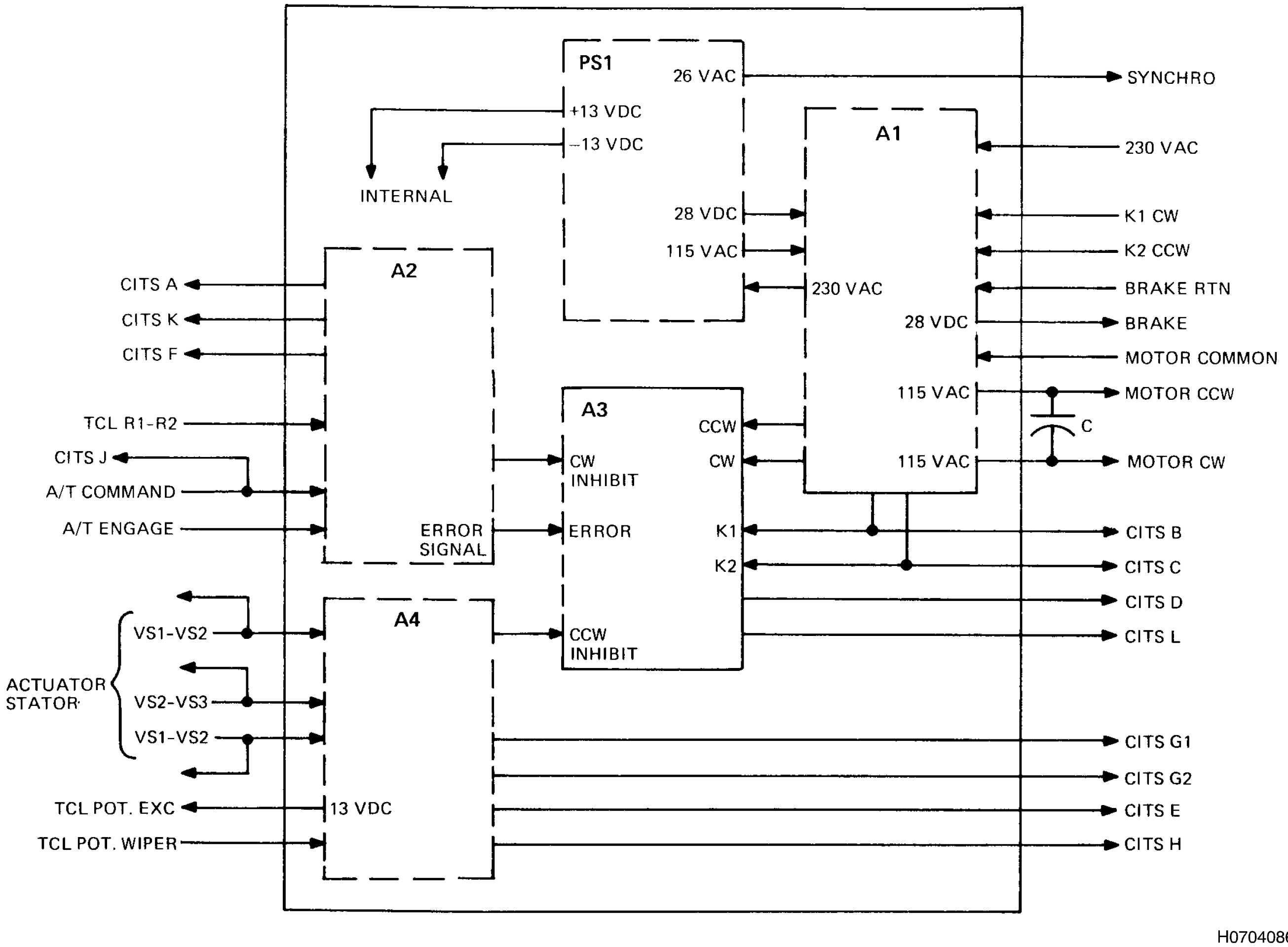

Figure 3-1. TCU Signal Interfaces

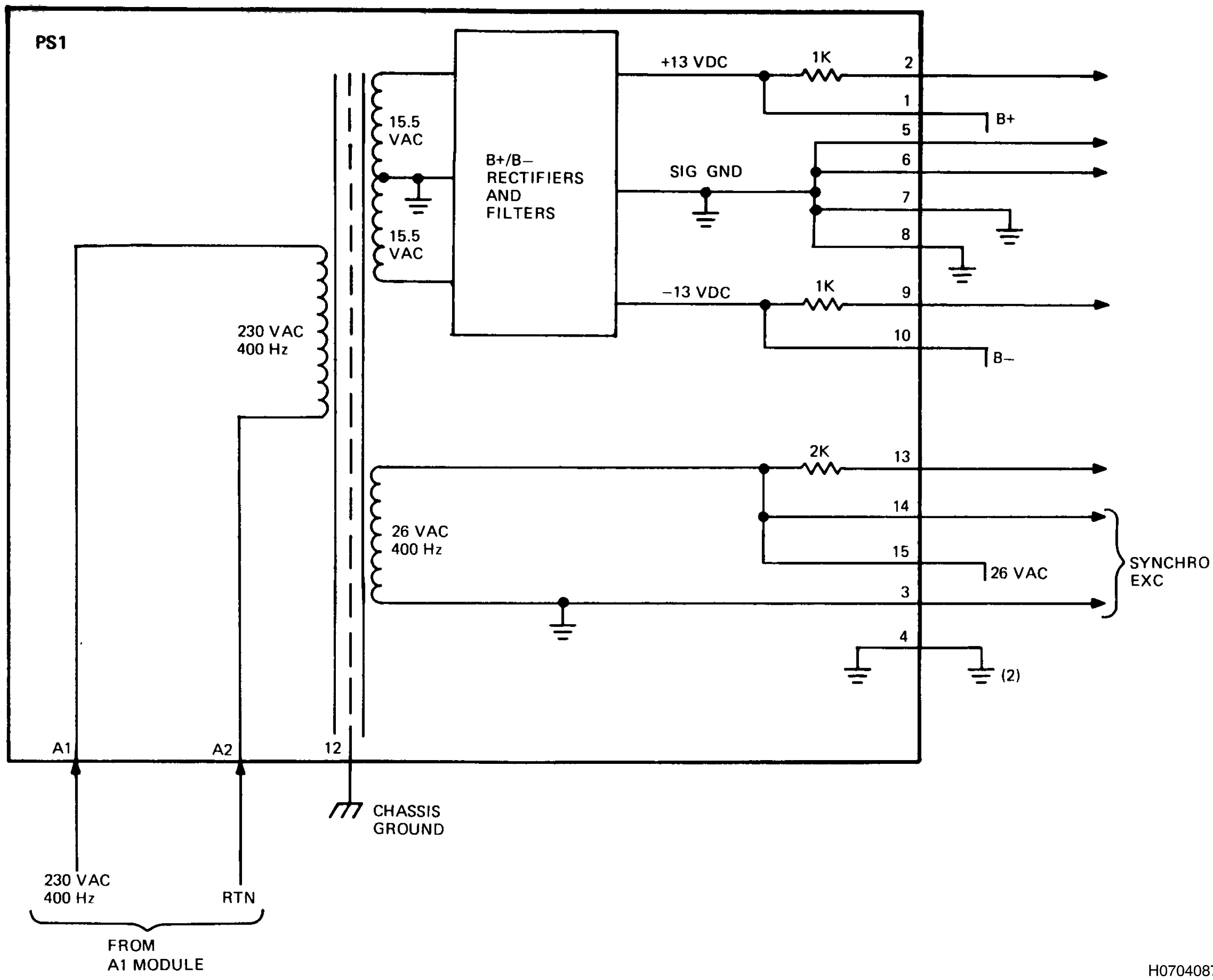

Figure 3-2. Power Supply Module (PS1) - Simplified Block Diagram

3.2.2 Component Module Assembly (A1). Connector J1 is installed on the front panel of component module assembly (A1) (Figure 3-3). 230 VAC 400 Hz to the TCU is applied to connector J1. The 230 VAC power is transmitted through fuse resistor FR1 to the A1 module stepdown transformer and to power supply PS1. The A1 stepdown transformer converts the 230 VAC to center-tapped 44.0 VAC (22.0 VAC per half) and 115 VAC.

3.2.2.1 The 44.0 VAC is full-wave rectified and capacitance filtered to produce +28 VDC. The 28 VDC is applied to TCU CCAs A2 and A3, and externally to the thrust control actuator. The 115 VAC is also applied externally to the thrust control actuator at the normal mode motor. Chassis mounted capacitor C1 provides 90 degrees of phase shift to provide the correct voltages for the two-phase motor.

3.2.2.2 The 10-ohm resistor in the motor common line is physically three 30-ohm resistors connected in parallel. It is used as a current-sense resistor. This current-sense resistor provides a voltage to CCA A3 to generate CITS L, the presence or absence of motor drive.

3.2.2.3 Relay K1 receives 28 VDC from pin A1-4. Relay K1 return line at A1-5 returns to a transistor switch driver on CCA A3. Pin A1-5 is low, to energize K1 when the actuator motor is to be driven CW.

3.2.2.4 Relay K2 receives 28 VDC from pin A1-12. Relay K2 return line at A1-13 is low to energize K2 when the actuator motor is to be driven CCW. When driving the actuator motor CCW, the 28 VDC power is removed from K2 when the motor reaches the idle position. This will prevent inadvertent engine shutdown.

3.2.2.5 When K1 and K2 are not energized, CITS B and CITS C are at +5 VDC. When K1 is energized, CITS B is at ground. When K2 is energized, CITS C is at ground. The K1 and K2 lines to the A3 CCA (A1-14 and A1-15) are used in AND gates to provide normal mode motor brake control.

3.2.2.6 A spare fuse resistor (25 ohms) is located on CCA A1. If 115 VAC, 28 VDC, +13 VDC, -13 VDC and 26 VAC are missing, check fuse resistor FR1. If FR1 is open, two wires can be unsoldered from FR1 and connected to spare fuse resistor below FR1.

3.2.3 Circuit Card Assembly (A2). CCA A2 provides the normal mode motor error amplifier circuits, five CITS outputs, and the CW inhibit signal for the motor (FO-2).

3.2.3.1 The pilots thrust control quadrant provides the synchro input to CCA A2 across pins A2-58 and A2-29. The synchro input voltage is designated VR1-R2 in (FO-2, Sheet 1). It is a 400 Hz sine wave that varies from 0 to 26 VAC. Its phase varies from 0 to -180° with respect to the 26 VAC reference input from power supply module (PS1).

3.2.3.2 Zener diodes VR1 and VR2 clip the 26 VAC reference voltage to <20 VP-P. Transistors Q1 and Q2 use the clipped 400 Hz sine wave to demodulate the synchro input. The clipped 400 Hz sine wave is also applied to CCA A4 and used as a demodulation voltage.

3.2.3.3 Amplifiers AR7 and AR6 amplify and filter the demodulated position error signal. The output voltage at pin A2-54 varies from 0 to ±12.5 VDC. The error amplifier provides a sum point for an autothrottle input in the Terrain Following and Landing Modes.

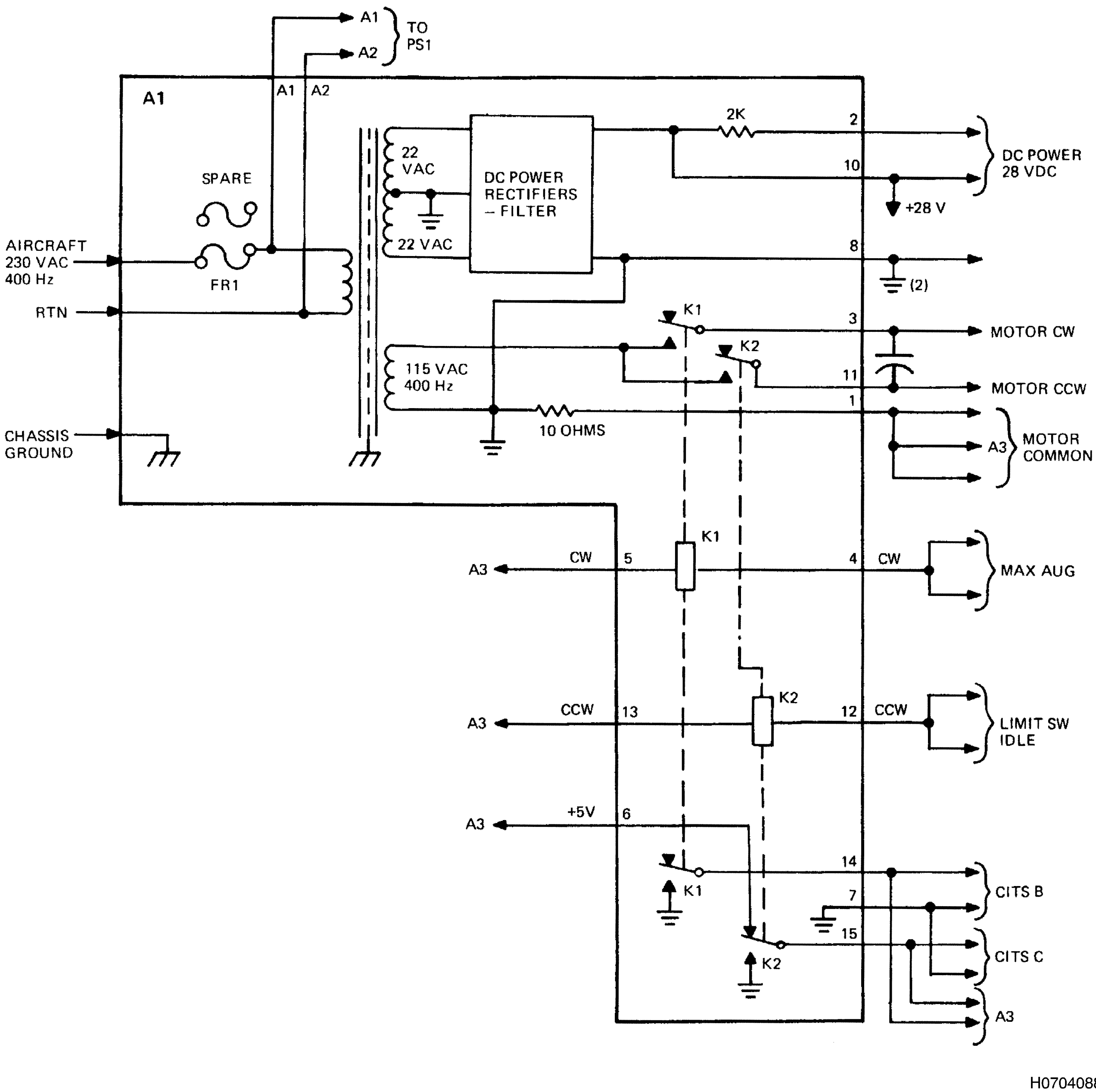

Figure 3-3. Component Module Assembly (A1) - Simplified Block Diagram

3.2.3.4 The TCL position is modified by the A/T Command signal from the Automatic Flight Control System (AFCS) applied to buffer AR13. The A/T Command signal across pins A2-33 and A2-2 varies between 0 VDC and ±10 VDC. CITS J monitors the AFCS A/T Command signal at pins A2-33 and A2-2.

3.2.3.5 The A/T Engage signal buffered by AR3 is a logic signal from the AFCS used to engage or disengage the A/T command input. The A/T Engage signal true is +3.0 to +3.5 VDC across pins A2-31 and A2-32, false is +1.5 to +2.0 VDC.

3.2.3.6 CITS A network resistors R3, R4, and R5 monitor the servo loop error signal at pin A2-54. The CITS A voltage across pins A2-11 and A2-14 is +2.5 ± 0.147 VDC when the error signal is within ±0.2°. The exact voltage at pin A2-11 varies as 0.1923 x [B+] +0.1222 x Vo, where B+ is +13 VDC from power supply module (PS1) and Vo is the voltage at pin A2-54. Vo can be plus or minus.

3.2.3.7 CITS K network resistors R1 and R2 monitor the Engine Thrust Control System (ETCS) Reference Voltage at pin A2-26. The CITS K voltage at A2-26 varies as 0.1914 x [B+].

3.2.3.8 CITS F buffer AR8 provides the A/T Limit by monitoring the A/T Range input from CCA A2 at pin A2-16. The CITS F voltage at pin A2-24 is 4.4 to 5.6 VDC when the A/T is ON and the actuator is >76.5°. The CITS F voltage at pin A2-24 is -0.1 to -0.7 VDC when either A/T is OFF or the actuator is <76.5°.

3.2.3.9 The CW inhibit logic consists of diodes CR8 and CR11 and -13 VDC bias resistor R48. The A/T Range and buffered A/T Engage are diode ORed and the biased result is output at pin A2-17. The CW inhibit signal output at pin A2-17 is +6.5 to +13.0 VDC when the motor is to be driven. When motor drive is to be inhibited (stopped), pin A2-17 is -6.5 to -8.7 VDC. The CW inhibit function is valid for an error signal at pin A2-54 up to +12.5 VDC.

a. If A/T Range at pin A2-16 is +7.6 to +13.0 VDC and A/T Engage across pins A2-31 and A2-32 is either true or false, pin A2-17 is +6.5 to +13.0 VDC and pin A2-24 is -0.1 to -0.7 VDC.

b. If A/T Range at pin A2-16 is -7.6 to -13.0 VDC and A/T Engage across pins A2-31 and A2-32 is false, pin A2-17 is +6.5 to +13.0 VDC and pin A2-24 is -0.1 to -0.7 VDC.

c. If A/T Range at A2-16 is -7.6 to -13.0 VDC and A/T Engage across pins A2-31 and A2-32 is true, pin A2-17 is -6.5 to -8.7 VDC (CW inhibit) and pin A2-24 is +4.4 to +5.6 VDC.

3.2.3.10 The Power Loss Detector function monitors the error amplifier signal at pin A2-55 and CCA A2's voltage sources. The Power Loss Detector output at pin A2-4 is +4.8 to +5.4 VDC if the +13V, -13V, +28V, and 26 VAC sources are present and the error signal has settled to less than 4 degrees in 4 seconds with A/T not engaged. If any of the voltage sources are not present or the error signal does not settle as described with the A/T not engaged, pin A2-4 will be at -0.1 to +0.1 VDC.

3.2.3.11 The detector filter AR11 filters the pin A2-55 error amplifier signal input to the timer/switch function. (FO-2, Sheet 2). The timer/switch signal input to the OR function indicates if the error signal does not settle correctly with the A/T not engaged.

3.2.3.12 The detector filter output drives comparator AR10. Comparator AR10 supplies timing pulses to timer circuit capacitor C16 and resistor R77. The timer circuit times out when the error signal does not settle correctly and the A/T is not engaged.

3.2.3.13 The A/T inhibit buffer AR2 buffers the CW inhibit logic signal at pin A2-17 from the timer/switch. The input from AR2 stops the timer circuit from timing out when the A/T is engaged.

3.2.3.14 Power loss detector output signal levels are determined by the setting of relay K1. Relay K1 is driven by OR function transistor Q6. Relay driver transistor Q6 receives its drive from the ±13V power loss function. The 26 VAC power loss input from AR12 and/or the timer/switch input from AR9 remove Q6's drive if their monitors detect a failure.

3.2.3.15 The +13V source supplied to the ±13V power loss function drives transistor Q6. Transistor Q5, biased by the -13V source, sinks the drive current for Q6 if the bias voltage source is lost.

3.2.3.16 Level detector AR12 output sinks Q6 drive current if the 26 VAC voltage rectified by diode CR14 on capacitor C14 drops due to loss of 26 VAC power. Comparator AR10 output sinks Q6 drive current if the voltage on capacitor C16 rises due to the timer circuit timing out.

3.2.3.17 A/T signal buffer AR13 provides -0.805 V/V gain from pin A2-35 to pin A2-33. (FO-2, Sheet 1). The line voltage pulse width multiplier (PWM) function, A/T non-linear function, and PWM function adjust the gradient of the buffered A/T Command before it is input to error amplifier AR7. These functions convert the A/T Command input DC voltage across pins A2-33 and A2-2 to a voltage that varies with the 26 VAC reference. The A/T Command input from the PWM function has the same voltage gradient as the TCL input from the position-error demodulator at the error amplifier summing point pin A2-9.

3.2.3.18 A/T Non-linear function amplifier AR5 and varactors VR5 and VR6 provide non-linear voltage gain from pin A2-7 to pin A2-35. The voltage gain is approximately -1.0 V/V until pin A2-7 voltage exceeds varactors VR5 and VR6 limits, at which point the gain changes.

3.2.3.19 The output of the A/T engage buffer at pin A2-34 is summed with the output of A/T signal buffer AR13 at the input to AR5. AR5's input to the PWM function is negatively saturated when the A/T engage buffer input at pin A2-31 is +1.5 to +2.0 VDC (A/T disengaged).

3.2.3.20 PWM transistor Q3 multiplies the A/T Non-linear function input to error amplifier AR7 with a voltage gain output by the line voltage PWM function.

3.2.3.21 Line Voltage PWM comparator AR14 supplies pulses to drive transistor Q3. One input to AR14 is the Demodulator reference voltage. The other input is the sum of the output of level detector AR4 which detects rectified 26 VAC and the output of the A/T engage buffer. AR14's drive to Q3 is held on and the output of the PWM function is held at ground when the A/T engage buffer input across pins A2-31 and A2-32 is +1.5 to +2.0 VDC (A/T disengaged).

3.2.3.22 The width of the pulses out of AR14 is dependent upon 26 VAC and the A/T Gain Trim potentiometer R51. R51 is adjusted at the factory under the following conditions:

A2-56	26.00 ±0.13 VAC
A2-31	+3.0 to +3.5 VDC
A2-33	+4.000 ±0.004 VDC
A2-58/29	6.43 ±0.010 VAC $\angle$-180° with respect to A2-56
A2-54	0.00 ±10 mVDC as adjusted by R51

3.2.3.23 With pin A2-31 at +1.5 to +2.0 VDC (A/T disengaged) and inputs applied at pin A2-58 to pin A2-29, the following outputs are obtained at pin A2-55 and pin A2-54:

Across Pins A2-58 and A2-29 (VAC)	Pin A2-55 (VDC)	Pin A2-54 (VDC)
1.000 $\angle$0°	-2.13 to -2.41	+5.80 to +6.55
0.000	0 ±0.042	0 ±0.040
1.000 $\angle$-180°	+2.13 to +2.41	-5.80 to -6.55

3.2.4 Circuit Card Assembly (A3). CCA A3 provides normal mode motor control and the corresponding CITS outputs. (Figure 3-4).

3.2.4.1 CCA A2 provides the position error signal input to CCA A3 at pin A3-57. The position error signal varies from 0.0 to ±12.5 VDC. Pin A3-57 provides the voltage to the +0.2° threshold CW comparator, the -0.2° threshold CCW comparator, and ±2.0° threshold brake comparator.

3.2.4.2 Resistors R1 and R2 establish the +0.2° threshold for CW comparator AR1. The output of AR1 at pin A3-30 provides the drive for motor driver CW transistors Q1 and Q2. The motor driver CW output signal (A1 CW) at pin A3-29 is applied to CCA A1s K1 relay.

a. K1 is de-energized (115 VAC not supplied to the motor) when pin A3-57 is ≤+0.50 VDC, pin A3-30 is at -6.5 to -13.0 VDC, and pin A3-29 is at +23.6 to +32.0 VDC.

b. The K1 relay is energized and 115 VAC is supplied to the normal mode motor when pin A3-57 is ≥+0.67VDC, pin A3-30 is at +6.5 to +13.0 VDC, and pin A3-29 is at +3.0 to +6.7 VDC.

3.2.4.3 CCA A2 provides the CW inhibit signal at pin A3-58 that is input to CW comparator AR1. The CW inhibit signal provides +6.5 to +13.0 VDC when the motor is to be driven CW and -6.5 to -8.7 VDC when motor drive is to be stopped. The CW inhibit stopping voltage stops motor CW drive for position error input less than +12.5 VDC at pin A3-57.

3.2.4.4 Resistor R9 establishes the -0.2° threshold for CCW comparator AR2. The output of AR2 at pin A3-27 provides the drive for Motor Driver CCW transistors Q3 and Q4. The Motor Driver CCW output signal (A1 CCW) at pin A3-26 is applied to CCA A1's K2 relay.

a. K2 is de-energized (115 VAC not supplied to the motor) when pin A3-57 ≥-0.48 VDC (more positive), pin A3-27 is at -6.5 VDC to -13.0 VDC, and pin A3-26 is at +23.6 to +32.0 VDC.

b. The K2 relay is energized and 115 VAC is supplied to the normal mode motor when pin A3-57 is ≤-0.69 VDC (more negative), pin A3-27 is at +6.5 to +13.0 VDC, and pin A3-26 is at +3.0 to +6.7 VDC.

3.2.4.5 CCA A4 provides the CCW inhibit signal at pin A3-28 that is input to CCW comparator AR2. The CCW inhibit signal provides signal ground when the motor is to be driven CCW and +8.3 VDC when motor drive is to be stopped. The CCW inhibit stopping voltage stops motor CCW drive for position error inputs greater (more positive) than -12.5 VDC at pin A3-57.

3.2.4.6 Zener diode VR2 generates from +13 V the +5 V CITS Reference voltage at pin A3-32. It is applied to CCA A1's K1 and K2 relay contacts. The voltage varies from +4.5 to +5.4 VDC.

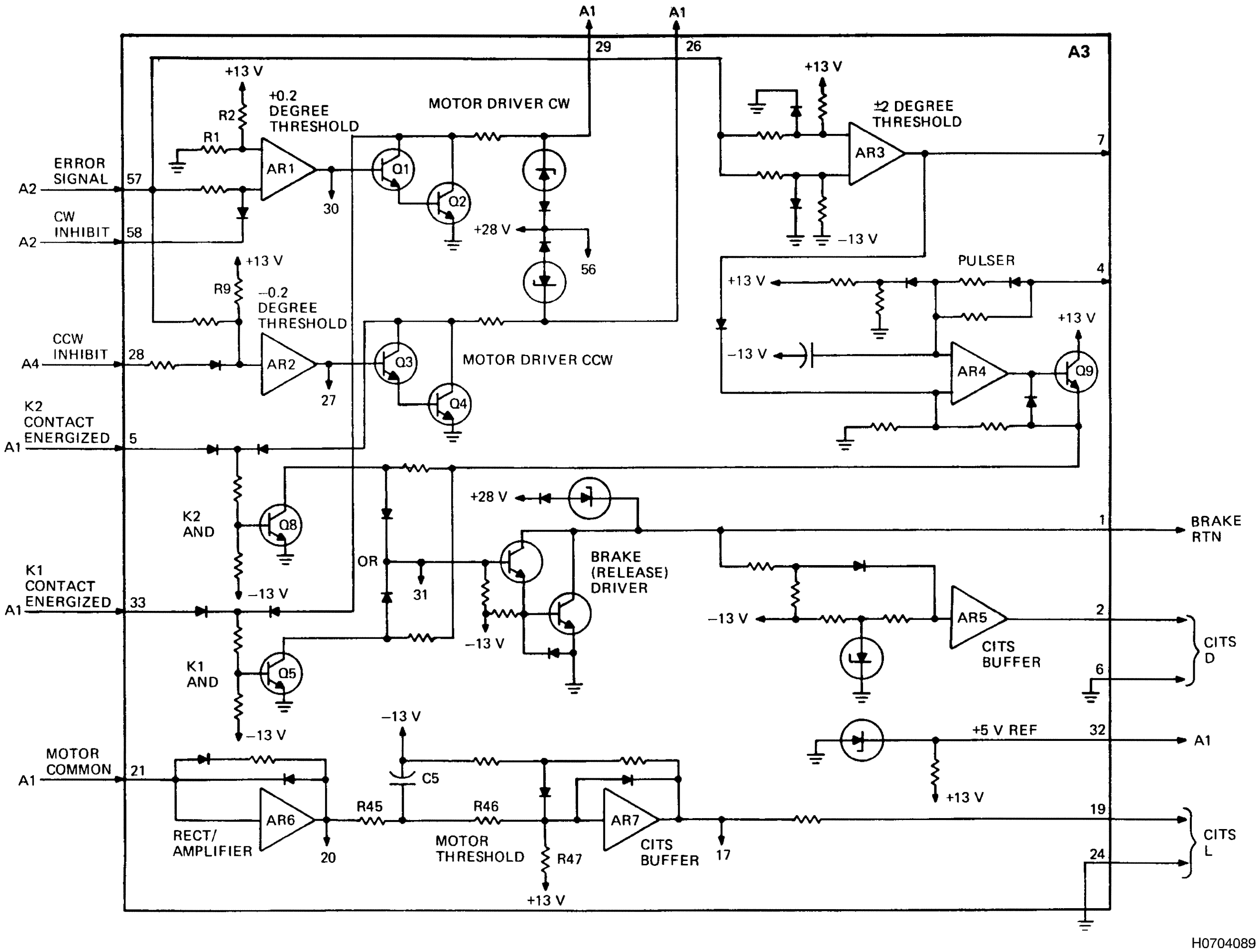

Figure 3-4. CCA A3 - Simplified Block Diagram

3.2.4.7 The ±2.0° threshold comparator AR3 signal output at pin A3-7 controls the pulser function. Pulser amplifier AR4 is a free-running oscillator. AR4's oscillations at pin A3-4 are input to the brake OR function at pin A3-31. The pulsed brake drive is output at pin A3-1 by brake driver transistors Q6 and Q7. Brake drive starts at ±2.0° of error signal at pin A3-57 and stops at ±0.2° of error signal (brake-on).

a. No oscillations are generated when the input at pin A3-57 is ≥+6.6 or ≤-6.6 VDC (more negative), the output at pin A3-7 is +9.0 to +13.0 VDC and the output at pin A3-4 is +8.0 to +12.6 VDC.

b. The pulser circuit is activated and oscillations are generated when the input at pin A3-57 is ≤+5.50 or ≥ -5.50 VDC (less negative), and the output at pin A3-7 is -9.5 to -13.2 VDC. The output at pin A3-4 is an asymmetrical square wave of +8.6 to +12.6 volts peak to -6.0 to -13.0 volts peak. The positive part of the waveform is 19.0 to 29.0 milliseconds and the negative part is 49.0 to 108.0 milliseconds.

3.2.4.8 The K2 and K1 AND circuits provide logic control for the brake release. The logic circuits remove the pulsed brake drive by sinking the current applied to the Brake OR at pin A3-31.

3.2.4.9 K1 AND transistor Q5 is driven by the Contact K1 Energized signal input at pin A3-33 or by the K1 Coil Energized signal at pin A3-29. K2 AND transistor Q8 is driven by the Contact K2 Energized signal input at pin A3- 5 or by the K2 Coil Energized signal at pin A3-26.

3.2.4.10 The voltage at pin A3-31 is +0.80 to +1.6 VDC, the brake driver transistors are turned on, the output at pin A3-1 is +6.25 VDC to 7.25 VDC and the CITS D output is +4.0 VDC to +5.4 VDC (brake energized and motor free to run) if either:

a. The motor is to be driven CW, the K1 relay is energized, the K1 contacts are closed, and the error signal at pin A3-57 is greater than +2.0°.

b. The motor is to be driven CCW, the K2 relay is energized, the K2 contacts are closed, and the error signal at pin A3-57 is less than -2.0° (more negative).

3.2.4.11 The voltage at pin A3-31 is -0.05 to -0.56 VDC and the brake driver transistors are turned off if any of the “AND” inputs for K1 and K2 are at the opposite state. For example, if coils are not energized, contacts are not closed, or error signal is less than the absolute value of ±0.2°, the output at pin A3-1 will be +23.9 to +32.0 VDC and the CITS D output of buffer AR5 will be at 0.0 to +0.5 VDC (brake on).

3.2.4.12 The voltage developed across the motor current sense resistor (10 ohms) of CCA A1 is input at pin A3-21. The rectifier-amplifier circuit AR6, R45, and C5 provides a voltage at pin A3-20 when motor current is present. Voltage greater than the voltage threshold provided by R46 and R47 is detected by comparator AR7 and output as the CITS L output from pin A3-19 to A3-24.

a. When the motor is not driven (assume 0.1 VAC of noise) the voltage at pin A3-20 is -0.4 to -0.5 VDC and the CITS L output is -1.0 to +0.5 VDC.

b. When the motor is being driven, the voltage at pin A3-21 is 0.34 VAC, the output at pin A3-20 is -1.42 to -1.64 VDC, and the CITS L output is +4.0 to +6.0 VDC.

3.2.5 Circuit Card Assembly (A4).

3.2.5.1 CCA A4 provides the circuits for A/T Range, CCW inhibit, CITS G1 and CITS G2 outputs, and CITS E and CITS H outputs. (Figure 3-5). A/T Range is applied to CCA A2 and combined with the A/T Engage function to produce the A/T limit function CITS F. CCW inhibit is applied to CCA A3 and used to inhibit motor drive below idle.

3.2.5.2 The Actuator provides three synchro inputs to CCA A4. Theses are designated synchro voltages: VS1-S3 at pin A4-8, VS2-S3 at pin A4-1, and VS1-S2 (signal ground) at pin A4-54. (Figure 3-5). CCA A2 provides a demodulation reference voltage (clipped sine wave of less than 20 volts peak-peak) used to demodulate VS1-S3 and VS2-S3.

3.2.5.3 Transistor Q1 and amplifier AR1 demodulate VS1-S3 and the result is filtered by AR2. The DC voltage output by AR2 at pin A4-35 is nominally -3.608 VDC if VS1-S3 is 12.0 VAC and in phase with the demodulation reference. An out-of-phase signal of 12.0 VAC at VS1-S3 produces +3.608 VDC at pin A4-35.

3.2.5.4 Transistor Q2 and amplifier AR5 demodulate VS2-S3 and the result is filtered by AR6. The DC voltage output by AR6 at pin A4-32 is nominally +3.608 VDC if VS1-S3 is 12.0 VAC and in phase with the demodulation reference. An out-of-phase signal of 12.0 VAC at VS2-S3 produces -3.608 VDC at pin A4-32.

3.2.5.5 The demodulated and filtered DC voltages are combined in comparators to produce the A/T Range, CCW inhibit, CITS G1, and CITS G2 outputs.

3.2.5.6 Comparator AR8 provides the A/T Range output signal at pin A4-24. The A/T Range output is at negative saturation (-7.6 to -13.0 VDC) with VS2-S3 at 10.0 VAC (out-of-phase) and VS1-S3 at 0.125 VAC (in phase). The A/T Range output is at positive saturation (+7.6 to +13.0 VDC) with VS1-S3 at 0.300 VAC (in phase).

3.2.5.7 Comparator AR7 provides the CCW inhibit output signal at pin A4-23. The CCW inhibit output is at negative saturation (-7.6 VDC to -13.0 VDC) with VS1-S3 at 11.8 VAC (in phase) and VS2-S3 at 3.83 VAC (in phase). The CCW inhibit output is at positive saturation (+7.6 to +13.0 VDC) with VS2-S3 at 4.16 VAC (in phase).

3.2.5.8 LDG Gear Warning Trim adjustment resistor R52 is a factory-set adjustment. It is set with VS2-S3 at 10.0 VAC (out-of-phase) and VS1-S3 at 1.93 VAC (in phase). R52 is adjusted to set the voltage at which comparator AR3 switches the CITS G1 output voltage level.

3.2.5.9 Buffer AR4 provides the CITS G1 output at pin A4-33. Buffer AR12 provides the CITS G2 output at pin A4-14. The CITS G1 output at pin A4-33 switches from the low state (-1.0 to +0.5 VDC) to the high state (+4.0 to +6.0 VDC) as indicated below:

a. CITS G1 is at +4.0 to +6.0 VDC and CITS G2 is at -1.0 to +0.5 VDC with VS2-S3 at 10.0 VAC (out-of-phase) and VS1-S3 at 2.14 VAC (in phase).

b. CITS G1 is at -1.0 to +0.5 VDC and CITS G2 is at +11.0 to +14.0 VDC with VS2-S3 at 10.0 VAC (out-of-phase) and VS1-S3 at 1.72 VAC (in phase).

3.2.5.10 The TCU provides +13 VDC excitation to the TCL potentiometer in the pilots thrust control quadrant. The TCL potentiometer’s wiper voltage input across pins A4-58 and A4-57 indicates TCL position. The wiper voltage varies from +0.8 to +3.7 VDC with the rig point of 25.7° equal to +2.35 VDC.

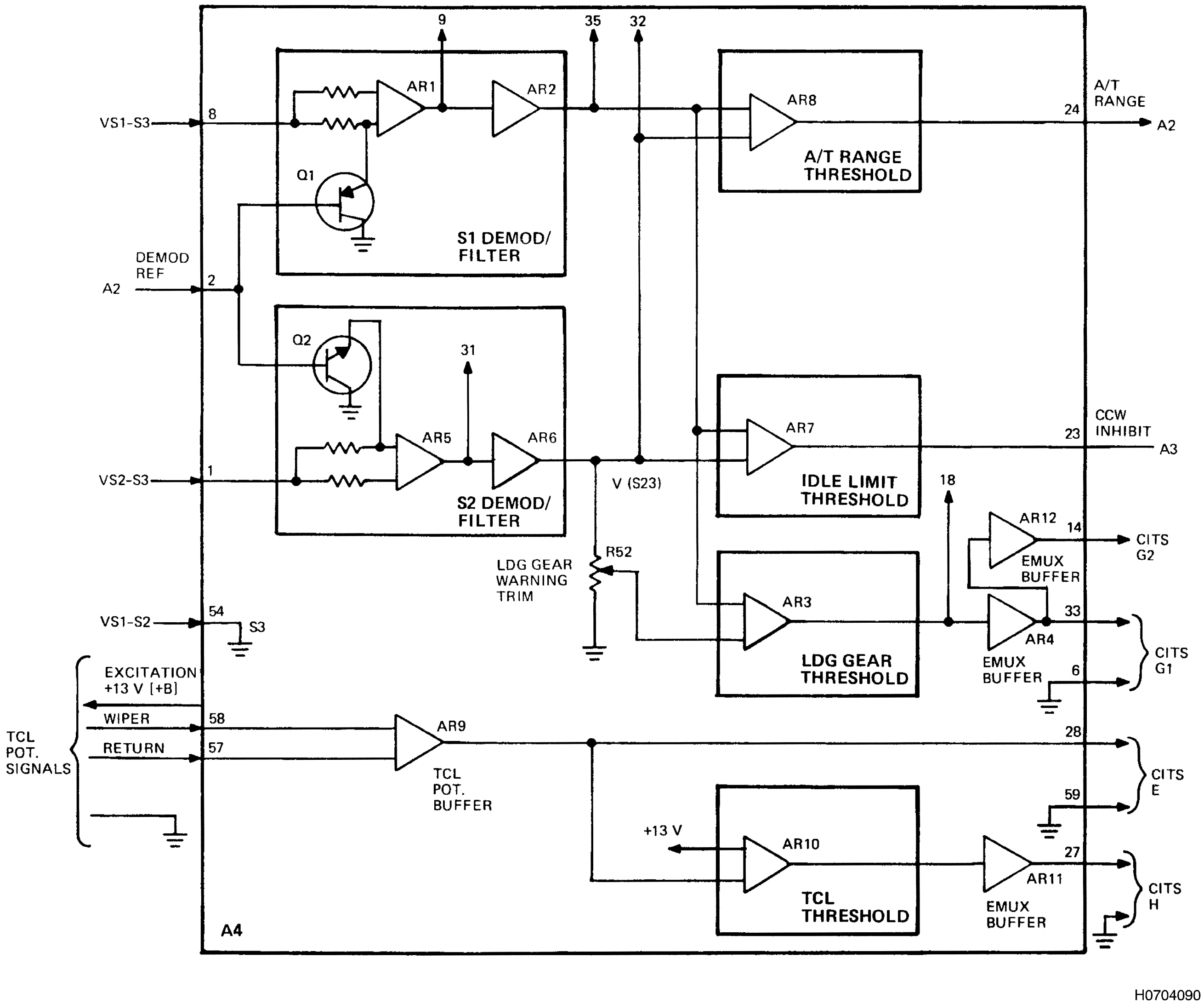

Figure 3-5. CCA A4 - Simplified Block Diagram

3.2.5.11 Buffer AR9 provides a gain of +0.870 to the wiper voltage. The result is output to comparator AR10 and to the CITS computer as CITS E at pin A4-28.

3.2.5.12 Comparator AR10 switches the CITS H output. CITS H is the EMUX A/T Disengage voltage. Buffer AR11 provides the CITS H output at pin A4-27. The CITS H output switches from the low state (-1.0 to +0.5 VDC) to the high state (+4.0 to +6.0 VDC) as indicated below:

a. CITS H is at +4.0 to +6.0 VDC when the TCL is below 26 to 28° (Actuator position at 790). CITS H is high when pin A4-58 is ≤ 0.212 x [B+], where B+ is +13 VDC.

b. CITS H is at -1.0 to +0.5 VDC when the TCL is greater than 26 to 28°. CITS H is low when pin A4-58 is ≥0.219 X [B+].

CHAPTER 4
DESCRIPTION OF SYSTEM, TIE-IN OF EQUIPMENT AND ACCESSORIES

4.1 GENERAL.

Refer to TO 1B-1B-2-76GS-00-1 for description of system tie-in of equipment and accessories.

CHAPTER 5
PREPARATION FOR MAINTENANCE

5.1 GENERAL.

This chapter includes instructions required for unpacking the Thrust Control Unit (TCU) and preparing it for maintenance.

5.2 UNPACKING.

Unpacking instructions for the TCU are as follows:

a. Remove TCU from packing case and retain all packing material for reuse for shipment or storage purposes.

b. Inspect the TCU for possible physical damage prior to test in accordance with Chapter 7.

5.3 CLEANING.

There are no special cleaning instructions required prior to maintenance.

CHAPTER 6
CHECKOUT AND TROUBLESHOOTING

6.1 GENERAL.

For checkout and troubleshooting the Thrust Control Unit (TCU) refer to TO 2JA8-28-8-1 when using the Intermediate Automatic Test Equipment (IATE) and TO 2JA8-28-8-6 when using the Advanced Digital Test Station (ADTS).

CHAPTER 7
MAINTENANCE/REPAIR INSTRUCTIONS

7.1 GENERAL.

Intermediate maintenance/repair instructions for the Thrust Control Unit (TCU) include limited maintenance (repair), inspection, malfunction isolation, disassembly, cleaning/corrosion prevention, lubrication, reassembly, and test.

7.2 LIMITED MAINTENANCE INSTRUCTIONS.

Refer to Table 7-1 for limited maintenance instructions for the thrust control unit.

7.2.1 Repair of Harness Assembly. Repair of the thrust control unit harness assembly consists of replacement of external and internal connectors and wiring. Refer to Table 7-2.

7.2.2 External Connectors J2 and J3. Several components will need to be partially removed from the chassis assembly (Figure 7-1, Sheet 2, 45) to allow access to connectors J2 and J3 (28 and 29). Proceed as follows:

a. Disassemble the thrust control unit in accordance with the procedures of Paragraph 7.4.

b. Remove screw (14) to free power inductor L1 (15). Do not desolder L1 wiring.

c. Loosen nut (16) and lock washer (17) to free radio interference filter FL1 (18). Do not desolder FL1 wiring.

d. Remove screw (19) to free capacitor assembly C1 (20). Do not desolder capacitor C1 wiring.

e. Remove screw (21), nut (22), lock washer (23) and flat washer (24) to free loop clamp (25). Removing loop clamp (25) will free the cabling between connectors J2 and J3 (28 and 29).

f. Disconnect protective cap (44) from connector J3 (29).

g. Remove screws (26) and flat washers (27) to free connectors J2 and J3 (28 and 29) from chassis assembly (45).

h. Push connectors J2 and J3 (28 and 29) into chassis. Carefully work the connectors, power inductor L1 (15), radio interference filter FL1 (18), capacitor assembly C1 (20) and the harness cabling to bottom of chassis assembly (45) to allow access to the rear of connectors J2 and J3 (28 and 29).

i. Loosen connector rings on backs of connectors J2 and J3 (28 and 29).

j. Using electrical connector maintenance kit (Table 2-1), remove contacts from connector to be replaced. Tag each wire with its location as it is removed.

k. When all contacts with wires attached have been removed from connector being replaced record the locations of unused contacts.

l. In replacement connector, insert contacts (Table 2-2, No. 12) and sealing plugs (30) into unused locations recorded in Step k.

m. Install rear connector ring over cabling.

Table 7-1. Thrust Control Unit Limited Maintenance Chart

Part	Procedure	Checkout/Alignment
Harness Assembly	Paragraph 7.2.1	Wire check in accordance with Table 7-2.
Connectors J2 and J3	Replace per Paragraph 7.2.2	Wire check (Table 7-2)
Connectors XA2, XA3, and XA4	Replace per Paragraph 7.2.3	Wire check (Table 7-2)
Connector XPS1	Replace per Paragraph 7.2.3.1	Wire check (Table 7-2)
Connector XA1	Replace per Paragraph 7.2.3.2	Wire check (Table 7-2)
Fuse Resistor FR1	Replace per Paragraph 7.2.3.3	Perform Fuse Diagnostic Testing per TO 2JA8-28-8-1 when using the Intermediate Automatic Test Equipment (IATE) and TO 2JA8-28-8-6 when using the Advanced Digital Test Station (ADTS).

Table 7-2. Control Unit Harness, Wire Run List

From	To	Color	AWG Size
C1-1	J2-1	GRA	24
C1-1	XA1-3	GRA	24
C1-2	J2-3	GRA	24
C1-2	XA1-11	GRA	24
E1	J2-66	WHT	26
E1	J3-19	BLK	24
E1	XA2-12	WHT	26
E1	XPS1-12	BLK	24
FL1-1	J3-28	RED	26
FL1-1	XA3-1	RED	26
FL1-2	J2-6	RED	26
J2-1	C1-1	GRA	24
J2-2	L1-1	BLK	24
J2-3	C1-2	GRA	24
J2-4	XPS1-14	GRA	24
J2-5	XPS1-3	WHT	26
J2-6	FL1-2	RED	26
J2-6	J3-33	RED	26
J2-7	J3-32	RED	26
J2-7	XA1-12	RED	26
J2-8	XA1-10	RED	26
J2-9	J3-31	RED	26
J2-9	XA1-4	RED	26
J2-10	J3-16	WHT	26
J2-10	XA2-58	WHT	26
J2-11	J3-17	WHT	26
J2-11	XA2-29	WHT	26
J2-17	J2-18	WHT	26
J2-17	J3-13	WHT	26
J2-18	J2-17	WHT	26
J2-18	XA4-8	WHT	26
J2-19	J2-20	WHT	26
J2-19	XA4-1	WHT	26
J2-20	J2-19	WHT	26
J2-20	J3-14	WHT	26
J2-21	J2-22	WHT	26
J2-21	J2-33	WHT	26
J2-22	J2-21	WHT	26
J2-22	J3-15	WHT	26
J2-23	XPS1-8	WHT	26
J2-24	XA4-47	RED	26
J2-25	XA3-24	WHT	26
J2-26	J2-32	WHT	26
J2-27	XA2-14	WHT	26
J2-28	J2-29	WHT	26
J2-29	J2-28	WHT	26
J2-29	J2-30	WHT	26
J2-30	J2-29	WHT	26
J2-30	XA1-7	WHT	26

Table 7-2. Control Unit Harness, Wire Run List - Continued

From	To	Color	AWG Size
J2-31	XA4-59	WHT	26
J2-32	J2-26	WHT	26
J2-32	XA2-25	WHT	26
J2-33	J2-21	WHT	26
J2-33	XA4-54	WHT	26
J2-34	XA3-19	WHT	26
J2-35	XA2-26	WHT	26
J2-36	XA2-11	WHT	26
J2-37	XA1-14	WHT	26
J2-38	XA1-15	WHT	26
J2-39	XA3-2	WHT	26
J2-40	XA4-28	WHT	26
J2-41	XA2-24	WHT	26
J2-42	XA4-27	WHT	26
J2-51	J3-35	RED	26
J2-51	XA2-4	WHT	26
J2-52	J3-8	WHT	26
J2-52	XA2-31	WHT	26
J2-53	J2-54	WHT	26
J2-53	XA2-33	WHT	26
J2-54	J2-53	WHT	26
J2-54	J3-21	WHT	26
J2-55	XA4-14	WHT	26
J2-56	J3-6	WHT	26
J2-56	XA4-58	WHT	26
J2-57	XA4-33	WHT	26
J2-58	J3-9	WHT	26
J2-58	XA2-32	WHT	26
J2-59	J2-60	WHT	26
J2-59	XA2-2	WHT	26
J2-60	J2-59	WHT	26
J2-60	J3-22	WHT	26
J2-62	J3-7	WHT	26
J2-62	XA4-57	WHT	26
J2-63	XA4-6	WHT	26
J2-65	XPS1-6	WHT	26
J2-66	F1	WHT	26
J3-1	XA1-2	RED	26
J3-2	XPS1-13	GRA	24
J3-3	XPS1-2	RED	26
J3-4	XPS1-9	VIO	26
J3-5	XA4-2	WHT	26
J3-6	J2-56	WHT	26
J3-7	J2-62	WHT	26
J3-8	J2-52	WHT	26
J3-9	J2-58	WHT	26
J3-10	XA3-58	WHT	26
J3-11	XA4-23	WHT	26

Table 7-2. Control Unit Harness, Wire Run List - Continued

From	To	Color	AWG Size
J3-12	XA3-57	WHT	26
J3-13	J2-17	WHT	26
J3-14	J2-20	WHT	26
J3-15	J2-22	WHT	26
J3-16	J2-10	WHT	26
J3-17	J2-11	WHT	26
J3-18	XPS1-5	WHT	26
J3-19	E1	BLK	24
J3-20	XA1-8	WHT	26
J3-21	J2-54	WHT	26
J3-22	J2-60	WHT	26
J3-23	XA2-3	WHT	26
J3-24	XA2-1	WHT	26
J3-25	XA4-24	WHT	26
J3-26	XA1-5	WHT	26
J3-27	XA1-13	WHT	26
J3-28	FL1-1	RED	26
J3-29	L1-2	BLK	24
J3-30	XA3-32	WHT	26
J3-31	J2-9	RED	26
J3-32	J2-7	RED	26
J3-33	J2-6	RED	26
J3-35	J2-51	RED	26
L1-1	J2-2	BLK	24
L1-2	J3-29	BLK	24
L1-2	XA1-1	BLK	24
XA1-A1	XPS1-A1	GRA	24
XA1-A2	XPS1-A2	GRA	24
XA1-1	L1-2	BLK	24
XA1-1	XA3-21	BLK	24
XA1-2	J3-1	RED	26
XA1-3	C1-1	GRA	24
XA1-4	J2-9	RED	26
XA1-5	J3-26	WHT	26
XA1-5	XA3-29	WHT	26
XA1-6	XA2-5	WHT	26
XA1-7	J2-30	WHT	26
XA1-7	XA3-6	WHT	26
XA1-8	J3-20	WHT	26
XA1-8	XPS1-4	WHT	26
XA1-10	J2-8	RED	26
XA1-10	XA3-56	RED	26
XA1-11	C1-2	GRA	24
XA1-12	J2-7	RED	26
XA1-13	J3-27	WHT	26
XA1-13	XA3-26	WHT	26
XA1-14	J2-37	WHT	26
XA1-14	XA3-33	WHT	26

Table 7-2. Control Unit Harness, Wire Run List - Continued

From	To	Color	AWG Size
XA1-15	J2-38	WHT	26
XA1-15	XA3-5	WHT	26
XA2-1	J3-24	WHT	26
XA2-2	J2-59	WHT	26
XA2-3	J3-23	WHT	26
XA2-4	J2-51	WHT	26
XA2-5	XA1-6	WHT	26
XA2-5	XA3-32	WHT	26
XA2-8	XA3-56	RED	26
XA2-11	J2-36	WHT	26
XA2-12	E1	WHT	26
XA2-14	J2-27	WHT	26
XA2-16	XA4-24	WHT	26
XA2-17	XA3-58	WHT	26
XA2-24	J2-41	WHT	26
XA2-25	J2-32	WHT	26
XA2-25	XPS1-8	WHT	26
XA2-26	J2-35	WHT	26
XA2-29	J2-11	WHT	26
XA2-31	J2-52	WHT	26
XA2-32	J2-58	WHT	26
XA2-33	J2-53	WHT	26
XA2-47	XA3-47	RED	26
XA2-47	XPS1-1	RED	26
XA2-49	XA3-49	VIO	26
XA2-49	XPS1-10	VIO	26
XA2-54	XA3-57	WHT	26
XA2-56	XPS1-15	GRA	24
XA2-57	XA4-2	WHT	26
XA2-58	J2-10	WHT	26
XA3-1	FL1-1	RED	26
XA3-2	J2-39	WHT	26
XA3-5	XA1-15	WHT	26
XA3-6	XA1-7	WHT	26
XA3-19	J2-34	WHT	26
XA3-21	XA1-1	BLK	24
XA3-24	J2-25	WHT	26
XA3-25	XPS1-7	WHT	26
XA3-26	XA1-13	WHT	26
XA3-28	XA4-23	WHT	26
XA3-29	XA1-5	WHT	26
XA3-32	J3-30	WHT	26
XA3-32	XA2-5	WHT	26
XA3-33	XA1-14	WHT	26
XA3-47	XA2-47	RED	26
XA3-49	XA2-49	VIO	26
XA3-55	XPS1-4	WHT	26
XA3-56	XA1-10	RED	26

Table 7-2. Control Unit Harness, Wire Run List - Continued

From	To	Color	AWG Size
XA3-56	XA2-8	RED	26
XA3-57	J3-12	WHT	26
XA3-57	XA2-54	WHT	26
XA3-58	J3-10	WHT	26
XA3-58	XA2-17	WHT	26
XA4-1	J2-19	WHT	26
XA4-2	J3-5	WHT	26
XA4-2	XA2-57	WHT	26
XA4-6	J2-63	WHT	26
XA4-8	J2-18	WHT	26
XA4-14	J2-55	WHT	26
XA4-23	J3-11	WHT	26
XA4-23	XA3-28	WHT	26
XA4-24	J3-25	WHT	26
XA4-24	XA2-16	WHT	26
XA4-25	XPS1-7	WHT	26
XA4-27	J2-42	WHT	26
XA4-28	J2-40	WHT	26
XA4-33	J2-57	WHT	26
XA4-47	J2-24	RED	26
XA4-47	XPS1-1	RED	26
XA4-49	XPS1-10	VIO	26
XA4-54	J2-33	WHT	26
XA4-57	J2-62	WHT	26
XA4-58	J2-56	WHT	26
XA4-59	J2-31	WHT	26
XPS1-A1	XA1-A1	GRA	24
XPS1-A2	XA1-A2	GRA	24
XPS1-1	XA2-47	RED	26
XPS1-1	XA4-47	RED	26
XPS1-2	J3-3	RED	26
XPS1-3	J2-5	WHT	26
XPS1-4	XA1-8	WHT	26
XPS1-4	XA3-55	WHT	26
XPS1-5	J3-18	WHT	26
XPS1-6	J2-65	WHT	26
XPS1-7	XA3-25	WHT	26
XPS1-7	XA4-25	WHT	26
XPS1-8	J2-23	WHT	26
XPS1-8	XA2-25	WHT	26
XPS1-9	J3-4	VIO	26
XPS1-10	XA2-49	VIO	26
XPS1-10	XA4-49	VIO	26
XPS1-12	E1	BLK	24
XPS1-13	J3-2	GRA	24

Table 7-2. Control Unit Harness, Wire Run List - Continued

From	To	Color	AWG Size
XPS1-14	J2-4	GRA	24
XPS1-15	XA2-56	GRA	24

NOTE

All wires are stranded, silver-plated copper with polytetrafluoroethylene insulation.

n. Install removed contacts into replacement connector using electrical connector maintenance kit (Table 2-1). If contacts are to be replaced use contacts (Table 2-2, No. 12).

o. Remove tags from connector wiring.

p. Tighten ring on rear of replaced connector.

q. Replace any cable lacing which may have been removed using lacing tape (Table 2-2, No. 21).

r. Carefully reinsert inductor L1 (15), radio interference filter FL1 (18), capacitor assembly C1 (20), the harness cabling, and connectors J2 and J3 (28 and 29) into chassis assembly (45).

s. Insert connectors J2 and J3 (28 and 29) into their mounting positions on chassis assembly (45). The master keyway of connector J2 (28) is positioned at 12 o'clock. The master keyway of connector J3 (29) is positioned at 1 o'clock.

t. Loosely attach connectors J2 and J3 (28 and 29) to chassis assembly (45) using screws (26) and flat washers (27).

u. Loosely secure harness cable in loop clamp (25) and loosely secure loop clamp (25) to chassis assembly (45) using screw (21), flat washer (24), lock washer (23), and nut (22).

v. Install radio interference filter FL1 (18) into its mounting bracket using lock washer (17) and nut (16).

w. Install capacitor assembly C1 (20) in chassis assembly (45) using screw (19).

x. Install power inductor L1 (15) in chassis assembly (45) using screw (14).

y. If necessary, arrange harness cabling in chassis assembly (45) to relieve any wiring stress. Tighten screw (21) to secure cable harness to chassis assembly (45).

z. Check that master keyway of connector J2 (28) is at 12 o'clock and the master keyway of connector J3 (29) is at 1 o'clock. Tighten eight screws (26).

aa. Install protective cap (44) on connector J3 (29).

ab. Wire check installation in accordance with Table 7-2.

ac. Reassemble the thrust control unit in accordance with the procedures of Paragraph 7.6.4.

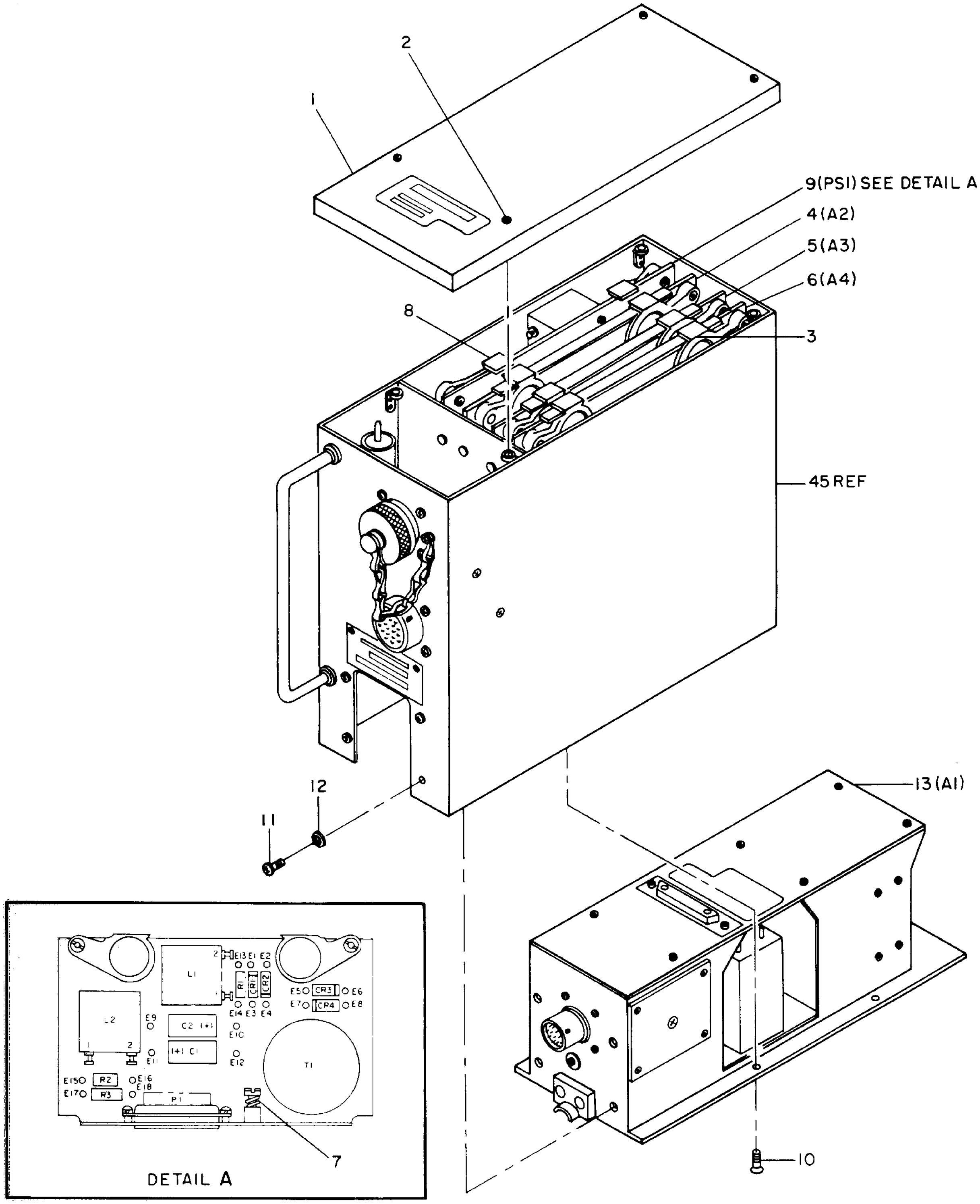

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Figure 7-1. Thrust Control Unit - Exploded View (Sheet 1 of 2)

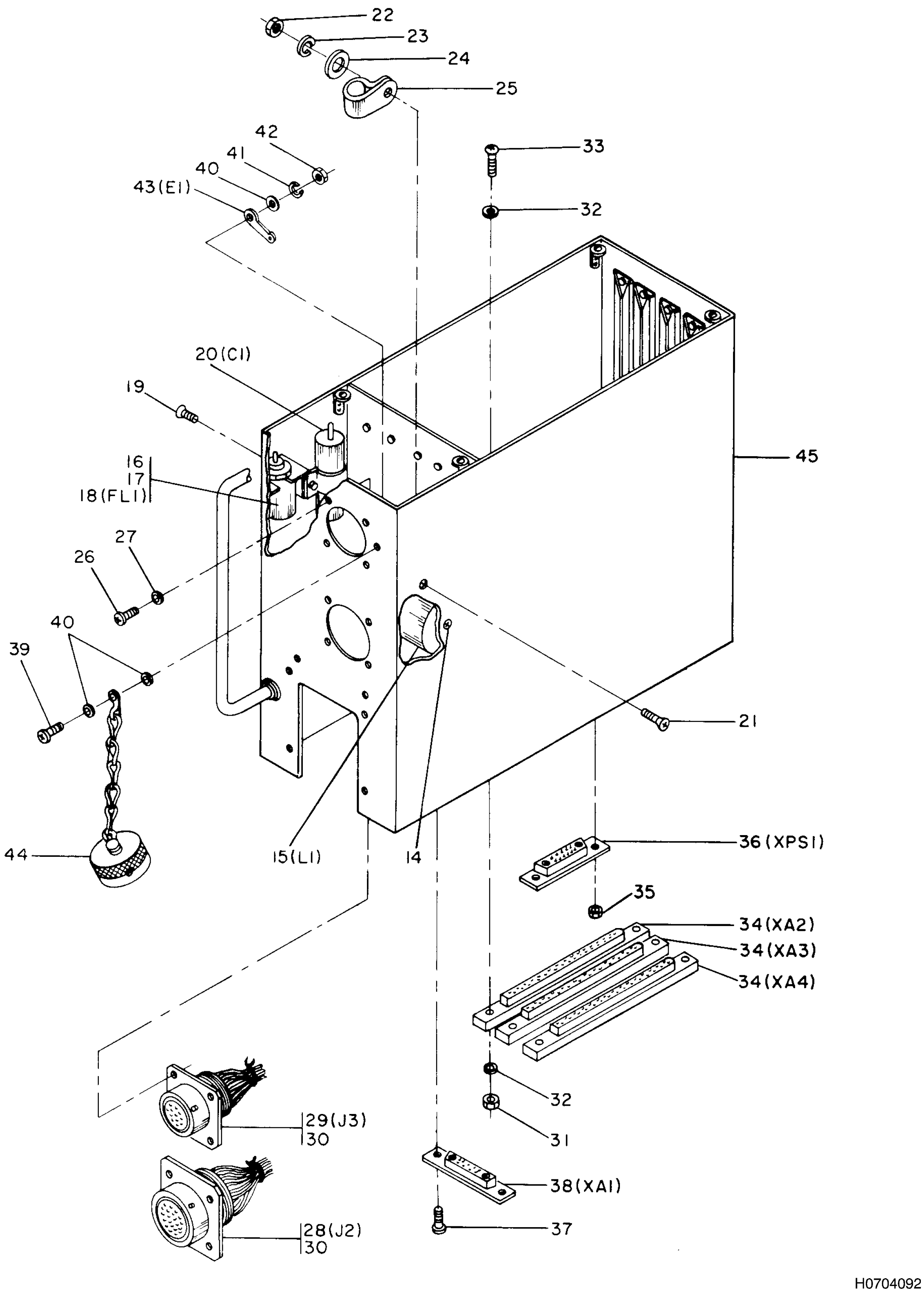

Figure 7-1. Thrust Control Unit - Exploded View (Sheet 2)

Legend for Figure 7-1:

1. Upper Cover Assembly
2. Captive Screw (4)
3. Card Extractor (6)
4. Circuit Card Assembly (A2)
5. Circuit Card Assembly (A3)
6. Circuit Card Assembly (A4)
7. Captive Screw
8. Card Extractor (2)
9. Power Supply Module (PS1)
10. Screw (5)
11. Screw (4)
12. Flat Washer (4)
13. Component Module Assembly (A1)
14. Screw
15. Inductor (L1)
16. Nut
17. Lock Washer
18. Radio Interference Filter (FL1)
19. Screw
20. Capacitor Assembly (C1)
21. Screw
22. Nut
23. Lock Washer
24. Flat Washer
25. Loop Clamp
26. Screw (8)
27. Flat Washer (8)
28. Connector (J2)
29. Connector (J3)
30. Sealing Plug
31. Nut (6)
32. Flat Washer (12)
33. Screw (6)
34. Connector (XA2, XA3, XA4)
35. Nut (2)
36. Connector (XPS1)
37. Screw (2)
38. Connector (XA1)
39. Screw
40. Flat Washer (3)
41. Lock Washer
42. Nut
43. Lug Terminal (E1)
44. Protective Cap
45. Chassis Assembly

7.2.3 Internal Connectors XA2, XA3, and XA4. Replace connectors **XA2, XA3,** or **XA4** (Figure 7-1, Sheet 2, 34) as follows:

a. Disassemble the thrust control unit in accordance with the procedures of Paragraph 7.4.

b. Tag and remove contacts from connector to be replaced using electrical connector maintenance kit (Table 2-1).

c. Remove nuts (31), flat washers (32), and screws (33) and remove connector (34).

d. Install connector keying pins for the connector (34) being replaced. Refer to Figure 7-2 for key plug installation.

e. Install connector (34) into chassis assembly (45) using screws (33), flat washers (32), and nuts (31).

f. Install removed contacts into replacement connector (34) using electrical connector maintenance kit (Table 2-1).

g. Remove tags from wires after contact installation.

h. Wire check installation in accordance with Table 7-2.

i. Reassemble the thrust control unit in accordance with the procedures of Paragraph 7.6.4.

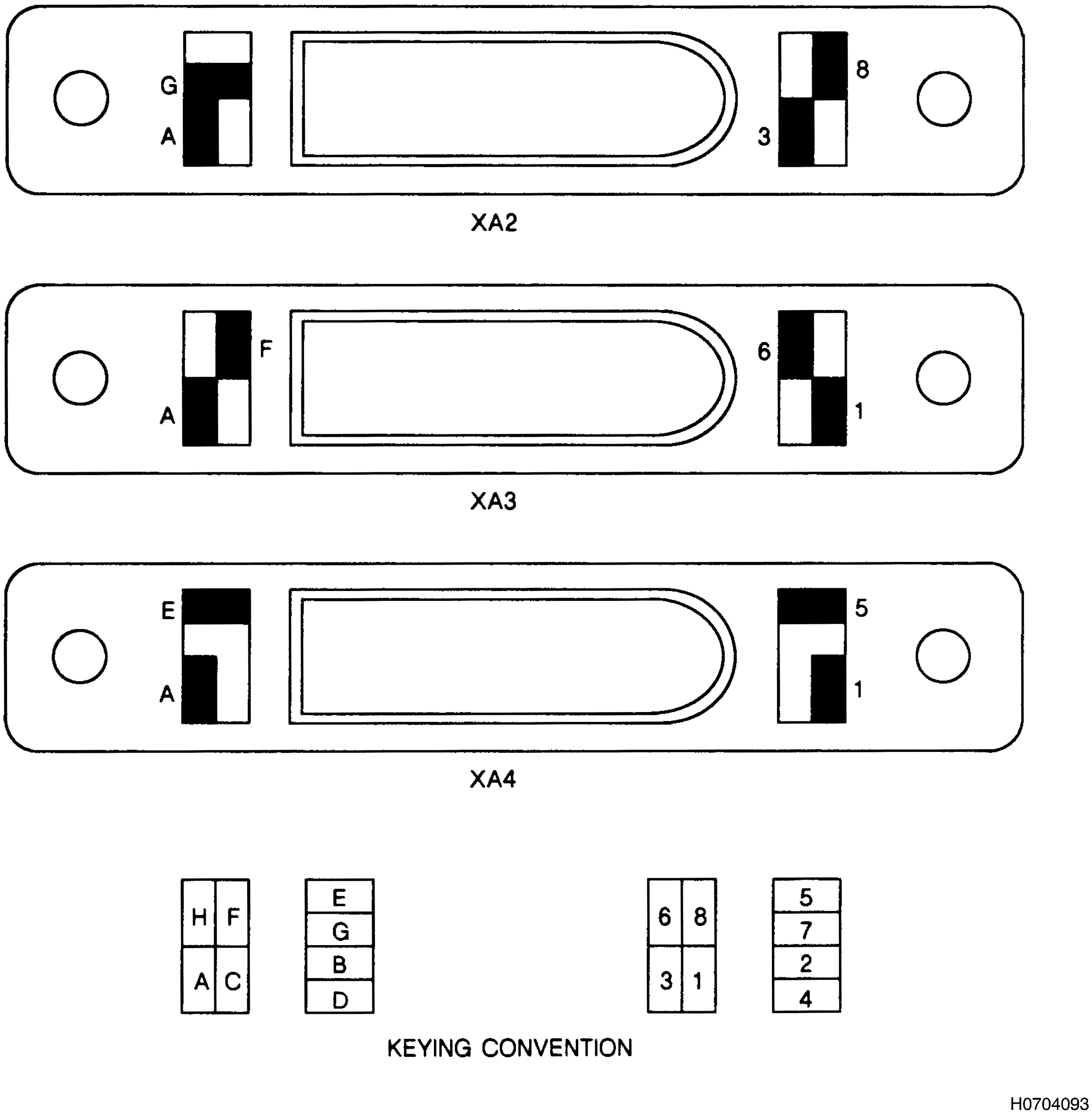

Figure 7-2. Connector Keying

7.2.3.1 Internal Connector XPS1. Replace connector XPS1 (Figure 7-1, Sheet 2, 36) as follows:

a. Disassemble the thrust control unit in accordance with the procedures of Paragraph 7.4.

b. Tag and remove wires from contacts of connector XPS1 (36). Remove heat shrunk sleeving from wire ends.

c. Remove nuts (35) and remove connector XPS1 (36).

d. Remove cable lacing, as necessary, to allow working room at connector.

e. Strip ends of connector wires 1/8-inch as required.

f. Slide a 5/16-inch length of heat shrinkable sleeving (Table 2-2, No. 16) over each connector wire. Push sleeving back away from stripped end.

g. Install replacement connector XPS1 (36) into chassis assembly (45) using nuts (35). When installed, XPS1 (36) must float to allow positive mating when its module is installed.

CAUTION

Use care when soldering to connector contacts A1 and A2. Blue portion of socket will burn if touched with hot iron.

h. Referring to TO 00-25-234, solder wiring to contacts of connector XPS1 (36). Use solder (Table 2-2, No. 18). Use care when soldering to A1 and A2 contacts.

NOTE

Contacts A1 and A2 can be pressed into connector by hand, but an extractor will be required to remove the A1 and A2 contacts.

i. Thoroughly clean all soldered connections using Isopropyl Alcohol (Table 2-2, No. 2). Allow to air dry.

j. Slide heat shrinkable sleeving (installed in Step f) over all soldered terminals and heat shrink sleeving.

k. Dress cabling in chassis assembly (45). Pull enough wire from cabling to form stress loops at connector XPS1 (36) contacts.

l. Spot tie cable, using lacing tape (Table 2-2, No. 21), and secure the cabling in the chassis.

m. Reassemble the thrust control unit in accordance with the procedures of Paragraph 7.6.4.

7.2.3.2 Internal Connector XA1. Replace connector XA1 (Figure 7-1, Sheet 2, 38) as follows:

a. Disassemble the thrust control unit in accordance with the procedures of Paragraph 7.4.

b. Remove screws (37) to free connector XA1 (38) from chassis.

c. Tag and remove wires from connector (38) contacts. Remove heat shrunk sleeving from wire ends.

d. Cut cable lacing to free XA1 wiring enough to make room to reconnect wiring to replacement connector XA1 (38).

e. Strip ends of connector XA1 wires 1/8-inch.

f. Slide a 5/16-inch length of heat shrinkable sleeving (Table 2-2, No. 16) over each connector wire. Push sleeving back away from stripped end.

CAUTION

Use care when soldering to connector contacts A1 and A2. Red portion of socket will burn if touched with hot iron.

g. Referring to TO 00-25-234, solder wiring to contacts of connector XA1 (38). Use solder (Table 2-2, No. 18). Use care when soldering to A1 and A2 contacts.

NOTE

Contacts A1 and A2 of connector XA1 (38) can be pressed into connector by hand, but an extractor will be required to remove the A1 and A2 contacts.

h. Thoroughly clean all soldered connections using Isopropyl Alcohol (Table 2-2, No. 2). Allow to air dry.

i. Slide heat shrinkable sleeving (installed in Step f) over all soldered terminals and heat shrink sleeving.

j. Dress cabling in chassis assembly (45). Pull enough wire from the cabling to form stress loops at connector XA1 (38) terminals when XA1 is held in its mounting position in the chassis assembly (45).

k. Install connector XA1 (38) in chassis assembly (45) using screws (37). When installed, connector XA1 (38) must float to allow positive mating when its module is installed.

l. Securely spot tie cabling in chassis assembly (45) using lacing tape (Table 2-2, No. 21).

m. Reassemble the thrust control unit in accordance with the procedures of Paragraph 7.6.4.

7.2.3.3 Fuse Resistor FR1. Replace fuse resistor FR1 (Figure 9-2, Sheet 2, 34) as follows:

a. Remove the component module assembly A1 in accordance with the procedures of Paragraph 7.4.4.

b. Remove screws (4), flat washers (5), and module cover (3).

NOTE

A spare fuse resistor is installed immediately below FR1. This spare may be utilized in lieu of replacing FR1.

c. Desolder FR1 as required from stud terminal (40).

d. Referring to TO 00-25-234, solder replacement FR1 in place as required to terminal (40). Use solder (Table 2-2, No. 18).

e. Position module cover (3). Install flat washers (5) and screws (4).

f. Install component module assembly A1 in accordance with the procedures of Paragraph 7.6.4.2.

7.2.3.3.1 Inspection. Visually inspect the TCU per Table 7-3. Do not disassemble solely for inspection purposes.

Table 7-3. Inspection

Part, Figure, and Index Number	Inspect For	Corrective Action
Upper cover (Figure 7-1, Sheet 1, 1), chassis assembly (Figure 7-1, Sheet 2, 45)	Deformation. Small dents allowed in flat areas. None allowed on mating or mounting surfaces.	Remove dents using clamps or straight bars. Replace part.
	Scratches.	Remove high metal. Touch up in accordance with Paragraph 7.6.
	Cracks.	Replace part.
Connectors J2 and J3 (Figure 7-1, Sheet 2, 28 and 29,)	Damaged shell.	Replace connector.
	Broken, bent, or missing contacts.	Replace.
	Corroded contacts.	Clean in accordance with TO 00-25-234. Replace as necessary.
Harness assembly	Loose, broken, or corroded module connectors.	Secure loose connectors. Replace if broken or corroded.
	Defective wiring.	Replace.
	Defective or loose cable lacing.	Replace, tighten, and secure.
Chassis mounted components FL1 (18), C1 (20), and L1 (15)	Loose components.	Secure.
	Evidence of overheating.	Replace.
CCAs A2, A3, and A4 (Figure 7-1, Sheet 1, 4, 5, and 6)	Loose, missing, or damaged components or printed wire runs. Damaged board.	Replace CCA.
	Damaged conformal coat.	Repair.
	Corroded edge connector contacts.	Replace CCA.
	Dirty edge connector contacts.	Clean.
Power supply module PS1 (9) or component module assembly A1 (13)	Loose, missing, or damaged components.	Replace module.
	Corroded or otherwise damaged connector.	Replace module.

7.3 MALFUNCTION ISOLATION PROCEDURES.

Malfunction isolation procedures are divided into prefunctional checks and functional checks.

7.3.1 Prefunctional Checks. There are no power-off checks of the TCU. Perform the inspection procedures of Paragraph 7.2.3.3.1.

7.3.2 Functional Checks. Perform the functional check of the TCU in accordance with Chapter 6.

7.4 DISASSEMBLY.

Determine the extent of disassembly required by preliminary visual inspection and test. No part shall be disassembled unless replacement or servicing is required, is unless removal is required to gain access to other parts requiring replacement or servicing. The disassembly procedures are described in detail unless self-evident upon inspection. Figure 7-1 shall be used as a reference for removal of all subassemblies and parts of the TCU. Disassemble the TCU as indicated in the following paragraphs. Refer to Paragraph 8.4.1 for Item Unique Identification (IUID) marking instructions.

7.4.1 Cover Assembly. Remove upper cover assembly (Figure 7-1, Sheet 1, 1) to gain access to components and assemblies by loosening captive screws (2).

7.4.2 Circuit Card Assemblies. Remove CCAs A2, A3, and A4 (Figure 7-1, Sheet 1, 4, 5, and 6) as follows:

a. Remove upper cover assembly in accordance with Paragraph 7.4.1.

b. Lift card extractors (3) and lever on wall of chassis assembly (45) to lift CCA A2, A3, or A4 (4, 5, or 6) out of chassis connector. Place in an electrostatic discharge (ESD) plastic bag (Table 2-2, No. 3).

7.4.3 Power Supply Module PS1. Remove power supply module PS1 (Figure 7-1, Sheet 1, 9) as follows:

a. Remove cover in accordance with Paragraph 7.4.1.

b. Loosen captive screw (7) at bottom of PS1.

c. Lift card extractors (8) and lever on wall of chassis assembly (45) to lift power supply module PS1 (9) out of chassis connector. Place power supply module in an ESD plastic bag (Table 2-2, No. 3).

7.4.4 Component Module Assembly A1. Remove component module assembly A1 (Figure 7-1, Sheet 1, 13) as follows:

CAUTION

Ensure that upper cover is installed per Paragraph 7.6.5 prior to placing chassis in a bottom-up position.

a. Place chassis assembly (45) bottom up on work surface.

b. Remove screws (10).

c. Remove screws (11) and flat washers (12).

d. Lift component module assembly A1 (13) from chassis and place in an ESD plastic bag (Table 2-2, No. 3).

7.4.5 Miscellaneous Parts. Remove the following parts as required:

a. Remove screw (39), flat washers (40), lock washer (41), and nut (42) to remove lug terminal E1 (43) and protective cap (44).

b. Desolder wires as required from lug terminal E1 (43). Refer to Table 7-2.

7.5 DETAIL PARTS AND ASSEMBLY CHECKOUT.

Check the components of the disassembled TCU in accordance with instructions in Table 7-1 and Chapter 8.

7.6 CLEANING/CORROSION PREVENTION.

Clean the TCU as directed in Paragraph 7.6.1. After cleaning, perform corrosion prevention procedures, as required, per Paragraph 7.6.2.

7.6.1 Cleaning.

a. Wipe all surfaces of cover, chassis, and component module assembly A1 with a clean dry lint-free cloth (Table 2-2, No. 8) to remove dirt, dust, oil, and grease. If necessary, moisten cloth with Isopropyl Alcohol (Table 2-2, No. 2) to remove foreign residue. Allow to air dry.

b. Use low pressure dry air to remove dust and dirt from inaccessible areas.

c. Clean CCAs using Isopropyl Alcohol (Table 2-2, No. 2). Allow to air dry.

7.6.2 Corrosion Prevention.

7.6.2.1 Refinishing Metal Surfaces. Aluminum and aluminum alloy parts have been chemically treated to form a chemical film finish per MIL-C-5541 chemical conversion coating. Touchup of repaired areas on previously treated parts shall be accomplished by using chemical conversion coating (Table 2-2, No. 9). Refer to TO 1-1-2.

7.6.2.2 Refinishing Painted Metal Surfaces. Painted surfaces are finished with lusterless lacquer (Table 2-2, No. 13). If touchup is required use the following procedures:

CAUTION

Area of bottom of component module A1 illustrated in Figure 7-3 should not be painted.

a. Prepare the surface using fine grade crocus cloth (Table 2-2, No. 7).

b. Treat surface in accordance with Paragraph 7.6.2.1.

c. Apply two coats of zinc chromate primer (Table 2-2, No. 15).

d. Finish surface with lusterless gray lacquer (Table 2-2, No. 13).

7.6.3 Lubrication. (Not applicable)

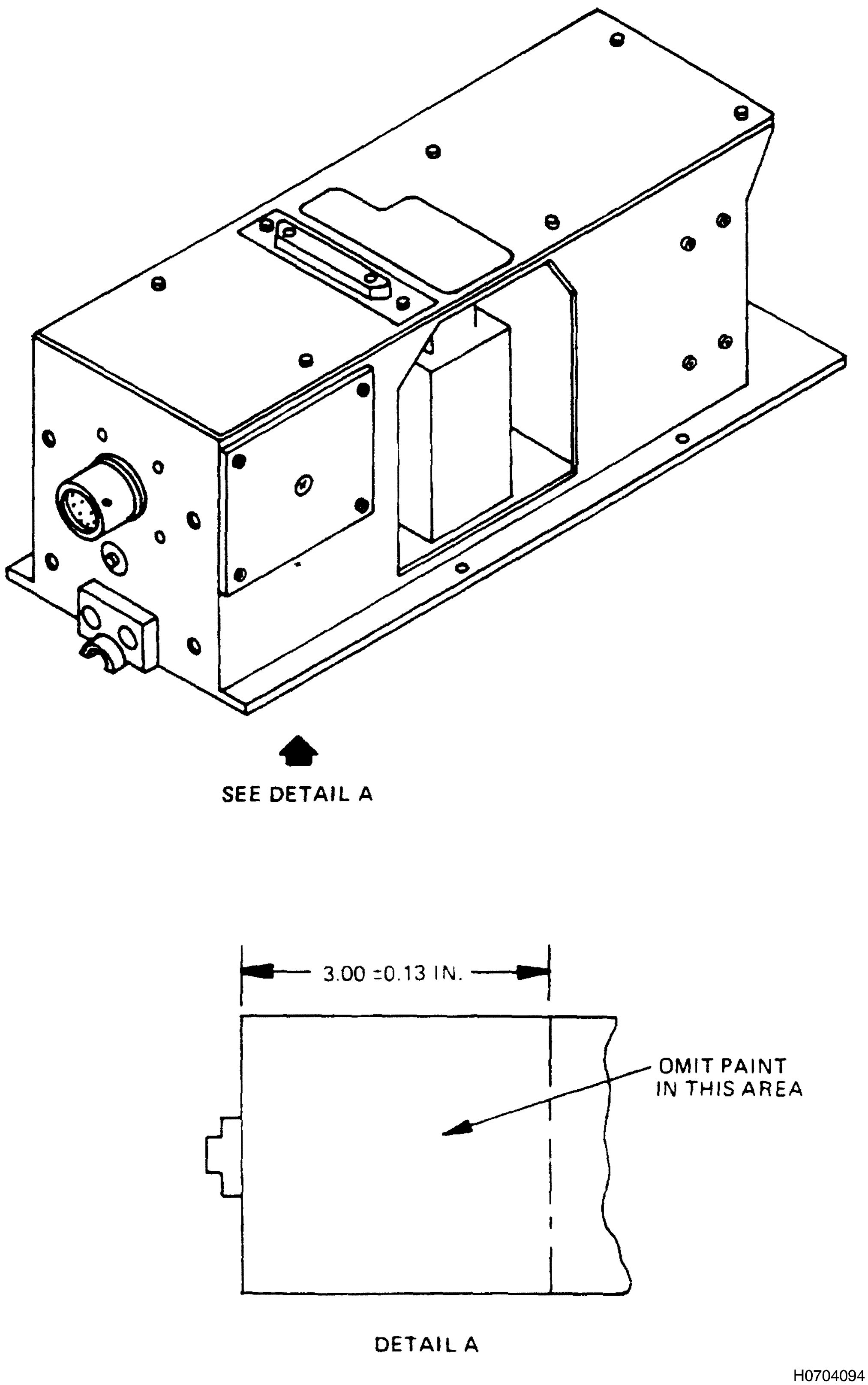

Figure 7-3. Unpainted Area of Bottom of Component Module A1

7.6.4 Reassembly. Reassemble the TCU as follows.

7.6.4.1 Miscellaneous Parts.

a. Referring to TO 00-25-234, solder wires as required to lug terminal E1 (Figure 7-1, Sheet 2, 43). Use solder (Table 2-2, No. 18). Refer to Table 7-2 for wire connections.

b. Assemble protective cap (44) and lug terminal (43) to chassis assembly (45) using screw (39), flat washers (40), lock washer (41), and nut (42).

7.6.4.2 Component Module Assembly A1.

CAUTION

Ensure that upper cover is installed per Paragraph 7.6.5 prior to placing chassis in a bottom-up position.

a. Place chassis assembly (Figure 7-1, Sheet 2, 45) bottom up on work surface.

b. Carefully insert component module assembly A1 (Figure 7-1, Sheet 1, 13) in chassis and mate module and chassis connector.

c. Loosely install screws (11) and flat washers (12).

d. Loosely install screws (10).

e. Tighten screws (10 and 11) previously installed.

7.6.4.3 Power Supply Module PS1.

a. Insert power supply module PS1 (Figure 7-1, Sheet 1, 9) into its chassis CCA guides. Carefully mate PS1 CCA connector with chassis connector. Press firmly until the PS1 card bottoms in the chassis.

b. Place card extractors (8) in their closed position.

c. Tighten captive screw (7) at bottom of PS1.

7.6.4.4 Circuit Card Assemblies.

a. Mate each CCA, A2, A3, and A4 (Figure 7-1, Sheet 1, 4, 5, and 6) in turn, with its chassis connector. Press carefully and firmly to fully seat each card.

b. Place card extractors (3) in their closed position.

7.6.5 Cover Assembly. Install upper cover (Figure 7-1, Sheet 1, 1) on chassis assembly (45) by tightening captive screws (2).

7.6.6 Alignment Checkout. (Not applicable)

7.6.7 Detailed Circuit Analysis. Detailed circuit analysis of the TCU is not required.

7.7 FINAL TEST.

Test the TCU in accordance with Chapter 6.

7.8 PREPARATION FOR SHIPMENT/STORAGE.

Prepare the TCU for shipment as follows:

a. Ensure that upper cover assembly (Figure 7-1, Sheet 1, 1) is securely fastened with captive screws (2).

b. Check that ESD dust caps (Table 2-2, No. 6) are installed on connector A1J1 (part of 13) and connector J2 (28).

c. Ensure that protective cap (Figure 7-1, Sheet 2, 44) is securely installed on connector J3 (29).

CHAPTER 8
OVERHAUL INSTRUCTIONS

SECTION I DISASSEMBLY

8.1 DISASSEMBLY.

8.1.1 General. This chapter contains disassembly instructions for the component module assembly A1 of the Thrust Control Unit (TCU). Refer to Chapter 7 for removal of component module assembly A1 from TCU.

8.1.2 Component Module Assembly A1. Disassembly of component module assembly A1 (Figure 8-1) consists of removal of a cover and several components to allow access to the remaining components of the module. Procedure for repair or replacement of components of component module assembly A1 are given in Section III. Proceed as follows:

a. Remove screws (Figure 8-1, Sheet 1, 1) and flat washers (2) and remove module cover (3) from chassis assembly (33).

b. Remove screws (4) and remove power inductors L1 and L2 (5) with screws (6) and chassis plates (7) attached.

c. Remove four screws (8) to free power transformer T1 (9).

d. Remove nut (10), lock washer (11), flat washer (12), and screw (13) to free loop clamp (14).

e. Remove screws (15) and flat washers (16) and remove armature relays K1 and K2 (17) with screws (18), flat washers (19), and spacer and insert assemblies (20) attached.

f. Remove instruction plate (21) by carefully peeling it off. Replace per Section III.

g. Remove screws (22) and flat washers (23) to free connector J1 (24).

h. Remove screws (26) and lock washers (27) to free connector P1 (28).

i. Remove screw (29), flat washer (30), and nut (31) to free lug terminal E14 (32).

j. Remove chassis mounted components (Figure 8-1, Sheet 2) in accordance with TO 00-25-234.

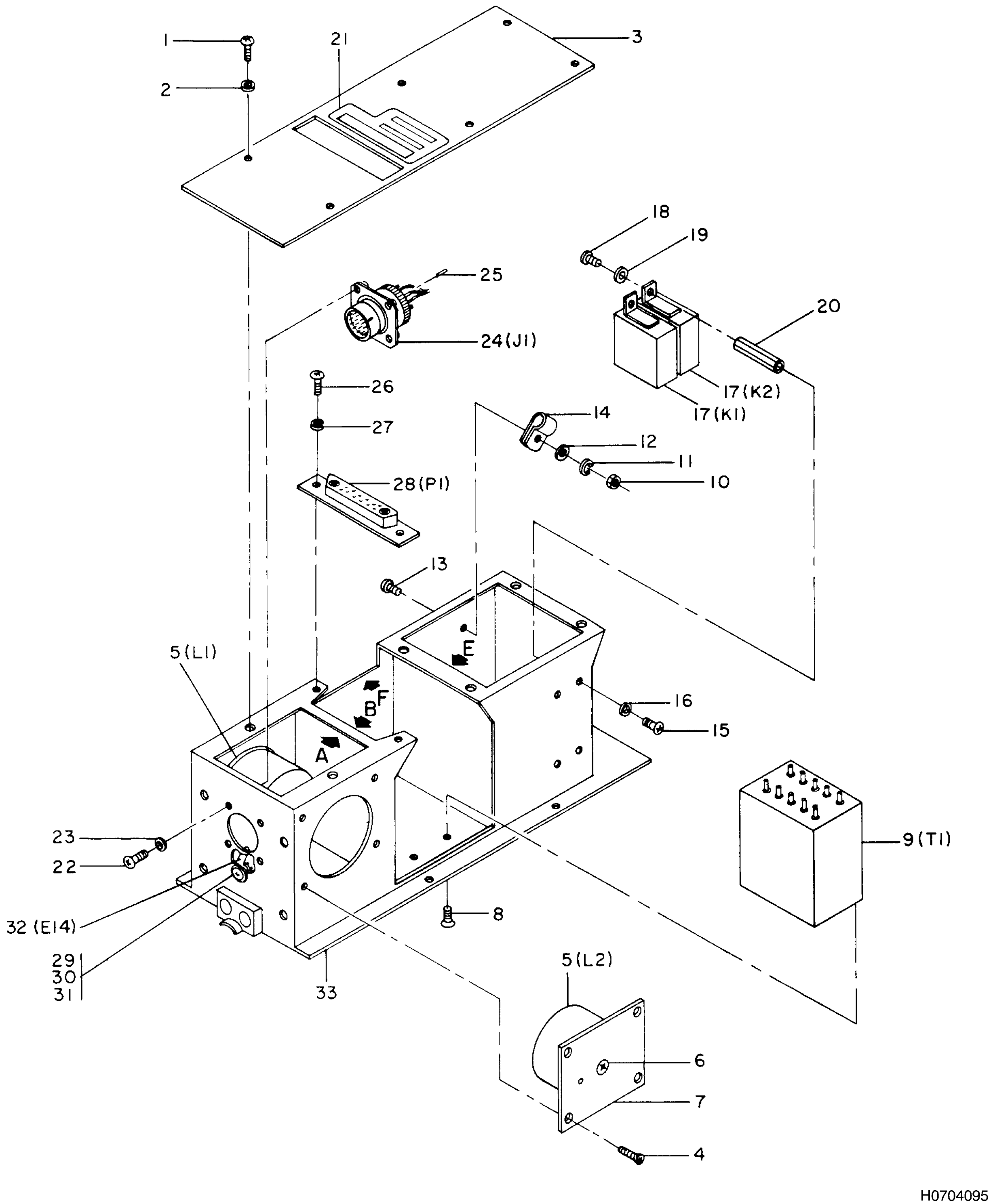

Figure 8-1. Component Module Assembly (A1) (Sheet 1 of 2)

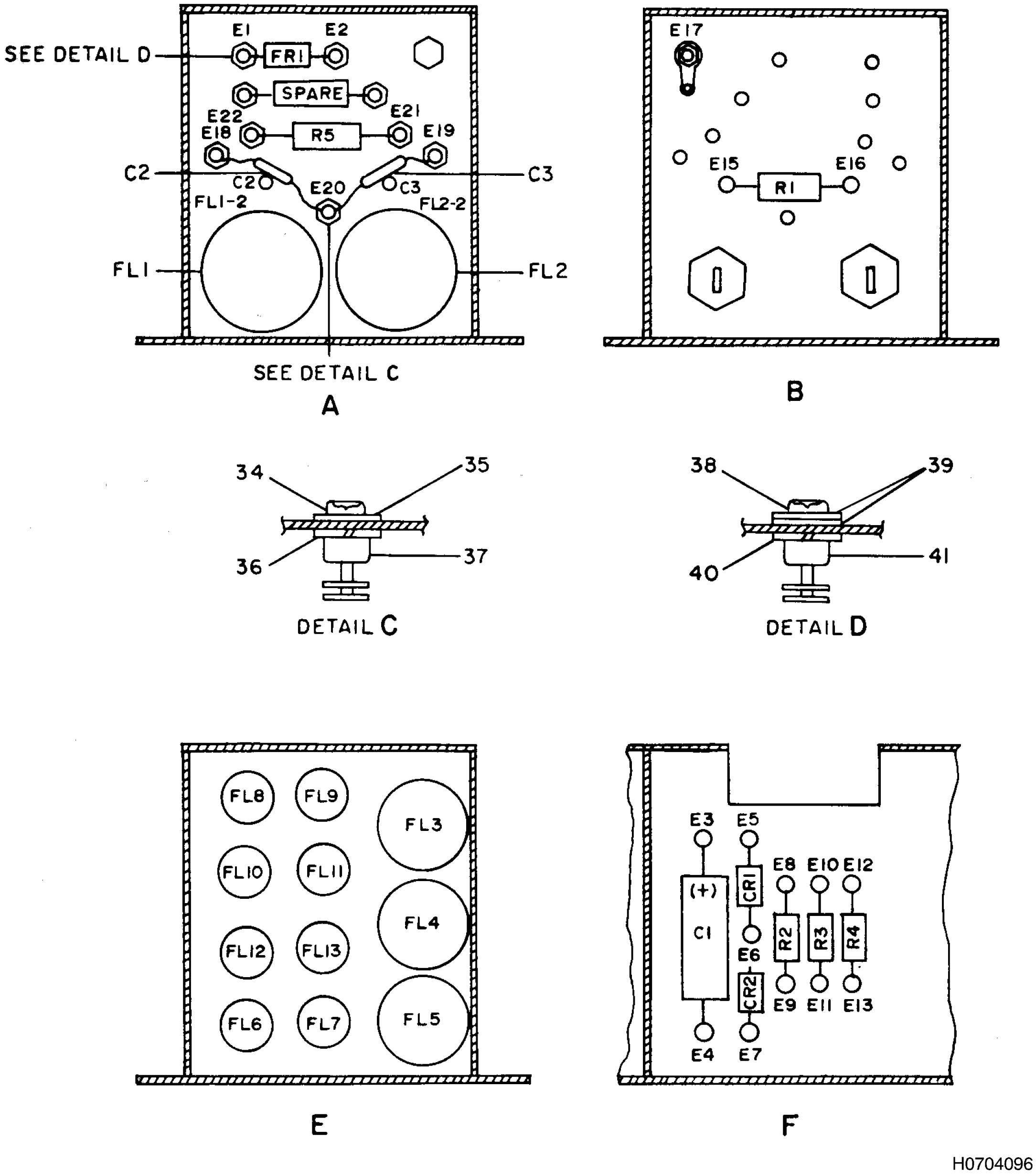

Figure 8-1. Component Module Assembly (A1) (Sheet 2)

Legend for Figure 8-1:

1 Screw (6)
2 Flat Washer (6)
3 Module Cover
4 Screw (8)
5 Power Inductor (2) L1, L2
6 Screw (2)
7 Chassis Plate (2)
8 Screw (4)
9 Power Transformer T1
10 Nut
11 Lock Washer
12 Flat Washer
13 Screw
14 Loop Clamp
15 Screw (4)
16 Flat Washer (4)
17 Armature Relay (2) K1, K2
18 Screw (4)
19 Flat Washer (4)
20 Spacer and Insert Assembly (4)
21 Instruction Plate
22 Screw (4)
23 Flat Washer (4)
24 Connector (J1)
25 Sealing Plug (10)
26 Screw (2)
27 Lock Washer (2)
28 Connector (P1)
29 Screw
30 Flat Washer
31 Nut
32 Lug Terminal (F14)
33 Chassis Assembly
34 Screw
35 Flat Washer
36 Lock Washer
37 Terminal (E20)
38 Screw (8)
39 Flat Washer (16)
40 Lock Washer (8)
41 Terminal (E1, E2, E15, E16, E18, E19, E21, E22)

SECTION II CLEANING

8.2 GENERAL.

Clean the Thrust Control Unit in accordance with TO 00-25-234.

SECTION III INSPECTION, REPAIR, AND REPLACEMENT

8.3 GENERAL.

This section provides detailed instructions for inspection, repair, and replacement of the assemblies of the Thrust Control Unit (TCU).

8.4 INSPECTION.

Visually inspect assemblies of the thrust control unit per Table 8-1. Inspection shall be performed in a clean work area free from dirt, metal particles, and other contamination.

8.4.1 Item Unique Identification (IUID). Visually inspect the identification plate/label/item for the Unique Item Identifier (UII) marking. If present, visually inspect for damage to the symbol. Ensure readability by using hand-held electronic reader, if symbol is not present or damaged mark/restore. REF: Drawing 200945069, Appendix A, Table A-1, of this TO for identification of parts to be marked, TO 00-25-260, and MIL-STD-130( ).

8.5 REPAIR.

8.5.1 Corrosion Prevention.

8.5.1.1 Refinishing Metal Surfaces. Touch up repaired areas on previously treated parts in accordance with Chapter 7.

8.5.1.2 Painted Metal Surfaces. Refinish painted metal surfaces in accordance with Chapter 7.

8.5.2 Repair of Conformal Coating. Repair conformal coating on printed circuit card assemblies as follows:

a. Remove conformal coating in immediate area of repair. Feather edges of conformal coating in area of repair.

b. Thoroughly clean area of repair with a trans-1, 2-dichloroethylene based solvent, such as Trans LC, or equivalent (Table 2-2, No. 10). Allow to air dry.

c. Remove any remaining solder flux residue with Isopropyl Alcohol (Table 2-2, No. 2). Allow to air dry.

d. Measure, by weight, the following materials and mix in a suitable container:

Conformal Coating (Table 2-2, No. 11):
50 parts of part A
35 parts of part B
Conap S-8 solvent (Table 2-2, No. 19):
17 parts

e. Let mixture stand for 30 minutes before use. Pot life of mixture is 8 hours.

f. Using fine camel hair brush (Table 2-2, No. 4), apply three thin coats of conformal coating mixture to area of repair. Let each coat air dry 10 minutes between coats.

g. Place assembly in oven and cure for 2 hours at 176 to 185°F (80 to 85°C) or air cure for 48 hours. Conformal coat will be tack-free in 12 hours.

h. Remove assembly from oven and allow to cool to room temperature.

8.6 REPLACEMENT.

There are two configurations of modules in the thrust control unit. The PS1 and A1 modules (Figure 7-1, Sheet 1, 9 and 13) are hard-wired modules. Modules A2, A3, and A4 (4, 5, and 6) are printed wire conformal coated circuit card assemblies. Remove and replace components in accordance with TO 00-25-234. The following component replacement procedures are peculiar to each type of module.

Table 8-1. Inspection

Part, Figure, and Index Number	Inspect for	Corrective Action
CCAs A2, A3, and A4 (Figure 7-1, Sheet 1, 4, 5, and 6)	Damaged components.	Replace.
	Missing components.	Replace.
	Loose components.	Repair as necessary.
	Loose, broken, or damaged printed wire runs.	Replace CCA.
	Damaged board.	Replace CCA.
	Damaged conformal coating.	Repair.
	Corroded edge connector contacts.	Replace CCA.

Table 8-1. Inspection - Continued

Part, Figure, and Index Number	Inspect for	Corrective Action
Power Supply Module PS1 (9) or Component Module Assembly A1 (13)	Loose components.	Secure or repair as necessary.
	Damaged or missing components.	Replace.
	Corroded connector.	Replace.
	Evidence of overheated insulation.	Replace defective wiring.

8.6.1 Component Module A1.

a. After wiring, shrink heat shrinkable sleeving (Table 2-2, No. 16) on the following terminals (Figure 8-1, Sheet 2):

(1) Terminal lug E17.

(2) Non-threaded terminals of radio interference filters FL1 through FL5.

b. After wiring, shrink heat shrinkable sleeving (Table 2-2, No. 16) over terminals of the following components (Figure 8-1, Sheet 1):

(1) Armature relays K1, K2 (17).

(2) Power transformer T1 (9), terminals 1 and 2.

(3) Connector P1 (28) contacts.

c. Form a stress loop in the wiring of the following components and secure the wiring to the body of the components using lacing tape (Table 2-2, No. 21).

(1) Radio frequency filters FL1 through FL13 (Figure 8-1, Sheet 2).

(2) Power transformer T1 (Figure 8-1, Sheet 1, 9).

d. Install fixed capacitor C1 as follows (Figure 8-1, Sheet 2):

(1) Using Isopropyl Alcohol (Table 2-2, No. 2) clean chassis area between terminal studs E3 and E4. Allow to air dry.

(2) Paint cleaned area of module chassis using primer (Table 2-2, No. 14). Allow to dry for 15 minutes.

(3) Bond capacitor C1 to module chassis using adhesive (Table 2-2, No. 1). Allow adhesive to cure for 12 hours.

e. Install capacitors C2 and C3 as follows (Figure 8-1, Sheet 2):

(1) Cut capacitor wires to proper length and sleeve their leads using insulation sleeving (Table 2-2, No. 17). Leave enough unsleeved lead to attach leads to terminals.

(2) Using Isopropyl Alcohol (Table 2-2, No. 2), clean chassis area between terminal lugs E18 and E20 and between E19 and E20. Allow to air dry.

(3) Paint area of chassis cleaned in Step e(2) using primer (Table 2-2, No. 14). Allow to air dry for 15 minutes.

(4) Solder capacitor C2 using solder (Table 2-2, No. 18) between stud terminals E18 and E20, being careful not to sear the lead sleeves.

(5) Solder capacitor C3, using solder (Table 2-2, No. 18), between stud terminals E19 and E20, being careful not to sear the lead sleeves.

(6) Clean soldered terminals using Isopropyl Alcohol (Table 2-2, No. 2) and a stiff bristle brush (Table 2-2, No. 5).

(7) Clean body of capacitors C2 and C3 using a lint-free cloth (Table 2-2, No. 8) moistened with Isopropyl Alcohol (Table 2-2, No. 2). Allow to air dry.

(8) Paint half of body of capacitors C2 and C3 (away from capacitor leads) using primer (Table 2-2, No. 14). Allow to dry for 15 minutes.

(9) Bond capacitors C2 and C3 to chassis using adhesive (Table 2-2, No. 1). Allow to cure for 12 hours.

f. Install instruction plate (Figure 8-1, Sheet 1, 21) on module cover (3) as follows:

(1) Clean area of module cover (3) to which instruction plate (21) is to be attached using Isopropyl Alcohol (Table 2-2, No. 2) and a lint-free cloth (Table 2-2, No. 8). Allow to air dry.

(2) Remove paper liner from back of instruction plate (21) and apply a uniform coating of Girder No. 3A solvent (Table 2-2, No. 20).

(3) Allow several seconds for the excess solvent to flash off then place the instruction plate (21) in position on module cover (3) and apply overall uniform pressure for 20 seconds.

8.6.2 Power Supply Module PS1.

a. Install capacitors C1 and C2, Figure 8-2, as follows:

(1) Using Isopropyl Alcohol (Table 2-2, No. 2) and a lint-free cloth (Table 2-2, No. 8) clean power supply bracket area between terminal studs E9 and E10 and E11 and E12. Allow to air dry.

(2) Clean body of capacitors C1 and C2 using Isopropyl Alcohol (Table 2-2, No. 2) and a lint-free cloth (Table 2-2, No. 8). Allow to air dry.

(3) Using primer (Table 2-2, No. 14), paint cleaned area of bracket and body of capacitors C1 and C2. Allow to dry for 15 minutes.

(4) Solder capacitor C1 leads to terminals E11 and E12 using solder (Table 2-2, No. 18). Solder capacitor C2 leads to terminals E9 and E10.

(5) Clean soldered terminals using Isopropyl Alcohol (Table 2-2, No. 2) and a stiff bristle brush (Table 2-2, No. 5). Allow to air dry.

(6) Apply adhesive (Table 2-2, No. 1) to bodies of capacitors C1 and C2 and to the power supply bracket to bond the capacitors to the bracket. Allow adhesive to cure for 12 hours.

b. When connector P1 is replaced, wired connector terminals 1 through 15 must be sleeved with heat shrinkable sleeving (Table 2-2, No. 16).

8.6.3 Circuit Card Assembly A2. CCA A2 contains vertically mounted resistors. These components must not interfere with other modules when installed in the TCU. Refer to Figure 8-3 when replacing vertically mounted resistors. The maximum height dimension must not be exceeded.

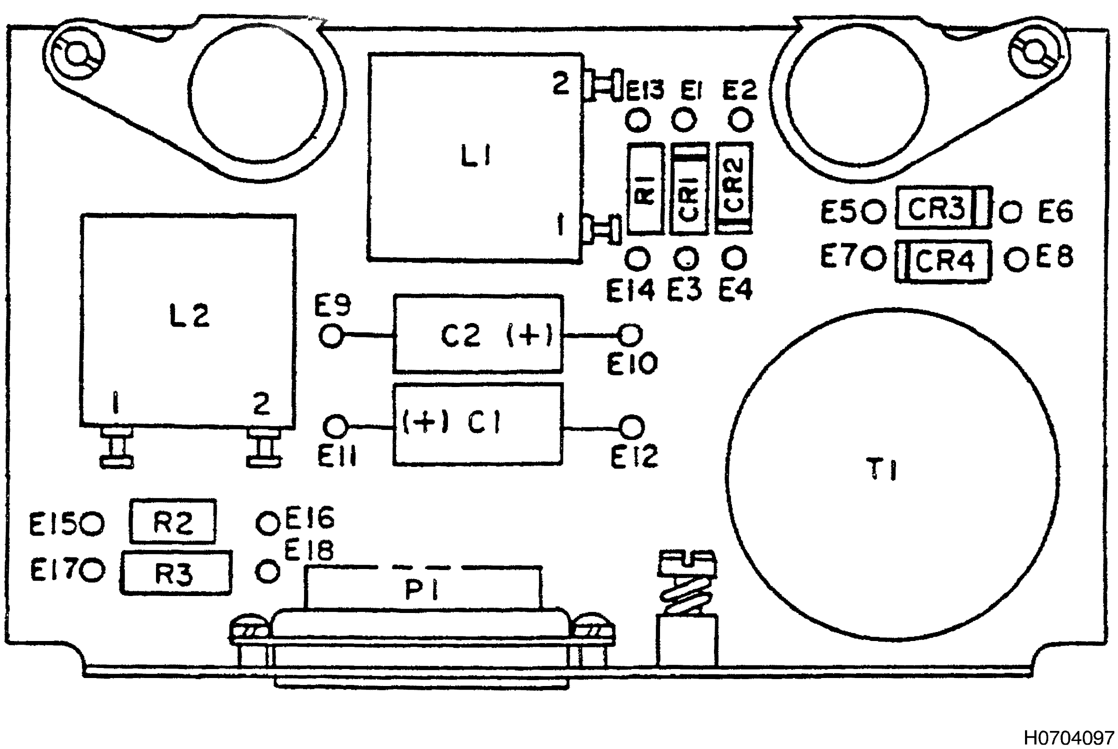

Figure 8-2. Power Supply Module (PS1)

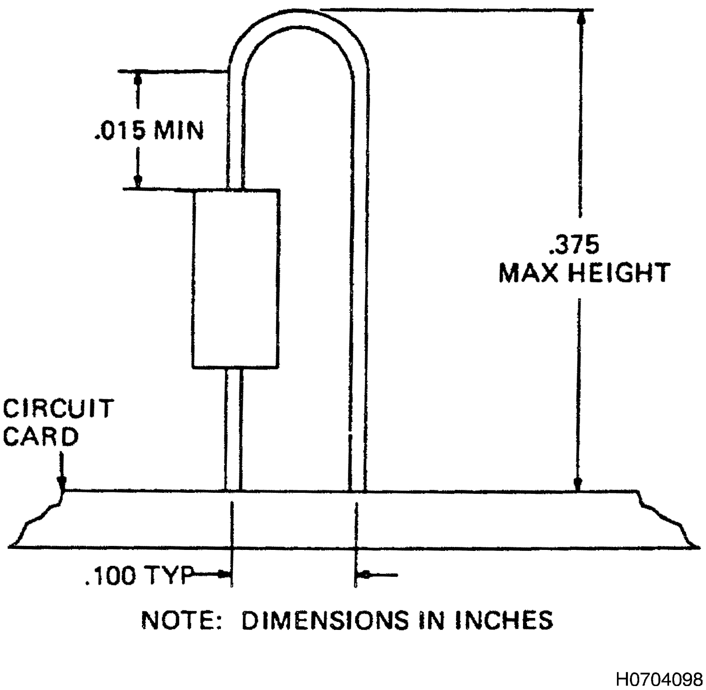

Figure 8-3. Vertically Mounted Components

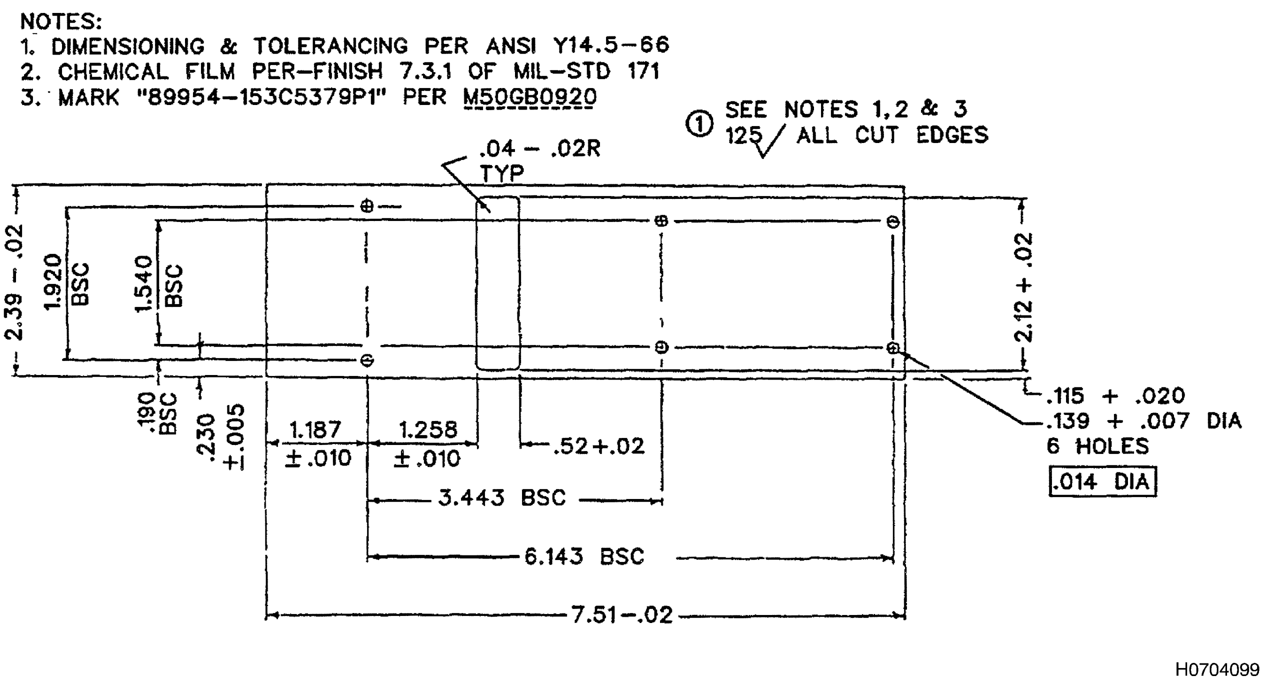

Figure 8-4. Module Cover

SECTION IV ASSEMBLY

8.7 GENERAL.

This section contains assembly instructions for component module assembly A1 of the Thrust Control Unit.

8.7.1 Component Module Assembly A1. Assemble component module assembly A1 as follows:

a. Replace chassis mounted components (Figure 8-1, Sheet 2) in accordance with TO 00-25-234.

b. Install lug terminal E14 (32) using screw (29), flat washer (30), and nut (31).

c. Install connector J1 (28) using screws (26) and flat washers (27).

d. Install connector J1 (24) using screws (22) and flat washers (23).

e. Install armature relays K1 and K2 (Figure 8-1, Sheet 1, 17), with screws (18), flat washers (19), and spacer and insert assemblies (20) attached, into rear component compartment of chassis assembly (33).

f. Dress cable of armature relays (17) in component compartment and install screws (15) and flat washers (16).

g. Install screw (13), loop clamp (14), flat washer (12), lock washer (11) and nut (10) to secure cable of armature relays (17).

h. Install screws (8) to secure power transformer T1 (9).

i. Install power inductors L1 and L2 (5), with screws (6) and chassis plates (7) attached, using screws (4).

j. Install module cover (3) using screws (1) and washers (2).

SECTION V ACCESSORIES

8.8 NOT APPLICABLE.

SECTION VI TESTING

8.9 GENERAL.

This section contains test instructions for the Thrust Control Unit (TCU) assemblies. Figure 8-5 through Figure 8-7, illustrate the TCU CCAs and their component locations. Table 8-2 and Table 8-3 give the wire running lists for modules PS1, and A1. Test the TCU in accordance with the procedure given in Chapter 6 of this manual. Test shop replaceable units A1, A2, A3, and A4 in accordance with the procedures specified in TO 2JA8-28-8-2. Test power supply module PS1 in accordance with the procedures given in Paragraph 8.9.1 through Paragraph 8.9.3 of this manual.

8.9.1 Testing Power Supply PS1. Test the TCU in accordance with the procedure given in Chapter 6 of this manual. Test shop replaceable units A1, A2, A3, and A4 in accordance with the procedures specified in TO 2JA8-28-8-2 using DATSA. Refer to TO 2JA8-28-8-7 to test circuit cards using the VDATS. Test power supply module PS1 in accordance with the procedures given in Paragraph 8.9.2 through Paragraph 8.9.4 of this manual.

8.9.1.1 Input Power Requirements. To test the power supply, an input power source with the following characteristics is required:

Voltage	226 ±6 VAC rms Single phase
Current	50 maAC min
Frequency	400 +20 Hz
Ripple	10V peak-to-peak max
Harmonic content	less than 5%
Source resistance	less than 100 ohms

8.9.2 Load Requirements. The following loads are required to test the power supply:

R_{L1}, R_{L2}	270 ±15 ohms, 1 watt or 90 ±5 ohms, 3 watts
R_{L3}	240 ± ohms, 4 watts or open circuit

8.9.3 Power Off Tests. Perform the following tests with power off and loads not connected:

a. Using multimeter (Table 2-1) verify continuity (less than 1 ohm) between the following pins: 3, 4, 5, 6, 7, 8.

b. Verify continuity (less than 1 ohm) between pins 14 and 15.

c. Verify resistance between the following pin pairs:

Pins	Resistance
1 and 2	1000 ±150 ohms
9 and 10	1000 ±150 ohms
13 and 15	2000 ±300 ohms

d. Verify an open circuit (greater than 5 megohms) between pins 3 and A1, and between each of these pins and the module frame.

8.9.4 Output Voltage Tests. Connect the power supply as shown in Figure 8-8. Apply input voltage as specified in Paragraph 8.9.1.1. Apply a load using decade box (Table 2-1) as specified in Table 8-4 and verify that output voltages are within the limits given in the table. Using an oscilloscope (Table 2-1) verify that the peak-to-peak ripple component of the voltage at pin 1 and at pin 10 is within the limits specified in Table 8-4.

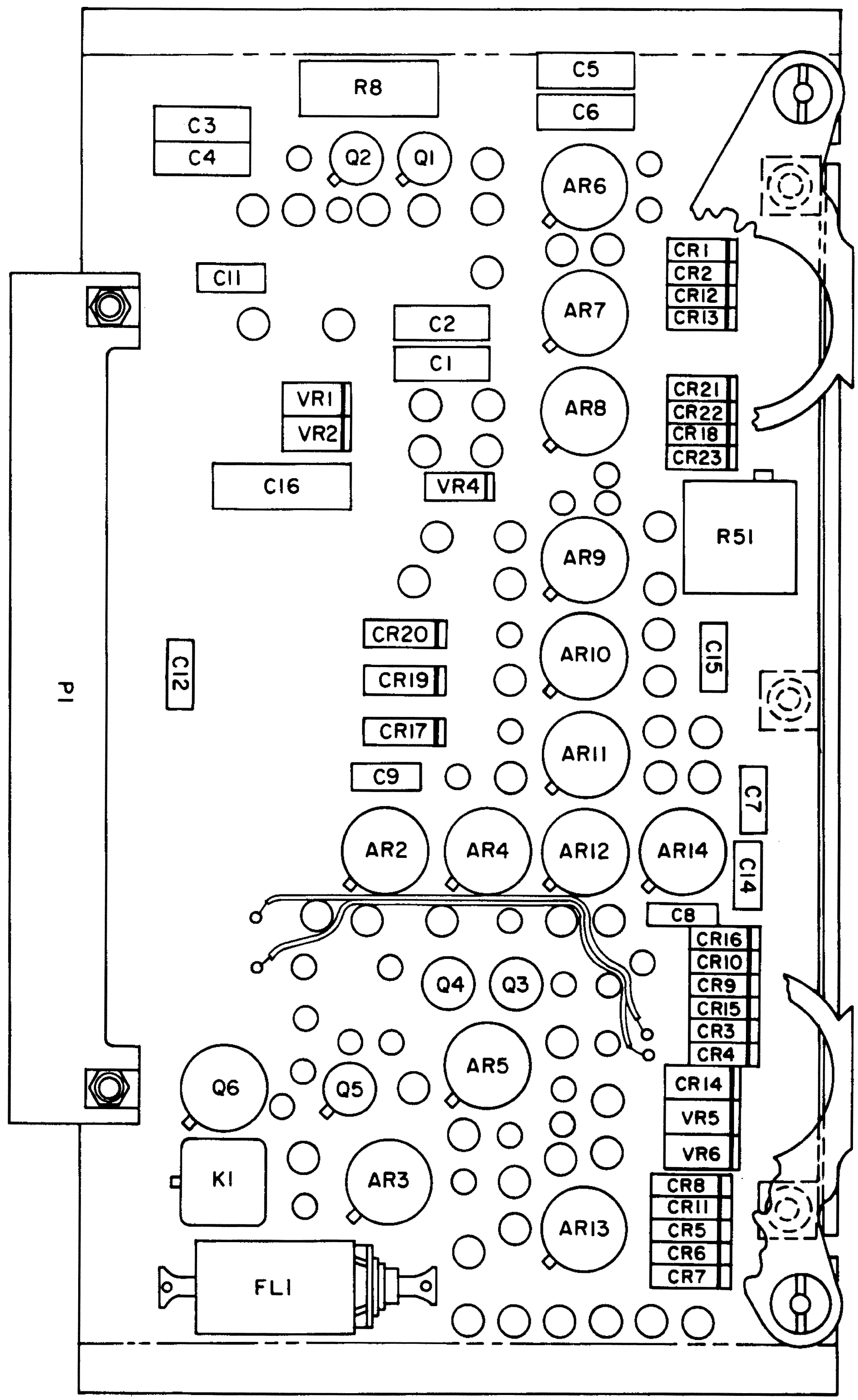

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Figure 8-5. Circuit Card Assembly (A2)

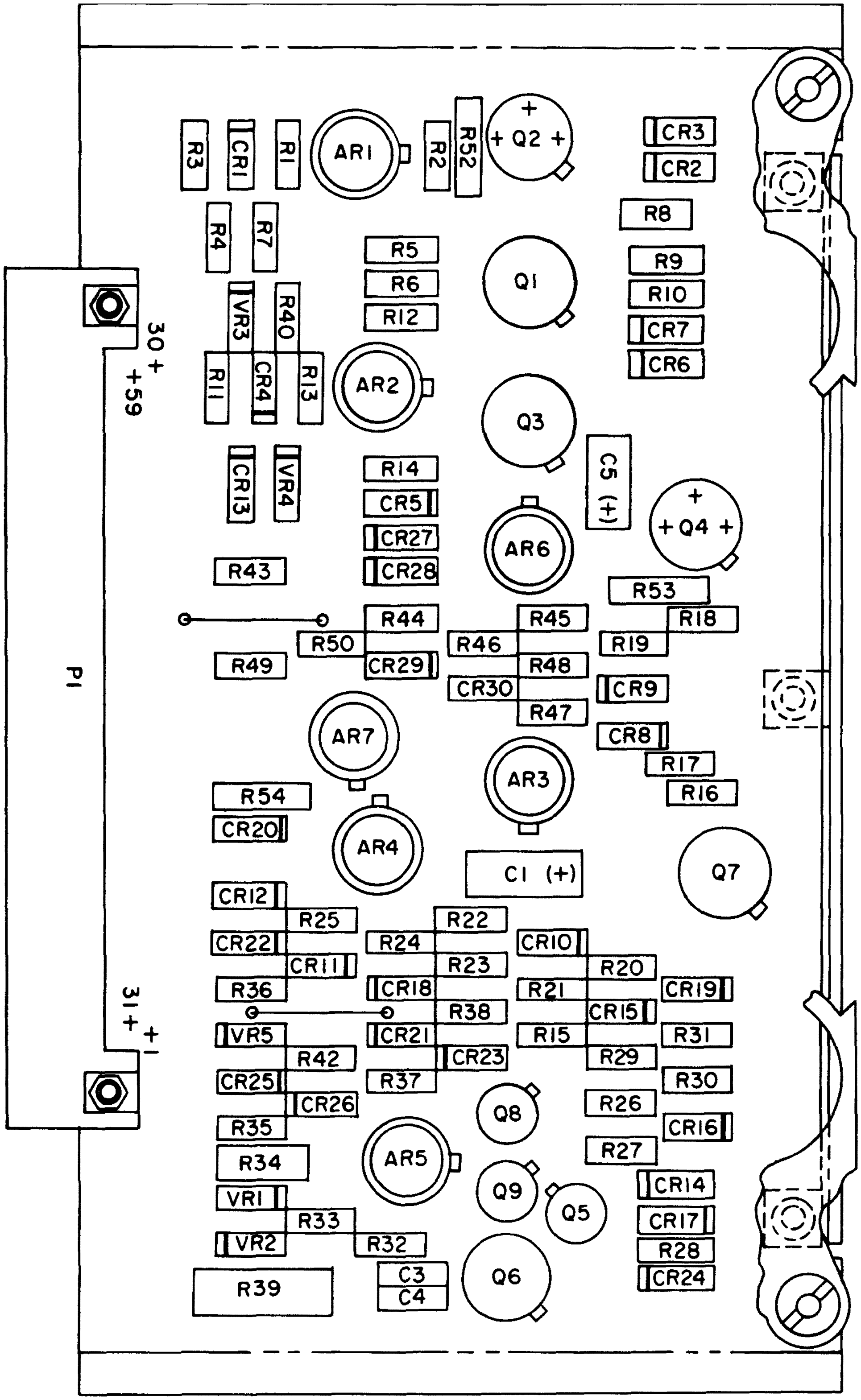

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Figure 8-6. Circuit Card Assembly (A3)

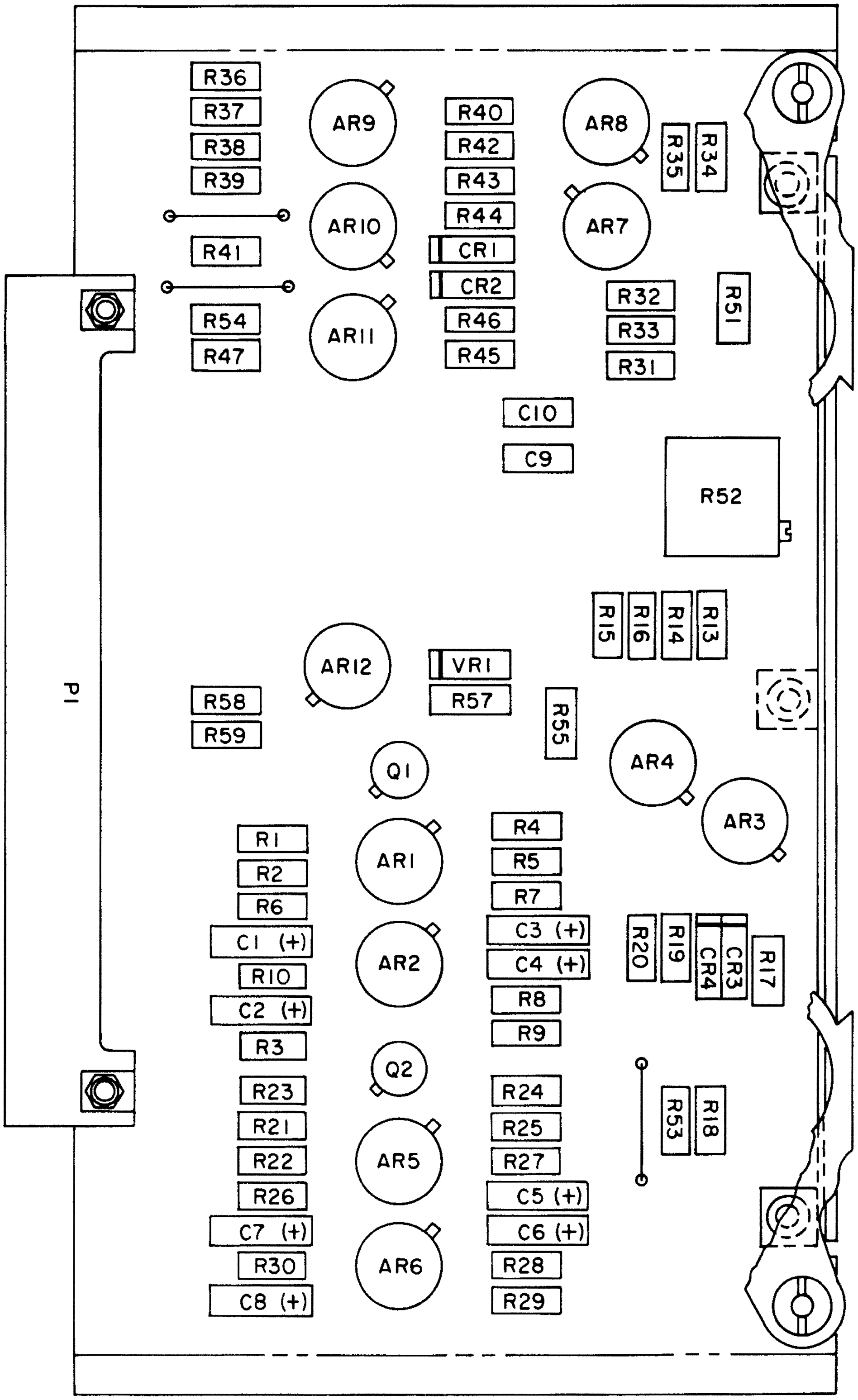

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Figure 8-7. Circuit Card Assembly (A4)

Table 8-2. Component Module (A1), Wire Run List

From	To	Color	AWG Size
E1	L1-1	GRA	24
E2	FL1-2	GRA	24
E3	E6	RED	26
E4	P1-8	WHT	26
E5	T1-5	RED	26
E6	E3	RED	26
E6	E15	RED	26
E7	T1-3	RED	26
E8	E10	BARE	24
E8	T1-9	WHT	26
E9	E11	BARE	24
E10	E8	BARE	24
E10	E12	BARE	24
E11	E9	BARE	24
E11	E13	BARE	24
E12	E10	BARE	24
E13	E11	BARE	24
E13	P1-1	GRA	24
E14	E22	BLK	24
E14	J1-12	BLK	24
E15	E6	RED	26
E15	P1-10	RED	26
E16	P1-2	RED	26
E17	T1-10	BLK	24
E18	L1-2	GRA	24
E19	L2-1	GRA	24
E20	E21	GRA	24
E21	E20	GRA	24
E22	E14	BLK	24
FL1-1	P1-A1	GRA	24
FL1-1	T1-1	GRA	24
FL1-2	E2	GRA	24
FL2-1	P1-A2	GRA	24
FL2-1	T1-2	GRA	24
FL2-2	L2-2	GRA	24
FL3-1	P1-3	GRA	24
FL3-2	K1-A2	GRA	24
FL4-1	P1-11	GRA	24
FL4-2	K2-A2	GRA	24
FL5-1	T1-7	GRA	24
FL5-2	K1-A1	GRA	24
FL5-2	K2-A1	GRA	24
FL6-1	P1-4	RED	26
FL6-2	K1-X1	RED	26
FL7-1	P1-5	RED	26
FL7-2	K1-X2	RED	26
FL8-1	P1-12	RED	26
FL8-2	K2-X1	RED	26
FL9-1	P1-13	RED	26

Table 8-2. Component Module (A1), Wire Run List - Continued

From	To	Color	AWG Size
FL9-2	K2-X2	RED	26
FL10-1	P1-14	WHT	26
FL10-2	K1-B2	WHT	26
FL11-1	P1-6	WHT	26
FL11-2	K1-B3	WHT	26
FL11-2	K2-B3	WHT	26
FL12-1	P1-7	WHT	26
FL12-2	K1-B1	WHT	26
FL12-2	K2-B1	WHT	26
FL13-1	P1-15	WHT	26
FL13-2	K2-B2	WHT	26
J1-11	L1-2	GRA	24
J1-12	E14	BLK	24
J1-13	L2-1	GRA	24
K1-A1	FL5-2	GRA	24
K1-A2	FL3-2	GRA	24
K1-B1	FL12-2	WHT	26
K1-B2	FL10-2	WHT	26
K1-B3	FL11-2	WHT	26
K1-X1	FL6-2	RED	26
K1-X2	FL7-2	RED	26
K2-A1	FL5-2	GRA	24
K2-A2	FL4-2	GRA	24
K2-B1	FL12-2	WHT	26
K2-B2	FL13-2	WHT	26
K2-B3	FL11-2	WHT	26
K2-X1	FL8-2	RED	26
K2-X2	FL9-2	RED	26
L1-1	E1	GRA	24
L1-2	E18	GRA	24
L1-2	J1-11	GRA	24
L2-1	E19	GRA	24
L2-1	J1-13	GRA	24
L2-2	FL2-2	GRA	24
P1-A1	FL1-1	GRA	24
P1-A2	FL2-1	GRA	24
P1-1	E13	GRA	24
P1-2	E16	RED	26
P1-3	FL3-1	GRA	24
P1-4	FL6-1	RED	26
P1-5	FL7-1	RED	26
P1-6	FL11-1	WHT	26
P1-7	FL12-1	WHT	26
P1-8	E4	WHT	26
P1-8	T1-4	WHT	26
P1-10	E15	RED	26
P1-11	FL4-1	GRA	24
P1-12	FL8-1	RED	26

Table 8-2. Component Module (A1), Wire Run List - Continued

From	To	Color	AWG Size
P1-13	FL9-1	RED	26
P1-14	FL10-1	WHT	26
P1-15	FL13-1	WHT	26
T1-1	FL1-1	GRA	24
T1-2	FL2-1	GRA	24
T1-3	E1	RED	26
T1-4	P1-8	WHT	26
T1-4	T1-9	WHT	26
T1-5	E5	RED	26
T1-7	FL5-1	GRA	24
T1-9	E8	WHT	26
T1-9	T1-4	WHT	26
T1-10	E17	BLK	24

NOTE

All wires are stranded silver-plated copper with polytetrafluoroethylene insulation.

Table 8-3. Power Supply Module (PS1), Wire Run List

From	To	Color	Awg Size
E1	E7	RED	26
E1	L1-2	RED	26
E2	E5	VIO	26
E3	E4	BARE	24
E4	E3	BARE	24
E4	T1	ORN	--
E4	E2	VIO	26
E5	L2-2	VIO	26
E6	E8	BARE	24
E7	E1	RED	26
E8	E6	BARE	24
E8	T1	GRN	--
E9	E15	VIO	26
E9	L2-1	VIO	26
E10	E12	WHT	26
E10	T1	YEL	--
E11	E14	RED	26
E11	P1-1	RED	26
E12	E10	WHT	26
E12	P1-8	WHT	26
E12	T1	BLU	--
E13	P1-2	RED	26
E14	E11	RED	26
E14	L1-1	RED	26

Table 8-3. Power Supply Module (PS1), Wire Run List - Continued

From	To	Color	Awg Size
E15	E9	VIO	26
E16	P1-9	VIO	26
E17	P1-14	GRA	24
E18	P1-13	GRA	24
L1-1	E14	RED	26
L1-2	E1	RED	26
L2-1	E9	VIO	26
L2-1	P1-10	VIO	26
L2-2	E5	VIO	26
P1-A1	T1	BRN	--
P1-A2	T1	RED	--
P1-1	E11	RED	26
P1-2	E13	RED	26
P1-3	P1-4	WHT	26
P1-4	P1-3	WHT	26
P1-4	P1-5	WHT	26
P1-5	P1-4	WHT	26
P1-5	P1-6	WHT	26
P1-6	P1-5	WHT	26
P1-6	P1-7	WHT	26
P1-7	P1-6	WHT	26
P1-7	P1-8	WHT	26
P1-8	E12	WHT	26
P1-8	P1-7	WHT	26
P1-9	E16	VIO	26
P1-10	L2-1	VIO	26
P1-12	T1	BLK	--
P1-13	E18	GRA	24
P1-14	E17	GRA	24
P1-14	P1-15	GRA	24
P1-15	P1-14	GRA	24
P1-15	T1	VIO	--
T1	E4	ORN	--
T1	E8	GRN	--
T1	E10	YEL	--
T1	E12	BLU	--
T1	P1-A1	BRN	--
T1	P1-A2	RED	--
T1	P1-12	BLK	--
T1	P1-15	VIO	--

NOTE

All wires are stranded silver-plated copper with polytetrafluoroethylene insulation.

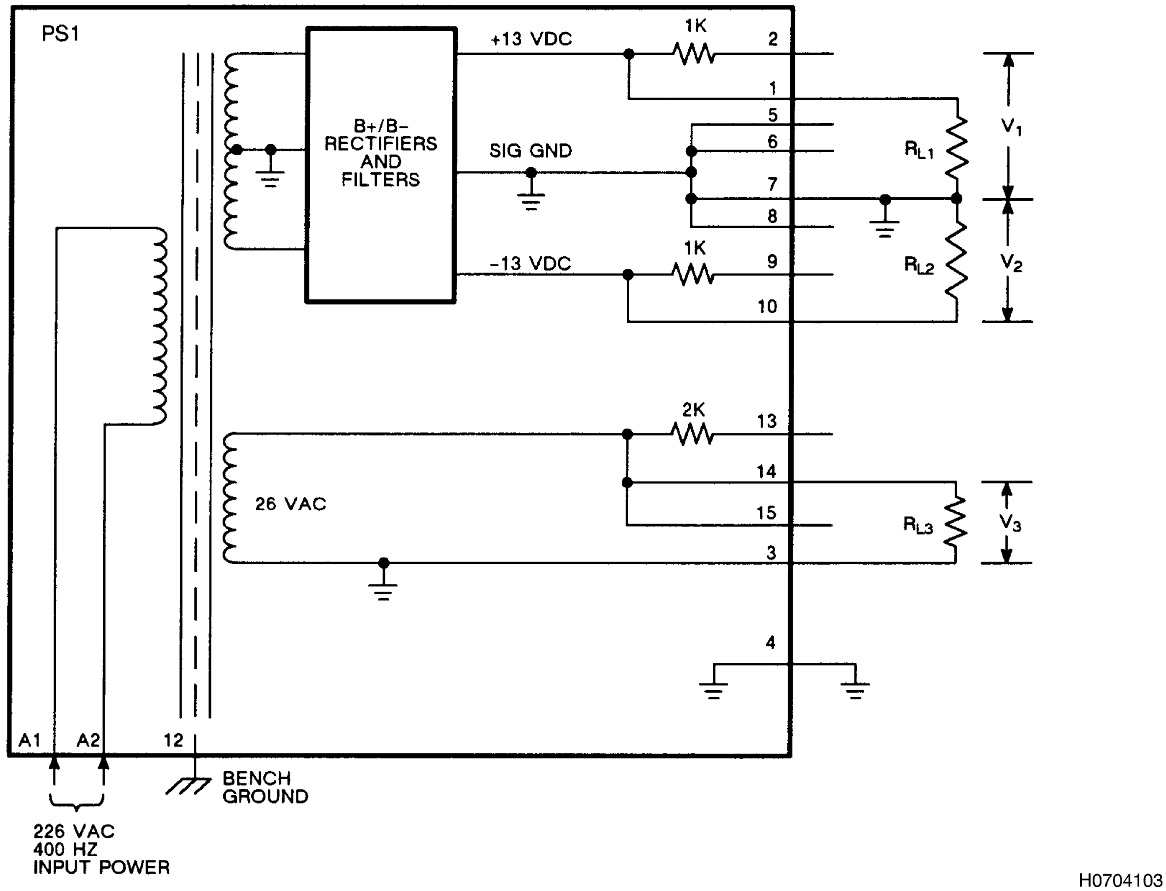


Figure 8-8. Test Connections for PS1 Test

Table 8-4. Output Voltage Tests

Load (ohms)	Output Voltage				
	V_3 (VRMS)	V_1 (VDC)	V_1 (mV P-P)	V_2 (VDC)	V_2 (mV P-P)
R_{L1} = 270 R_{L2} = 270 R_{L3} = open circuit	24.9 to 27.0	12.3 to 14.1	100 max	-12.3 to -14.1	100
$R_{L\ 1}$ = 90 R_{L2} = 90 R_{L3} = 240	24.4 to 27.0	11.2 to 13.8	100 max	-11.2 to -13.8	100 max

SECTION VII TABLE OF LIMITS

8.10 GENERAL.

There are no wear or service tolerance limits for the Thrust Control Unit.

CHAPTER 9
ILLUSTRATED PARTS BREAKDOWN

SECTION I INTRODUCTION

9.1 GENERAL.

This Illustrated Parts Breakdown (IPB) is prepared to assist supply and maintenance personnel in requisitioning, storing, issuing, identifying parts, and for illustrating assembly and disassembly relationships. This IPB is not intended to be used as an authority for procedure of disassembly or assembly. Maintenance or repair should be performed by authorized personnel using the applicable maintenance instructions.

9.2 MODELS COVERED.

This IPB lists and describes items necessary for support of Thrust Control Unit, Part Number 282E261G4, and 282E261G5.

9.3 FINDING PART NUMBERS, ILLUSTRATION, DESCRIPTION.

Refer to the illustrations at end of this section for explanation of: How to find a part number or description using the TOC and illustration titles or How to find the illustration or description if a part number or reference designation is known.

9.3.1 Maintenance Parts List. The MPL is separated into figures by main groups or assemblies and keyed to associated illustrations by figure and index numbers. In general, assemblies and parts installed at time end item(s) was manufactured are listed and identified in this manual. When an assembly or part (including vendor items), which is different from original, was installed during manufacture of later items, series or blocks, all assemblies and parts are listed (and usable on coded). However, when original assembly or part does not have continued application (no spares of original were procured or such parts are no longer authorized for replacement), only preferred assembly or part is listed. Also, when an assembly or part was installed during modification and original does not have continued application, only preferred item is listed. When a standard part can be replaced with an oversize or undersize part, the latter parts, showing sizes, are also listed.

9.3.2 Figure and Index Number Column. The figure number shall precede the first index number in each listing and the first index number on each page of a multi-page listing, but shall not be repeated before each subsequent index number. Index numbers shall be listed in sequence. When the same part appears more than once on an illustration, and is part of different assemblies or subassemblies, or it is otherwise necessary to list it more than once, each listing shall be assigned a different index number. An index number shall not be assigned to an assembly when the detail parts are indexed unless the assembly is shown completely assembled on the same illustration, or bracket or circle shows which parts comprise the assembly. When a change to an end item requires addition of parts to a listing and illustration, the index number shall be the same as the preceding index number with an added decimal number, e.g. 22.1, 22.2, etc.

9.3.3 Part Number Column. This column contains part numbers, including dash numbers, assigned to each listed part in accordance with current industry engineering practices. The appropriate numbers for the type, model, specification, specification control drawing, and the source control drawing shall follow the nomenclature. When the drawing number for a part is different than the part number (excluding dash number differences), the drawing number shall follow the description. When a part is authorized for local manufacture, and a specific drawing is required for such accomplishment, the drawing number shall follow the description. When only a model or type number identifies component, such a number shall be used in lieu of a part number.

9.3.3.1 Commercial Hardware. Commercial hardware procurable from normal commercial sources is identified with the letters COML in the part number column. When ordering the part, the complete description must be given.

9.3.3.2 Parts Not Illustrated. Parts, which are indexed in the listing but are not shown on the illustration, shall have a dash (-) preceding the index number.

9.3.3.3 Alternate Parts. Alternate parts shall have an equal sign (=) following the part number.

9.3.3.4 Marking. Decalcomanias, metalcals, vinyl film markings, etc., shall be considered as parts. The identifying drawing number for each marking shall appear in the Part Number column. The part number for each marking shall be followed (flush right) by an asterisk (*) symbol and means “requisition this marking in accordance with the requirements of DoDI 5330.03”. The nomenclature or title of each marking shall appear in the description column. The location of each marking shall be illustrated, but a separate readable illustration is not necessary.

9.3.3.5 Government Furnished Equipment (GFE) and Contractor Furnished Equipment (CFE) Covered by Separate Manuals. Following the part number of a Government Furnished Equipment (GFE) or Contractor Furnished Equipment (CFE) item, a number (#) symbol shall be inserted flush right. This symbol means that detail parts are listed in a separate manual.

9.3.4 CAGE Column. When a CAGE code for the appropriate design activity or government agency is not published in the current issues of the H4/H8 Cataloging Handbooks, the word NONE shall be inserted in the CAGE column directly opposite the part, model or type number listed in the part number column. When the word NONE is listed in the CAGE column, the complete name and address of the design activity or government agency, whose part, model or type number appears in the part number column, shall be shown in parentheses at the end of the description. When several CAGE codes are not available, or the design activity's name and address must be repeated several times, numerically identified footnotes may be used in the CAGE column and identified at the end of the MPL.

9.3.5 Description Column. This column contains the description of each part listed. The identifying noun shall be the first word. The size, dimensions, material, and tolerances for a part shall be indicated to make the description complete. When the listing is an assembly or installation, the word ASSEMBLY or INSTALLATION shall follow the noun. A comma shall separate these words from the balance of the description. When units of measure are the same, they shall not be repeated with each dimension; example: 1/8 by 3/4 inch (not 1/8 inch by 3/4 inch). To avoid confusion, the dash shall be used between a whole number and a fraction; example: 1-1/16. When a decimal with a value of less than 1.0 is given, a zero shall precede the decimal point; example: 0.002 inch. The appropriate numbers for the type, model, specification, specification control drawing, and the source control drawing shall follow the nomenclature. When the drawing number for a part is different than the part number (excluding dash number differences), the drawing number shall follow the description. When a part is authorized for local manufacture, and a specific drawing is required for such accomplishment, the drawing number shall follow the description. Reference to another illustration for information on the detail parts of an assembly and reference to another illustration for information on the NHA shall be made. Attaching parts shall be identified by the abbreviation (AP) following the description of the part.

9.3.5.1 Indentation. Parts listed in the MPL are indented to indicate item relationship or Next Higher Assembly (NHA). Nomenclature of each assembly is followed in the list (except for attaching parts) by nomenclature of its component indented one column to the right. This indentation indicates relationship of component to assembly. To determine NHA of a part or assembly, note column in which the first word of nomenclature appears. First item directly above, which appears one column to the left (except for attaching parts), is NHA.

Description

1 2 3 4 5 6 7

END ITEM, COMPONENT, MAJOR ASSEMBLY

HCI DETAIL PARTS FOR END ITEM, COMPONENT, MAJOR ASSEMBLY

· ASSEMBLY

· Attaching parts for assembly (AP)

· · Detail parts for assembly

· · Subassemblies

· · Attaching parts for subassemblies

· · · Detail parts for subassemblies

· · · Sub-subassemblies

· · · Attaching parts for sub-subassemblies (AP)

· · · · Detail parts for sub-subassemblies

9.3.5.2 Cross-Reference.

a. The notation (See Figure _____ for detail bkdn) following description of a part number indicates that further breakdown of part will be shown on figure noted.

b. The notation (See Figure _____ for NHA) following description of a part number indicates that the correct assembly relationship of the part will be shown on the figure and index noted.

c. The notation (Intchg with _____) is used when two or more parts are dimensionally and functionally interchangeable and none are preferred over the other. Notation is shown in the description column of all such interchangeable parts.

d. The notation (Use only on (part number)) following description of a part number indicates that part is used only on part number indicated.

e. The notation (When exh, use _____) following description of a part number indicates an improved or redesigned part has been developed and replaced part is still authorized for continued use until stock is depleted.

9.3.6 Units Per Assembly Column. Quantity shown in this column represents units required for each detail part in an assembly, each assembly in the NHA, and each attaching part to attach one unit or one assembly. The abbreviation AR (as required) is used when quantity required must be determined when parts are installed. The abbreviation REF (reference) indicates that item has been previously listed under its NHA. The SEE FIGURE notation in description of item will indicate figure and index number at which units per assembly can be determined.

9.3.7 Usable On Code Column. This column contains codes to indicate the configuration of the end items to which listed assemblies and parts apply. No code is required in this column if a part applies to all configurations or the manual covers only one configuration. The coding system shall consist of letters of the alphabet beginning with single letters; example: A, B, C, and continues with double letters AA, AB, AC, when necessary.

9.3.8 Source, Maintenance, and Recoverability (SMR) Code Column. This column shall contain the five-digit Joint Military Services Uniform SMR codes.

9.3.9 Numerical Index. The numerical index is required when the MPL contains 200 or more different part numbers. The numerical index contains the part numbers of all parts listed in the MPL, including model and type numbers for components not assigned part numbers.

9.3.9.1 Part Number Column. Part number arrangement shall begin at the extreme left and continue from left to right, one position at a time. For the first character of the part number, the letters A through N and P through Z take precedence over the numerals, 0 through 9. (Alphabetic Os are considered numeric zeros.) For the second and succeeding characters of a part number, precedence is as follows: (1) diagonal, (2) period, (3) dash, (4) letters A through N and P through Z, and (5) numerals, 0 through 9.

9.3.9.2 Figure and Index Number Column. This column contains the figure and index numbers for all parts listed, if applicable. When an assembly or part has not been assigned an index number, the figure and index number of the preceding part in the MPL shall be used with the letter F before the figure number, such as F7-6. The letter F means follows.

9.3.10 Reference Designator Index. This index is required only when reference designators have been established for any parts listed in the MPL. When the MPL contains less than 200 different part numbers, a reference designator index is not required, and the applicable reference designators have been inserted on the illustrations.

9.3.10.1 Reference Designator Column. This column contains all applicable reference designators shown on schematic and wiring diagrams and those cited in applicable operation and maintenance manuals. Reference designators are arranged in alphanumeric sequence. When the block or unit system of designations is employed, identical items having reference designators that are consecutive, and are at the same location, may be grouped.

9.3.10.2 Figure and Index Number Column. This column contains the figure and index numbers applicable to all reference designations listed. When an assembly or part has a reference designator, but not an index number, the figure and index number of the preceding part in the MPL shall be used with the letter F before the figure number, such as F7-6. The letter F means follows.

9.4 SIMILAR ASSEMBLIES.

Similar assemblies, which contain a majority of identical parts, are combined and listed as follows; otherwise, assemblies are listed separately:

a. Similar assembly part numbers are listed first, followed by detail parts.

b. A part common to all assemblies in the same quantity is listed once.

c. A part common to all assemblies in differing quantities is listed once for each quantity and identified with Use only on _____ note to which assembly it pertains.

d. Peculiar parts are listed once and identified with Use only on _____ note to which assembly it pertains.

9.5 QUICK CHANGE UNITS/REPAIR PARTS KITS.

When replaceable parts of an assembly are available in the form of kits, the words REPAIR KIT AVAILABLE shall follow the description of the applicable assembly. The detail parts of the assembly that are contained in the kit shall be assigned kit source codes and these codes shall appear in the description column. Illustrations of kits are not required.

9.6 SHEET NUMBER EXPLANATION.

If multiple sheet illustrations are used, the sheet number follows the index number and is separated with a slash (/). Figure 12, index 1, on sheet two of the illustration would be listed as 12 1/2. When an indexed item appears on more than one sheet of the illustration, the first and last sheet number on which the item appears shall be listed.

9.7 USABLE ON CODES.

When two or more assemblies are listed in the same parts list, a code letter (A, B, etc.) is assigned to each main assembly. All subcomponents, peculiar to a particular assembly, are identified by the same code letter as the main assembly. If parts are common to all assemblies covered, the Usable on Code column is left blank.

9.8 SOURCE, MAINTENANCE, AND RECOVERABILITY (SMR) CODES.

This manual contains Joint Military Service Uniform Source, Maintenance, and Recoverability (SMR) codes. Definitions of these SMR codes are available in TO 00-25-195.

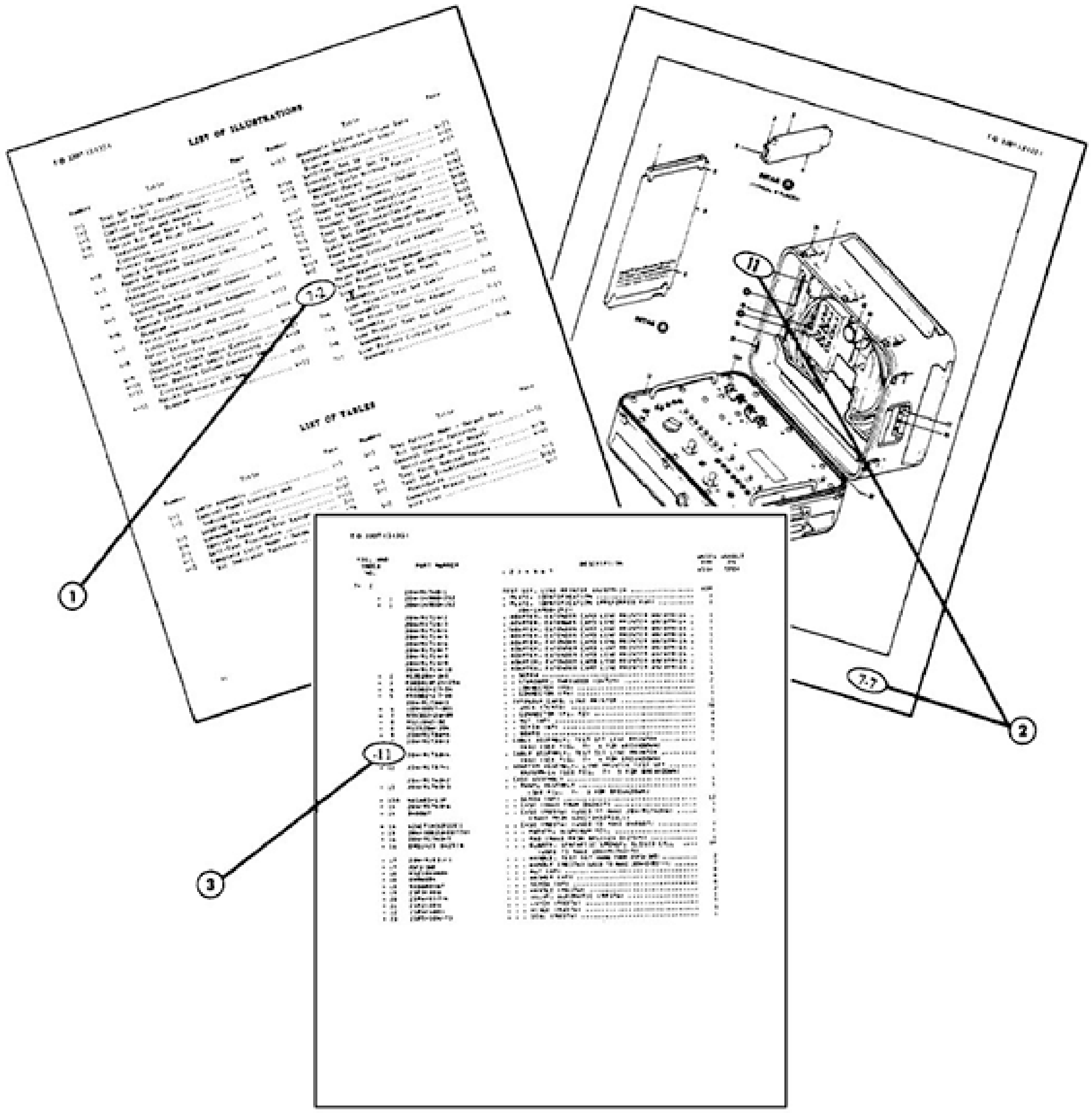
9.9 PARTS STANDARDIZATION.

Authority for use of a part number different than the part number listed in this IPB is established by the Department of Defense (DoD) Interchangeability and Substitution (I&S) Program. Refer to the D043B Master Item Identification Base for Air Force I&S information. The maintenance technician has final responsibility and authority for determining acceptability of substitute parts.

9.10 COMMERCIAL AND GOVERNMENT ENTITY (CAGE).

The cage codes are in accordance with H4/H8 Cataloging Handbooks, Commercial and Government Entity.

HOW TO USE THE ILLUSTRATED PARTS BREAKDOWN



WHEN THE PART NUMBER IS NOT KNOWN

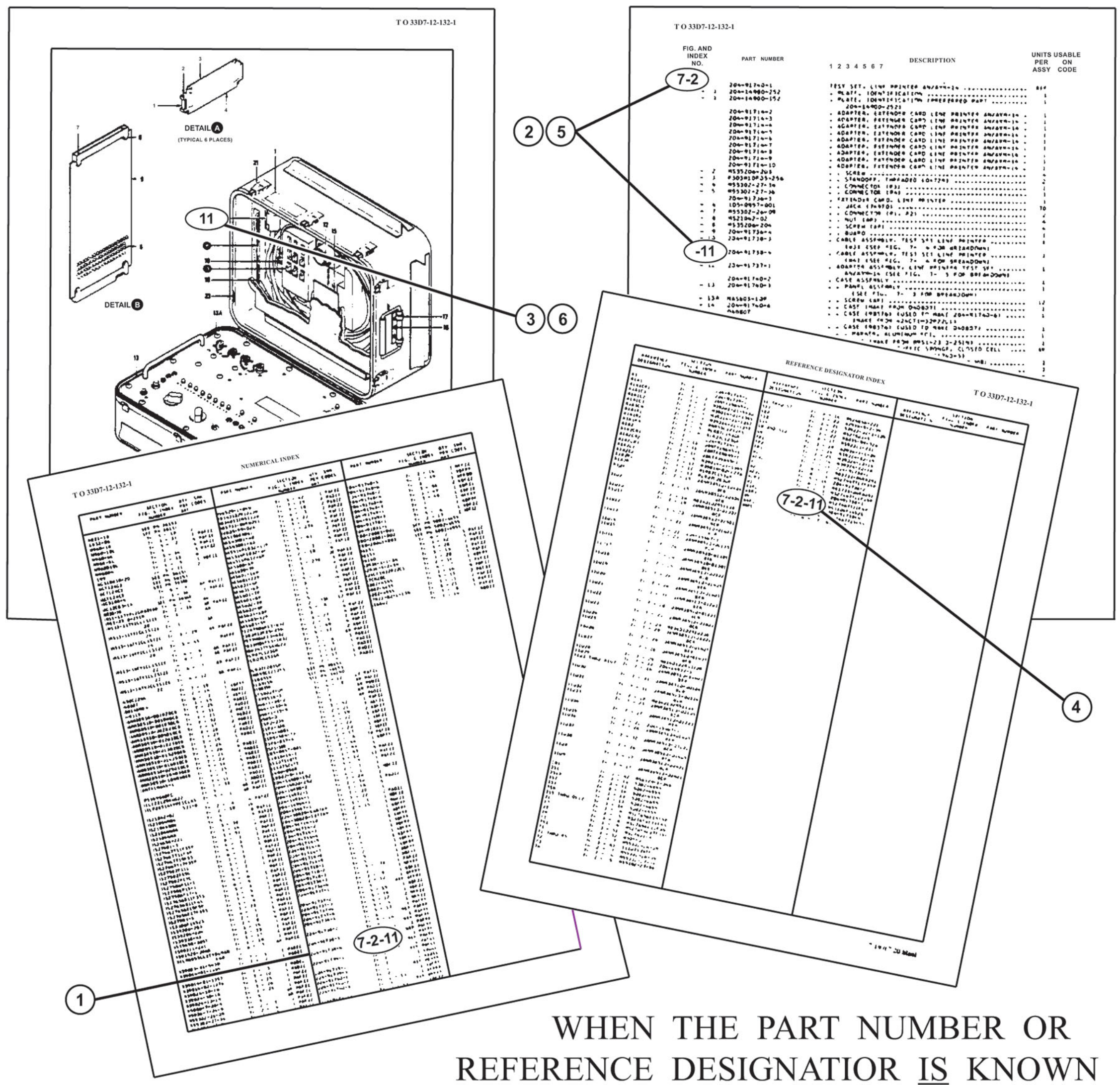
1. Determine the function and application of the part required. Turn to the List of Illustrations and select the most appropriate title. Note the illustration page number.

2. Turn to the page indicated and locate the desired part on the illustration.

3. From the illustration, obtain the index number assigned to the part desired. Refer to the accompanying description for specific information regarding the part.

STDTXHU1

HOW TO USE THE ILLUSTRATED PARTS BREAKDOWN



WHEN THE PART NUMBER OR REFERENCE DESIGNATIOR <u>IS</u> KNOWN

1. When the part number is known, locate the part number in the Numerical Index. Note the figure and index number assigned to the part number.
2. Turn to the figure number indicated; locate the index number referenced in the Numerical Index.
3. If a pictorial representation or location of the part is desired, refer to the same index number on the accompanying illustration.
4. When the referenced designatior is known, locate the reference designatior number in the Reference Designatior Index. Note the figure and index number, and the part number assigned.
5. Turn to the figure indicated; locate the index number referenced in Reference Designatior Index.
6. If a pictorial representation or location of the part is desired, refer to the same index number on the accompanying illustration.

STDTXHU2

SECTION II MAINTENANCE PARTS LIST

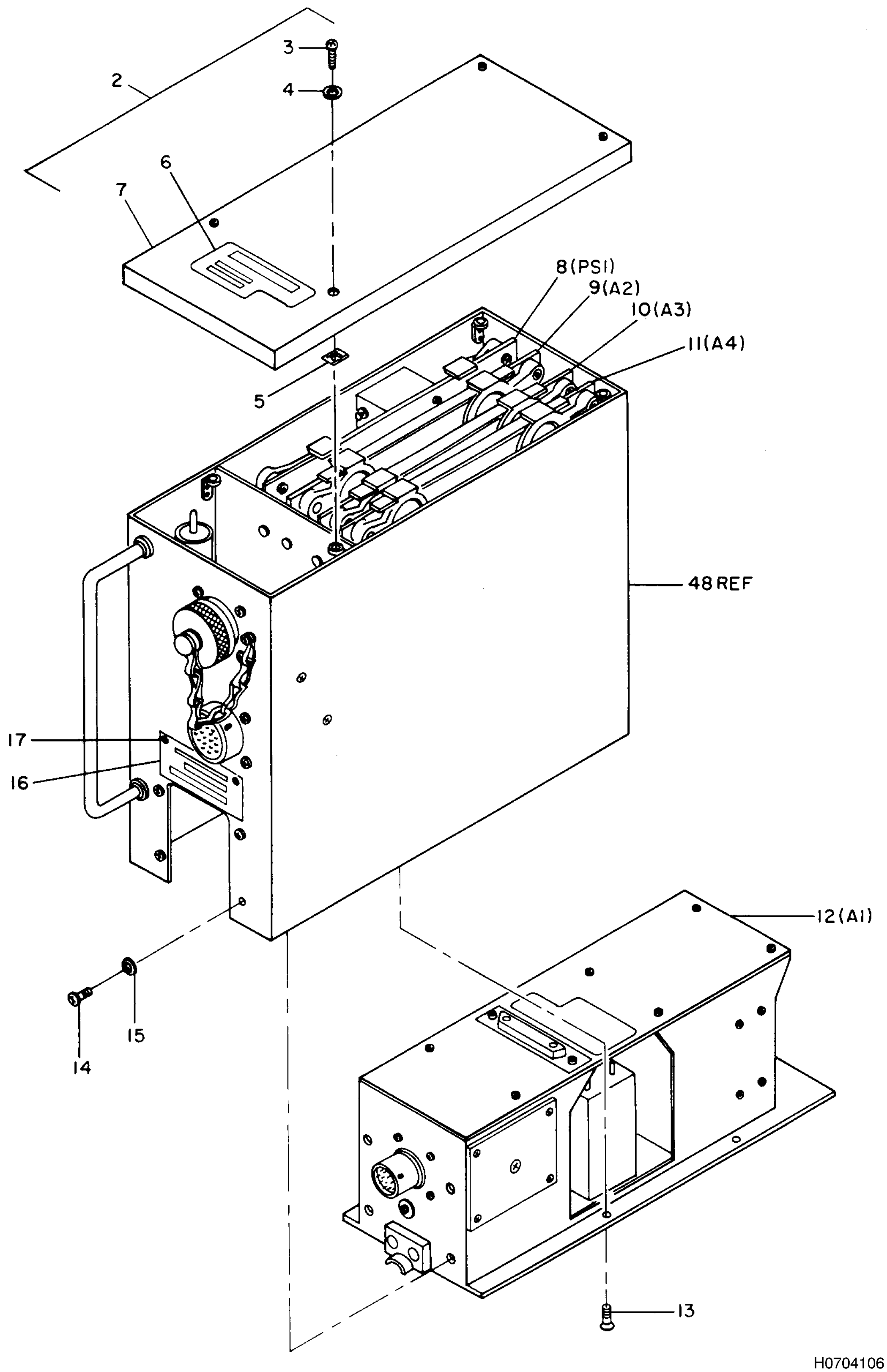


Figure 9-1. Control Unit, Thrust (Sheet 1 of 2)

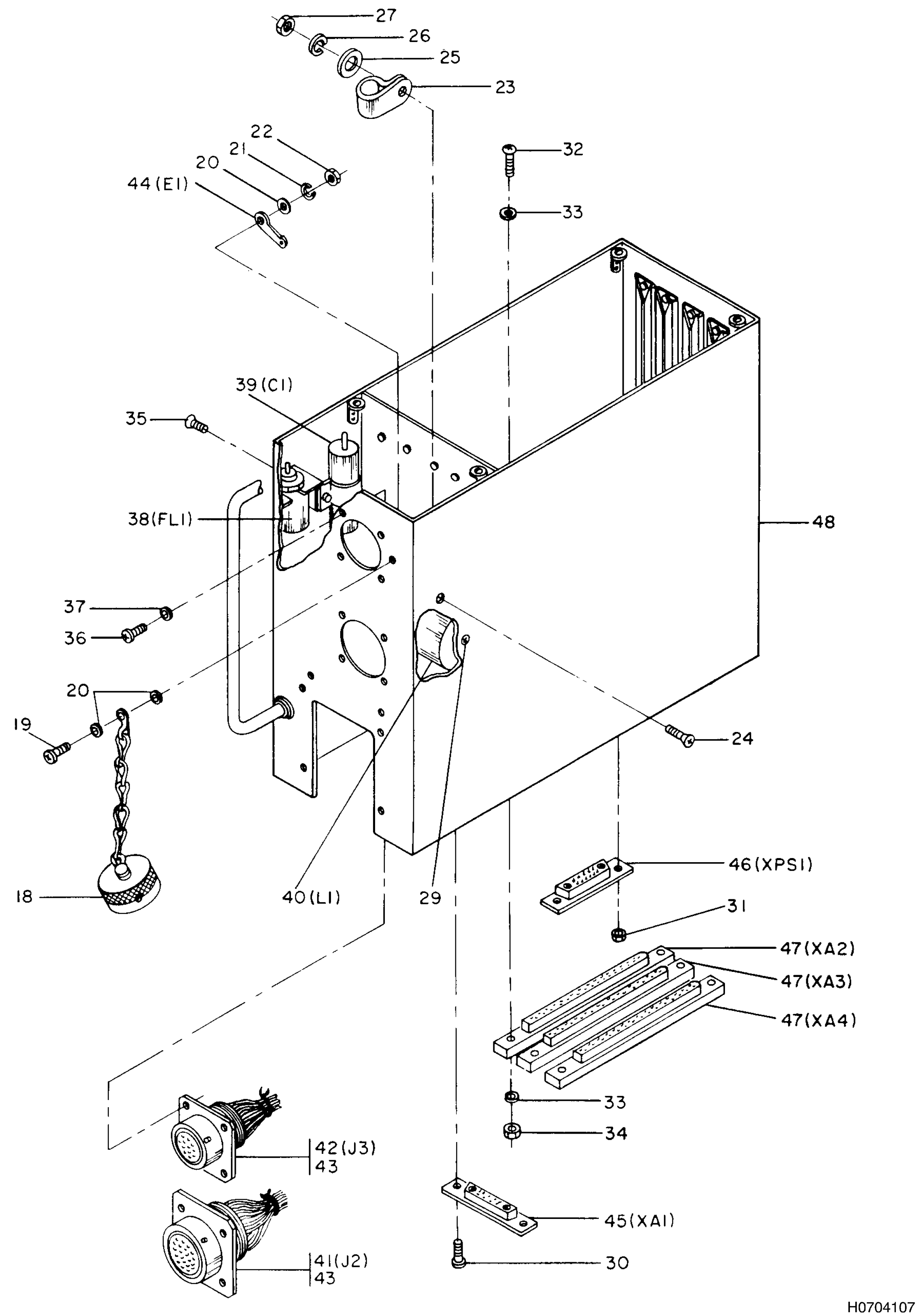


Figure 9-1. Control Unit, Thrust (Sheet 2)

FIGURE & INDEX/ SHEET NO.	PART NUMBER		CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-1-				CONTROL UNIT, THRUST			
-1	282E261G4		89954	CONTROL UNIT, THRUST	REF		PAOLD
	282E261G5		89954	CONTROL UNIT, THRUST	REF		PAOLD
2/1	12305211G1		89954	. COVER ASSEMBLY, UPPER	1		XB
3/1	16488465P1		89954	. . SCREW (ALTERED FROM MS51957-14)	4		PADZZ
4/1	NAS620C4L		80205	. . WASHER, FLAT (AP)	1		PADZZ
5/1	C7000-436-67		78553	. . NUT (AP)	1		PADZZ
6/1	569A730P1	$	89954	. . PLATE, INSTRUCTION (ADH-BACKED AL . . . . FOIL, RED WITH AL LTR, 0.003-0.005 INCH THICK; 1.98-2.07 x 0.98-1.02)	1		MDF
7/1	123D5211P1		89954	. . COVER, UPPER	1		XB
8/1	123D5213G1		89954	. MODULE ASSEMBLY, POWER SUPPLY . . . . . . (PS1) (FOR DETAILS Figure 9-6)	1		PADLD
9/1	188C1254G2		89954	. CIRCUIT CARD ASSEMBLY (A2) (FOR DE- . . . . TAILS Figure 9-3)	1		PADLD
	188C1254G1	=	89954				PADLD
10/1	123D7316G1		89954	. CIRCUIT CARD ASSEMBLY (A3) (FOR DE- . . . . TAILS Figure 9-4)	1		PADLD
11/1	123D7314G1		89954	. CIRCUIT CARD ASSEMBLY (A4) (FOR DE- . . . . TAILS Figure 9-5)	1		PADLD
12/1	282E265G4		89954	. MODULE ASSEMBLY (A1), COMPONENT . . . . . (FOR DETAILS Figure 9-2)	1	A	PADLD
	282E265G5		89954	. MODULE ASSEMBLY (A1), COMPONENT . . . . . (FOR DETAILS Figure 9-2)	1	B	PADLD
13/1	MS24693S3		96906	. SCREW (AP)	5		PADZZ
14/1	MS35206-215		96906	. SCREW (AP)	4		PADZZ
15/1	MS27183-3		96906	. WASHER, FLAT (AP)	4		PADZZ
16/1	188C1624P2	$	89954	. PLATE, IDENT (AL ALLOY PER QQ-A-250/ . . . . 11, TEMP T4, 0.032 INCH STOCK, 2.015 X 1.015 INCH -0.030 INCH)	1	A	MDF
-16A	200945069-501		98748	. PLATE, IDENTIFICATION (Reference draw- . . . . ing 200945069 and Appendix A of this TO when manufacturing plate)	1	B	MDD
	188C1624P8	$	89954	. PLATE, IDENT (AL ALLOY PER QQ-A-250/ . . . . 11, TEMP T4, 0.032 INCH STOCK, 2.015 X 1.015 INCH -0.030 INCH)	1	B	MDF
17/1	926C171P213		89954	. SCREW (ALTERED FROM MS35206-213) . . . . . (AP)	2		PADZZ
18/2	MS27511F14C		96906	. CAP, PROTECTIVE	1		PADZZ
19/2	MS35206-216		96906	. SCREW (AP)	1		PADZZ
	MS35206-215	=	96906				PADZZ
20/2	MS27183-4		96906	. WASHER, FLAT (AP)	3		PADZZ
21/2	MS35338-40		96906	. WASHER, LOCK (AP)	1		PADZZ
22/2	MS35649-242		96906	. NUT (AP)	1		PADZZ
23/2	NAS1397P5		96906	. CLAMP, LOOP	1		PADZZ
	MS25281F5	=	96906				PADZZ
24/2	MS24693S6		96906	. SCREW (AP)	1		PADZZ
25/2	MS27183-4		96906	. WASHER, FLAT (AP)	1		PADZZ
26/2	MS35338-40		96906	. WASHER, LOCK (AP)	1		PADZZ
27/2	MS35649-242		96906	. NUT (AP)	1		PADZZ
-28	123D5212G3		89954	. HARNESS ASSEMBLY, CONTROL UNIT	1		XB
29/2	MS24693S2		96906	. SCREW (AP)	1		PADZZ
30/2	MS35206-203		96906	. SCREW (AP)	2	A	PADZZ
	MS21090-0202		96906	. SCREW (AP)	2	B	PADZZ
	MS35206-203	=	96906			B	PADZZ
31/2	H14-02		15653	. NUT (GE SPEC CONT DWG 10788156P1) . . . . (AP)	2		PADZZ
32/2	MS35206-218		96906	. SCREW (AP)	6		PADZZ
33/2	MS27183-3		96906	. WASHER, FLAT (AP)	12		PADZZ
34/2	H10-04		15653	. NUT (GE SPEC CONT DWG 10788151P1) . . . . (AP)	6		PADZZ
35/2	MS24693S3		96906	. SCREW (AP)	1		PADZZ
36/2	MS35206-214		96906	. SCREW (AP)	8		PADZZ
37/2	MS27183-3		96906	. WASHER, FLAT (AP)	8		PADZZ
38/2	A2225A		23939	. . FILTER, RADIO INTERFERENCE (GE . . . . SPEC CONT DWG 144A8714P1)	1		PADZZ
39/2	172B5197G1		89954	. . CAPACITOR ASSEMBLY	1		PADZZ

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-1- 40/2	TW63883	02640	. . INDUCTOR, POWER (GE SPEC CONT . . . . . . DWG 144A8717P1)	1		PADZZ
	K4812 =	51181				PADZZ
41/2	164B7943G3	89954	. . CONNECTOR ASSEMBLY, ELEC	1		PADZZ
42/2	164B7943G2	89954	. . CONNECTOR ASSEMBLY, ELEC	1		PADZZ
43/2	MS27488-22	96906	. . PLUG, SEALING (IN UNUSED CONTACT . . . . POSITIONS OF J2 AND J3)	18		PADZZ
44/2	2103-04-00	78189	. . TERMINAL, LUG (GE SPEC CONT DWG . . . . . 539D600P4)	1		PADZZ
	1414-4 =	83330				PADZZ
45/2	DBMMR17W2P	71468	. . CONNECTOR (SUPPLIED WITH 2 RED . . . . . PINS DM51158-3) (GE SPEC CONT DWG 144A8720P1)	1		PADZZ
46/2	DBMMF17W2S	71468	. . CONNECTOR (SUPPLIED WITH 2 BLUE . . . . . SOCKETS DM51156-3) (GE SPEC CONT DWG 144A8720P3)	1		PADZZ
47/2	10-285416-4S	77820	. . CONNECTOR, RCPT, ELEC. (GE SPEC . . . . . CONT DWG 854D896P12)	3		PADZZ
48/2	937E678G1	89954	. CHASSIS ASSEMBLY, ELEC EQPT	1	A	XB
	937E67BG2	89954	. CHASSIS ASSEMBLY, ELEC EQPT	1	B	XB
			CODES USABLE ON			
			A 282E261G4			
			B 282E261G5			

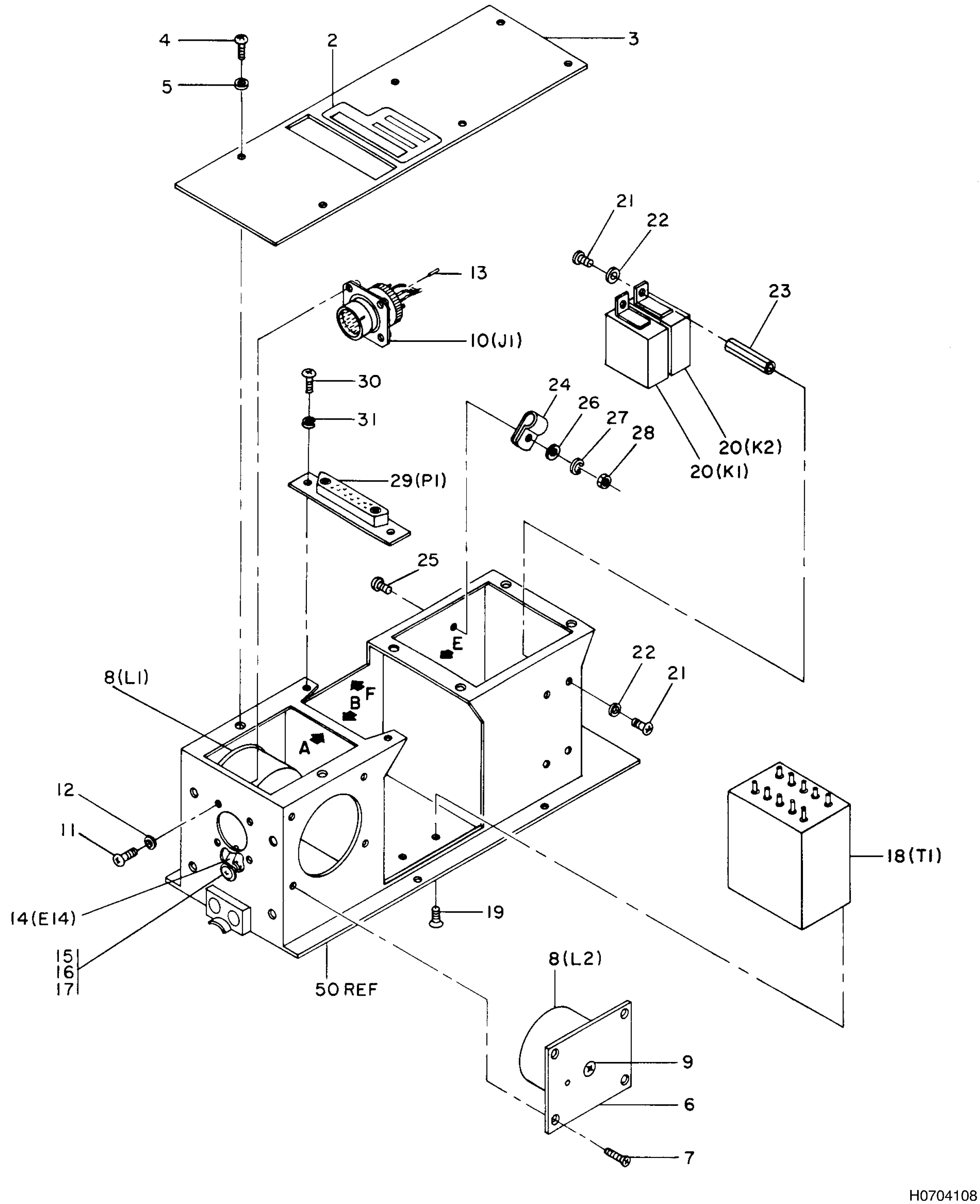


Figure 9-2. Module Assembly (A1), Component (Sheet 1 of 2)

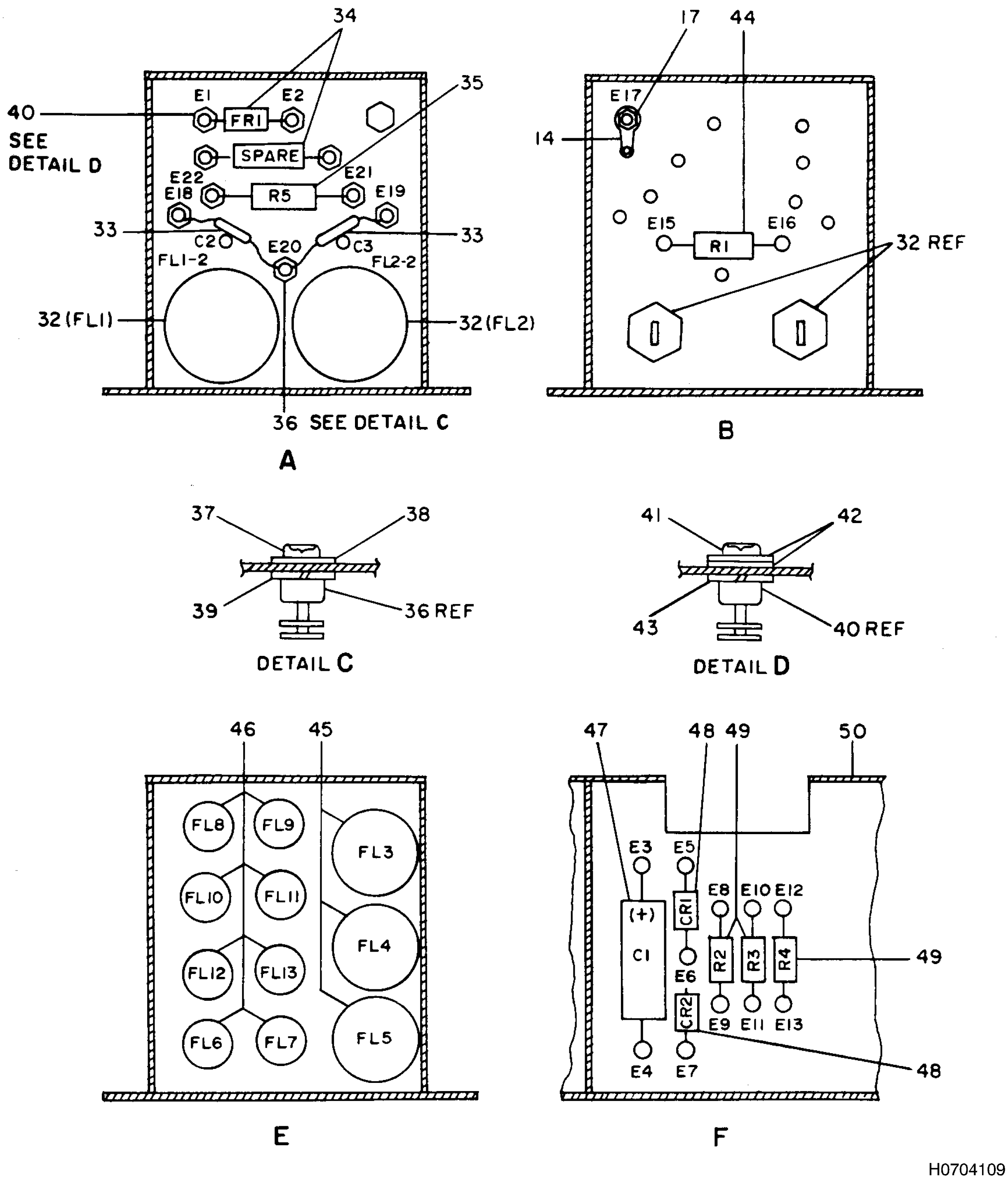
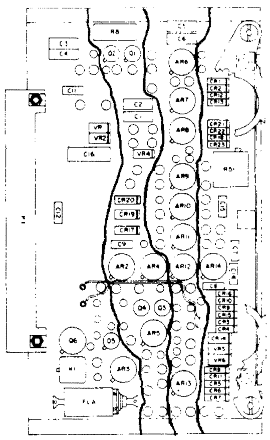


Figure 9-2. Module Assembly (A1), Component (Sheet 2)

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-2-			MODULE ASSEMBLY (A1), COMPONENT			
-1	282E265G4	89954	MODULE ASSEMBLY (A1), COMPONENT . . . . . . (FOR NHA Figure 9-1)	REF		PADLD
	282E265G5	89954	MODULE ASSEMBLY (A1), COMPONENT . . . . . . (FOR NHA Figure 9-1)	REF		PADLD
2/1	569A730P1 $	89954	. PLATE, INSTRUCTION (ADH-BACKED AL . . . . . FOIL, RED WITH AL LTR, 0.003-0.005 INCH THICK; 1.98-2.07 x 0.98-1.02)	1		MDF
3/1	153C5379P1	89954	. COVER, MODULE	1		MDD
4/1	MS35206-213	96906	. SCREW (AP)	6		PADZZ
5/1	NAS620-4L	80205	. WASHER, FLAT (AP)	6		PADZZ
6/1	937E677P2	89954	. PLATE, CHASSIS ELEC	2		XB
7/1	MS24693S2	96906	. SCREW (AP)	4		PADZZ
8/1	K5295	51181	. INDUCTOR, POWER (GE SPEC CONT . . . . DWG 144A9535P1)	2		PADZZ
	412001 =	72149				
9/1	MS35190-235	96906	. SCREW (AP)	1		PADZZ
10/1	164B7943G1	89954	. CONNECTOR ASSEMBLY, ELEC	1		PADZZ
11/1	MS35206-213	96906	. SCREW (AP)	4		PADZZ
12/1	NAS620-4L	80205	. WASHER, FLAT (AP)	4		PADZZ
13/1	MS27488-22	96906	. PLUG, SEALING (IN UNUSED CONTACT . . . . . POSITIONS OF J1)	10		PADZZ
14/1/2	2103-04-00	78189	. TERMINAL, LUG (GE SPEC CONT DWG . . . . . 539D600P4)	2		PADZZ
	1414-4 =	83330				PADZZ
15/1	MS35206-214	96906	. SCREW (AP)	1		PADZZ
16/1	MS27183-4	96906	. WASHER, FLAT (AP)	1		PADZZ
17/1/2	H10-04	15653	. NUT (GE SPEC CONT DWG 107B8151P1) . . . . (AP)	2		PADZZ
18/1	TW63881	02640	. TRANSFORMER, POWER (GE SPEC CONT . . . DWG 144A8718P1)	1		PADZZ
	K4941 =	51181				PADZZ
	B573 =	26522				PADZZ
19/1	MS24693S2	96906	. SCREW (AP)	4		PADZZ
20/1	MS27401-13	35344	. RELAY, ARMATURE (GE REF DWG . . . . 144A8697P1)	2		PADZZ
21/1	MS35206-213	96906	. SCREW (AP)	4		PADZZ
22/1	NAS620-4L	80205	. WASHER, FLAT (AP)	4		PADZZ
23/1	113B3626G16	89954	. SPACER AND INSERT ASSEMBLY (AP)	2		PADZZ
24/1	HP2N	09922	. CLAMP, LOOP (GE SPEC CONT DWG . . . . 672B452P1)	1		PADZZ
25/1	MS35206-216	96906	. SCREW (AP)	1		PADZZ
26/1	MS27183-4	96906	. WASHER, FLAT (AP)	1		PADZZ
27/1	MS35338-40	96906	. WASHER, LOCK (AP).	1		PADZZ
28/1	MS35649-242	96906	. NUT	1		PADZZ
29/1	DBMM17W2S	71468	. CONNECTOR, (SUPPLIED WITH 2 BLUE . . . . . SOCKETS DM51156-3) (GE SPEC CONT DWG 144A8720P2)	1		PADZZ
30/1	MS35206-202	96906	. SCREW (AP)	2		PADZZ
31/1	MS35338-39	96906	. WASHER, LOCK (AP).	2		PADZZ
32/2	A1936	23939	. FILTER, RADIO INTERFERENCE (GE SPEC . . . CONT DWG 144A8713P1)	2		PADZZ
	5004-6347 =	14158				PADZZ
33/2	CK63AY103M	81349	. CAPACITOR	2		PADZZ
34/2	A425	11502	. RESISTOR, FUSE (1 SPARE) (GE SPEC. . . . . . CONT DWG 144A8722P1)	2		PADZZ
35/2	RCR32G220JR	81349	. RESISTOR	1		PADZZ
36/2	DL13449-21GCETCI	27494	. TERMINAL, STUD (GE SPEC CONT DWG . . . . 926C139P1)	1		PADZZ
37/2	MS16997-9	96906	. SCREW (AP)	1		PADZZ
38/2	MS27183-3	96906	. WASHER, FLAT (AP)	1		PADZZ
39/2	MS35338-40	96906	. WASHER, LOCK (AP).	1		PADZZ
40/2	DL13550-38CETCI	27494	. TERMINAL, STUD (GE SPEC CONT DWG . . . . 926C139P2)	8		PADZZ
41/2	MS16997-1	96906	. SCREW (AP)	1		PADZZ
42/2	MS27183-2	96906	. WASHER, FLAT (AP)	2		PADZZ
43/2	MS35338-39	96906	. WASHER, LOCK (AP).	1		PADZZ
44/2	RCR20G202JR	81349	. RESISTOR	1		PADZZ

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-2- 45/2	A2225A	23939	. FILTER, RADIO INTERFERENCE (GE SPEC . . . CONT DWG 144A8714P1)	3		PADZZ
46/2	1JX2811X	56289	. FILTER, RADIO INTERFERENCE (GE SPEC . . . CONT DWG 144A8715P1)	8		PADZZ
	8414-1005 =	14158				PADZZ
	A1973 =	23939				PADZZ
47/2	40EW277C050WIC	26769	. CAPACITOR, FXD (GE SPEC CONT DWG . . . . 144A8721P2)	1		PADZZ
48/2	JAN1N4944	81350	. SEMICONDUCTOR	2		PADZZ
49/2	RCR32G300JR	81349	. RESISTOR	3		PADZZ
50/2	937E677G1	89954	. CHASSIS ASSEMBLY, ELEC EQPT	1	A	XB
	937E677G2	89954	. CHASSIS ASSEMBLY, ELECT EQPT	1	B	XB
-51	200945069-501	98748	. PLATE, IDENTIFICATION (Reference draw- . . . . ing 200945069 and Appendix A of this TO when manufacturing plate)	1	B	MDD
			CODES / USABLE ON			
			A 282E265G4			
			B 282E265G5			



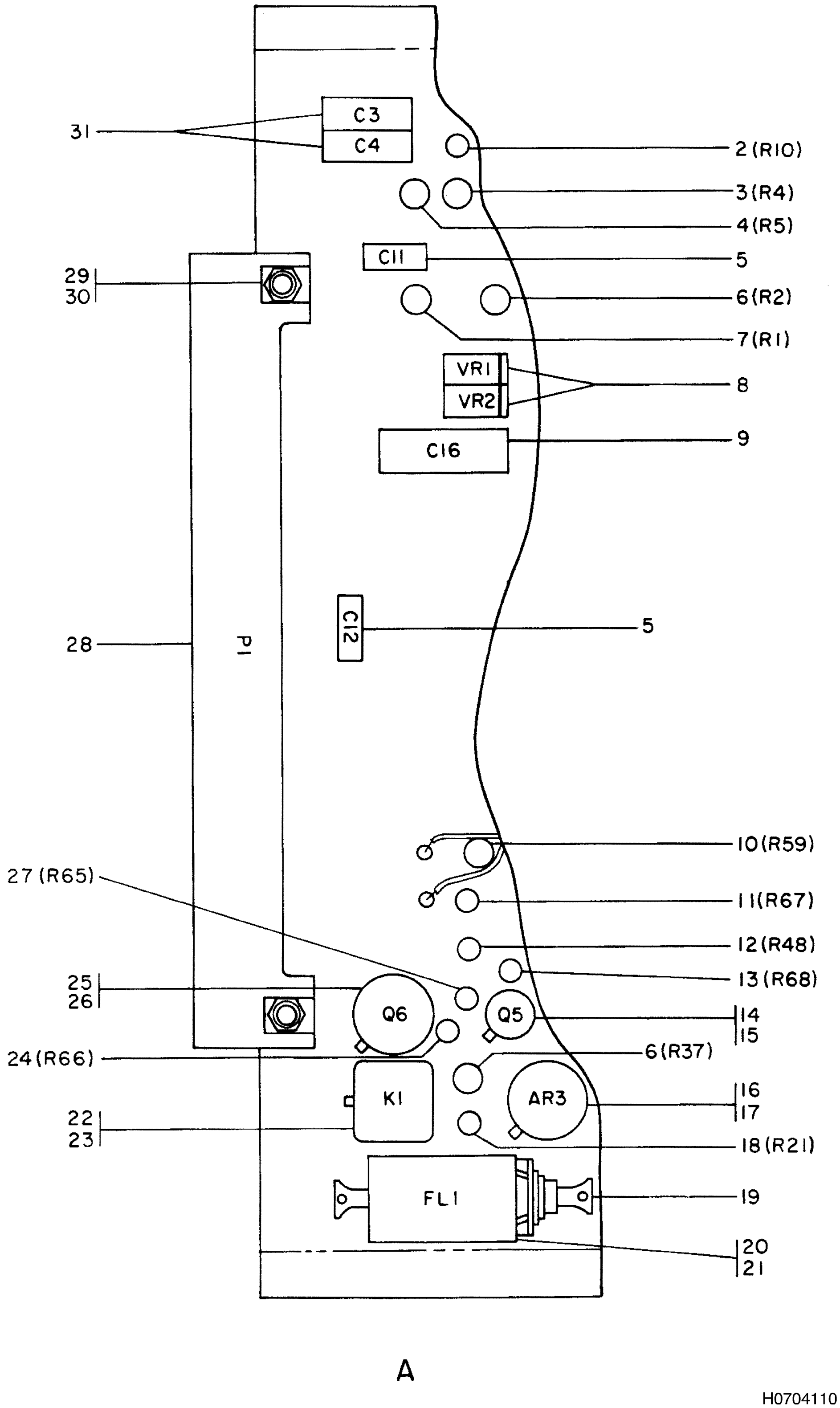


Figure 9-3. Circuit Card Assembly (A2) (Sheet 1 of 3)

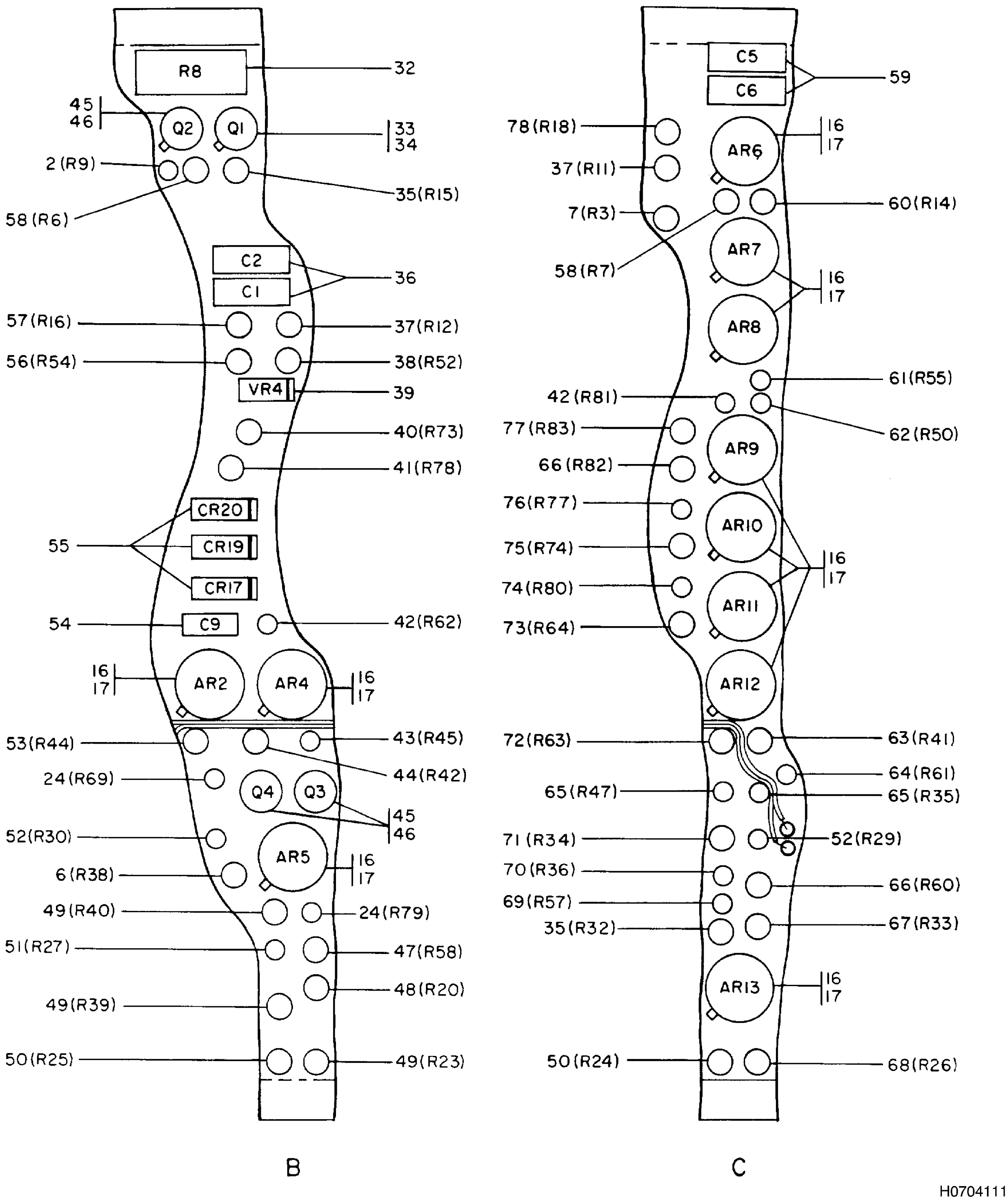


Figure 9-3. Circuit Card Assembly (A2) (Sheet 2)

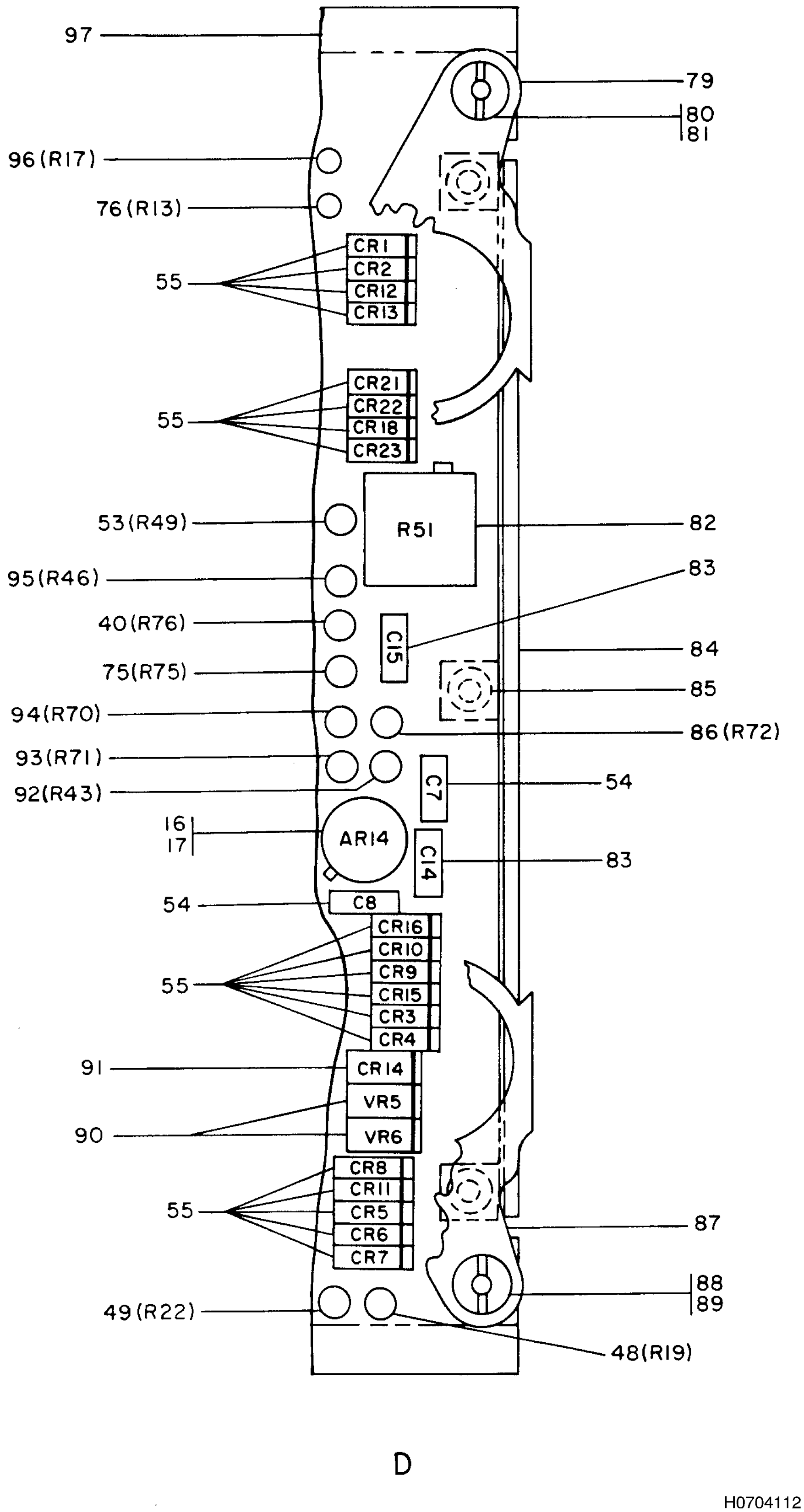


Figure 9-3. Circuit Card Assembly (A2) (Sheet 3)

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-3-			CIRCUIT CARD ASSEMBLY (A2)			
-1	188C1254G1	89954	CIRCUIT CARD ASSEMBLY (A2) (FOR NHA . . . . Figure 9-1)	REF		PADLD
	188C1254G2	89954	CIRCUIT CARD ASSEMBLY (A2) (FOR NHA . . . . Figure 9-1)	REF		PADLD
2/1/2	RCR07G432JS	81349	. RESISTOR	2		PADZZ
3/1	RNC55H2371FM	81349	. RESISTOR	1		PADZZ
4/1	RNC55H1332FM	81349	. RESISTOR	1		PADZZ
5/1	M39014-02-0218	81349	. CAPACITOR	2		PADZZ
6/1/2	RNC55H2001FM	81349	. RESISTOR	3		PADZZ
7/1/2	RNC55H8451FM	81349	. RESISTOR	2		PADZZ
8/1	JANTX1N755A	81349	. SEMICONDUCTOR	2		PADZZ
9/1	M39003-01-5044	81349	. CAPACITOR	1		PADZZ
10/1	RNC55H1782FS	81349	. RESISTOR	1		PADZZ
11/1	RLR07C1001GR	81349	. RESISTOR	1		PADZZ
12/1	RCR07G682JS	81349	. RESISTOR	1		PADZZ
13/1	RLR07C6801GR	81349	. RESISTOR	1		PADZZ
14/1	JANTX2N2222A	81349	. TRANSISTOR	1		PADZZ
15/1	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
16/1/2/3	JM38510-10101BGX	81349	. MICROCIRCUIT	13		PADZZ
	LM741H/883 =					
17/1	10336DAP	07047	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 144A8597P1) (AP)	1		PADZZ
18/1	RCR07G223JS	81349	. RESISTOR	1		PADZZ
19/1	1JX2811X	56289	. FILTER, RADIO INTERFERENCE (GE SPEC . . . CONT DWG 144A8715P1)	1		PADZZ
	8414-1005 =	14158				PADZZ
	A1973 =	23939				PADZZ
20/1	172B5192P1	89954	. BRACKET, FL	1		PADZZ
21/1	MS20470B3-3		. RIVET (AP)	2		PADZZ
22/1	M39016-18-036M	81349	. RELAY	1		PADZZ
23/1	172B4918P1	89954	. INSULATOR (AP)	1		
24/1/2	RLR07C2001GR	81349	. RESISTOR	3		PADZZ
25/1	JANTX2N3501	81349	. SEMICONDUCTOR	1		PADZZ
26/1	M38527-5-02D	81349	. PAD, MOUNTING (AP)	1		PADZZ
27/1	RLR07C1202GR	81349	. RESISTOR	1		PADZZ
28/1	10-285418-5P	77820	. CONNECTOR, PLUG (GE SPEC CONT . . . . DWG 854D896P22)	1		PADZZ
29/1	68-1660-26	72962	. NUT (GE SPEC CONT DWG 10788220P3) . . . . (AP)	2		PADZZ
30/1	MS35206-205	96906	. SCREW (AP)	2		PADZZ
31/1	M39003-01-2283	81349	. CAPACITOR	2		PADZZ
32/2	RCR32G102JR	81349	. RESISTOR	1		PADZZ
33/2	JANTX2N2945A	81349	. TRANSISTOR	1		PADZZ
34/2	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
35/2	RNC55H5761FM	81349	. RESISTOR	2		PADZZ
36/2	M39003-01-2590	81349	. CAPACITOR	2		PADZZ
37/2	RNC55H6191FM	81349	. RESISTOR	2		PADZZ
38/2	RNC55H2002FM	81349	. RESISTOR	1		PADZZ
39/2	JANTX1N823	81349	. SEMICONDUCTOR	1		PADZZ
40/2/3	RNC55H1472FS	81349	. RESISTOR	2		PADZZ
41/2	RNC55H4993FS	81349	. RESISTOR	1		PADZZ
42/2	RCR07G474JS	81349	. RESISTOR	2		PADZZ
43/2	RCR07G203JS	81349	. RESISTOR	1		PADZZ
44/2	RNC55H7502FM	81349	. RESISTOR	1		PADZZ
45/2	JANTX2N2432A	81349	. TRANSISTOR	3		PADZZ
46/2	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
47/2	RNC55H2372FM	81349	. RESISTOR	1		PADZZ
48/2/3	RNC55H1002BM	81349	. RESISTOR	2		PADZZ
49/2/3	RNC55H1002FM	81349	. RESISTOR	4		PADZZ
50/2	RNC55H8061BM	81349	. RESISTOR	2		
51/2	RCR07G472JS	81349	. RESISTOR	1		PADZZ
52/2	RCR07G302JS	81349	. RESISTOR	2		PADZZ

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-3- 53/2/3	RNC55H1432FM	81349	. RESISTOR	2		PADZZ
54/2/3	M39014-02-0230	81349	. CAPACITOR	3		PADZZ
55/2/3	JANTX1N4148-1	81350	. SEMICONDUCTOR	22		PADZZ
56/2	RNC55H2432FM	81349	. RESISTOR	1		PADZZ
57/2	RNC55H5621FM	81349	. RESISTOR	1		PADZZ
58/2	RNC55H1212FM	81349	. RESISTOR	2		PADZZ
59/2	M39003-01-2596	81349	. CAPACITOR	2		PADZZ
60/2	RNC55H7682FM	81349	. RESISTOR	1		PADZZ
61/2	RCR07G510JS	81349	. RESISTOR	1		PADZZ
62/2	RCR07G821JS	81349	. RESISTOR	1		PADZZ
63/2	RNC55H4642FM	81349	. RESISTOR	1		PADZZ
64/2	RLR07C1003GR	81349	. RESISTOR	1		PADZZ
65/2	RCR07G242JS	81349	. RESISTOR	2		PADZ
66/2	RNC55H4991FS	81349	. RESISTOR	2		PADZ
67/2	RNC55H7151FM	81349	. RESISTOR	1		PADZZ
68/2	RNC55H5491FM	81349	. RESISTOR	1		PADZZ
69/2	RCR07G512JS	81349	. RESISTOR	1		PADZZ
70/2	RCR07G752JM	81349	. RESISTOR	1		PADZZ
71/2	RNC55H6491FM	81349	. RESISTOR	1		PADZZ
72/2	RNC55H8451FS	81349	. RESISTOR	1		PADZZ
73/2	RNC55H2001FS	81349	. RESISTOR	1		PADZZ
74/2	RCR07G103JS	81349	. RESISTOR	1		PADZZ
75/2/3	RNC55H7151FS	81349	. RESISTOR	2		PADZZ
76/2/3	RCR07G332JS	81349	. RESISTOR	2		PADZZ
77/2	RNC55H8061FS	81349	. RESISTOR	1		PADZZ
78/2	RNC55H3092FM	81349	. RESISTOR	1		PADZZ
79/3	135C4148P2	89954	. EXTRACTOR	1		PADZZ
80/3	144A5514P1	89954	. NUT (AP).	1		PADZZ
81/3	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
82/3	RJ22CP102	81349	. RESISTOR	1		PADZZ
83/3	M39014-02-1407	81349	. CAPACITOR	2		PADZZ
84/3	153C5392P2	89954	. STIFFENER	1		XB
85/3	MS2047083-3		. RIVET (AP)	3		PADZZ
86/3	RNC55H1652FS	81349	. RESISTOR	1		PADZZ
87/3	135C4148P1	89954	. EXTRACTOR	1		PADZZ
88/3	144A5514P1	89954	. NUT (AP).	1		PADZZ
89/3	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
90/3	JANTX1N751A	81349	. SEMICONDUCTOR	2		PADZZ
91/3	JANTX1N649	81349	. SEMICONDUCTOR	1		PADZZ
92/3	RNC55H4022FM	81349	. RESISTOR	1		PADZZ
93/3	RNC55H4992FS	81349	. RESISTOR	1		PADZZ
4/3	RNC55H3012FM	81349	. RESISTOR	1		PADZZ
95/3	RNC55H5111FM	81349	. RESISTOR	1		PADZZ
96/3	RCR07G822JS	81349	. RESISTOR	1		PADZZ
97/3	188C1253P1	89954	. BOARD, PRINTED WIRING	1	A	XA
	188C1777P1	89954	. BOARD, PRINTED WIRING	1	B	XA
			CODES / USABLE ON			
			A 188C1254G1			
			B 188C1254G2			

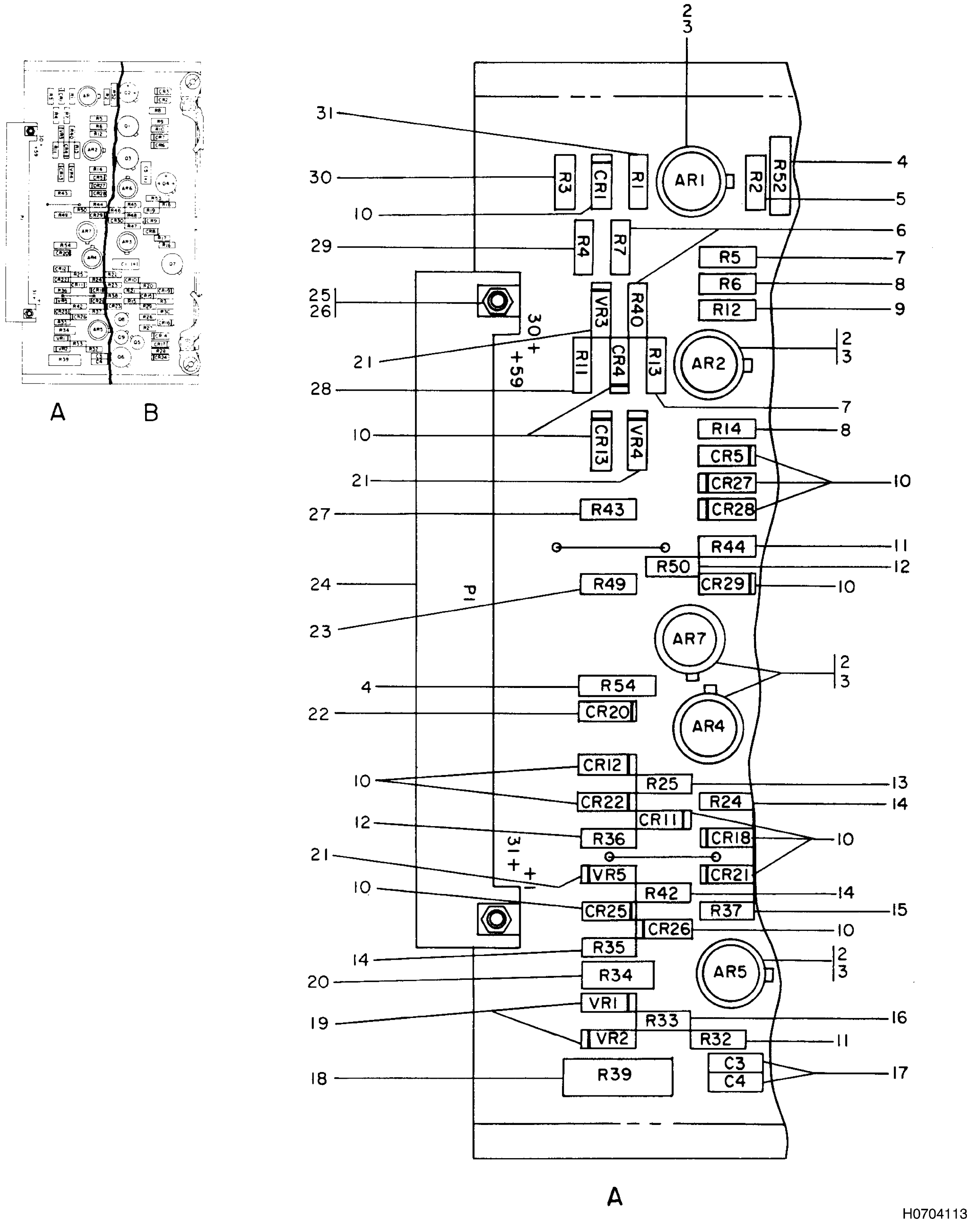


Figure 9-4. Circuit Card Assembly (A3) (Sheet 1 of 2)

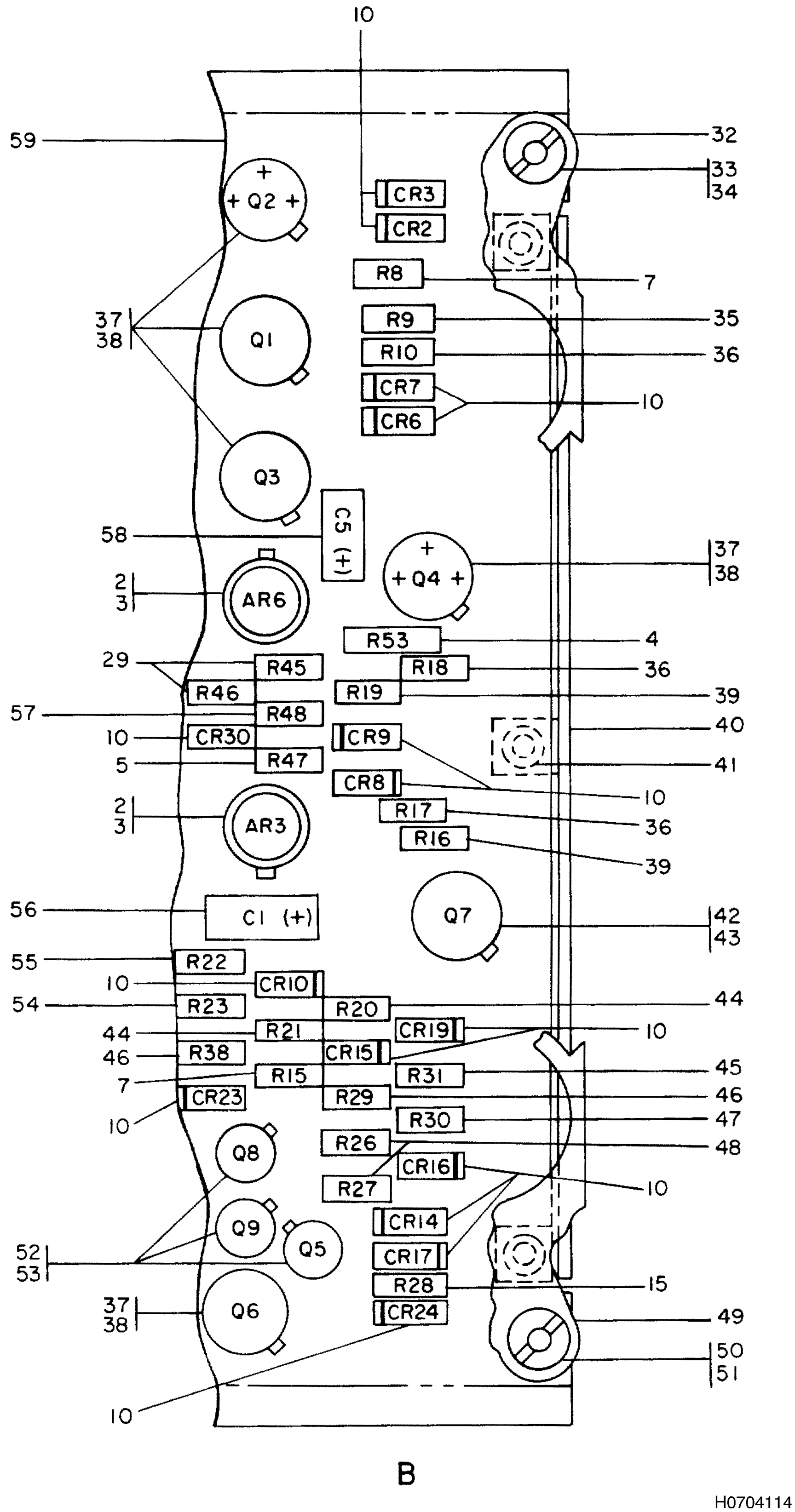
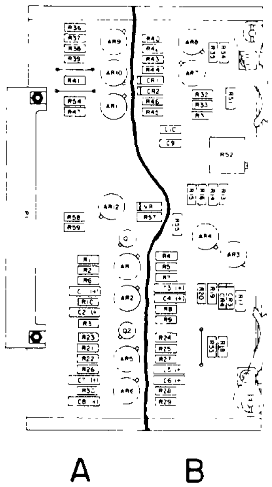


Figure 9-4. Circuit Card Assembly (A3) (Sheet 2)

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-4-			CIRCUIT CARD ASSEMBLY (A3)			
-1	123D7316G1	89954	CIRCUIT CARD ASSEMBLY (A3) (FOR NHA . . . . Figure 9-1)	REF		PADLD
2/1/2	JM38510-10101BGX	81349	. MICROCIRCUIT	7		PADZZ
	LM741H/883 =					
	14489511P1 =	89954				PADZZ
3/1	10336DAP	07047	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 144A8597P1) (AP)	1		PADZZ
4/1/2	RWR80S24R3P	81349	. RESISTOR	3		PADZZ
	RWR80S24R3FR =					
6/1	RCR07G470JS	81349	. RESISTOR	2		PADZZ
7/1/2	RCR07G103JS	81349	. RESISTOR	4		PADZZ
8/1	RCR07G202JS	81349	. RESISTOR	2		PADZZ
9/1	RCR07G472JS	81349	. RESISTOR	1		PADZZ
10/1/2	JANTX1N4148-1	81350	. SEMICONDUCTOR	29		PADZZ
11/1	RNC55H3012FM	81349	. RESISTOR	2		PADZZ
12/1	RCR07G510JS	81349	. RESISTOR	2		PADZZ
13/1	RNC55H2002FM	81349	. RESISTOR	1		PADZZ
14/1	RNC55H4022FM	81349	. RESISTOR	3		PADZZ
15/1/2	RCR07G122JS	81349	. RESISTOR	2		PADZZ
16/1	RNC55H6192FM	81349	. RESISTOR	1		PADZZ
17/1	M39014-02-0218	81349	. CAPACITOR	2		PADZZ
18/1	RCR32G301JR	81349	. RESISTOR	1		PADZZ
19/1	JAN1N751A	81349	. SEMICONDUCTOR	2		PADZZ
20/1	RCR07G681JR	81349	. RESISTOR	1		PADZZ
21/1	JAN1N970B	81349	. SEMICONDUCTOR	3		PADZZ
22/1	JAN1N4944	81350	. SEMICONDUCTOR	1		PADZZ
23/1	RNC55H2001FM	81349	. RESISTOR	1		PADZZ
24/1	10-285418-5P	77820	. CONNECTOR, PLUG (GE SPEC CONT . . . . DWG 854D896P22)	1		PADZZ
25/1	68-1660-26	72962	. NUT (GE SPEC CONT DWG 10788220P3) . . . . (AP)	2		PADZZ
26/1	MS35206-205	96906	. SCREW (AP)	2		PADZZ
27/1	RNC55H3011FM	81349	. RESISTOR	1		PADZZ
28/1	RCR07G182JS	81349	. RESISTOR	1		PADZZ
29/1/2	RNC55H1502FM	81349	. RESISTOR	3		PADZZ
30/1	RNC55H1432FM	81349	. RESISTOR	1		PADZZ
31/1	RNC55H7151FM	81349	. RESISTOR	1		PADZZ
32/2	135C4148P2	89954	. EXTRACTOR	1		PADZZ
33/2	144A5514P1	89954	. NUT (AP)	1		PADZZ
34/2	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
35/2	RNC55H1003FM	81349	. RESISTOR	1		PADZZ
36/2	RNC55H4531FM	81349	. RESISTOR	3		PADZZ
37/2	JAN2N2219A	81349	. TRANSISTOR	5		PADZZ
38/2	101270AP	07047	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C160P1) (AP)	1		PADZZ
	7717-79DAP =	13103				PADZZ
39/2	RNC55H1132FM	81349	. RESISTOR	2		PADZZ
40/2	153C5392P2	89954	. STIFFENER	1		XB
41/2	MS2047083-3	96906	. RIVET (AP)	3		PADZZ
42/2	JAN2N3507	81349	. TRANSISTOR	1		PADZZ
43/2	10127DAP	07047	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C160P1) (AP)	1		PADZZ
	7717-79DAP =	13103				PADZZ
44/2	RNC55H1001FM	81349	. RESISTOR	2		PADZZ
45/2	RCR07G123JS	81349	. RESISTOR	1		PADZZ
46/2	RCR07G393JS	81349	. RESISTOR	2		PADZZ
47/2	RCR07G104JS	81349	. RESISTOR	1		PADZZ
48/2	RCR07G302JS	81349	. RESISTOR	2		PADZZ
49/2	135C4148P1	89954	. EXTRACTOR	1		PADZZ
50/2	144A5514P1	89954	. NUT (AP)	1		PADZZ
51/2	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
52/2	JAN2N2222A	81349	. TRANSISTOR	3		PADZZ
53/2	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
54/2	RNC55H1212FM	81349	. RESISTOR	1		PADZZ

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-4- 55/2	RNC55H2262FM	81349	. RESISTOR	1		PADZZ
56/2	M39003-01-2608	81349	. CAPACITOR	1		PADZZ
57/2	RNC55H5761FM	81349	. RESISTOR	1		PADZZ
58/2	M39003-01-2522	81349	. CAPACITOR	1		PADZZ
	M39003-01-2596 =	81349				PADZZ
59/2	144A9516P1	89954	. BOARD, PRINTED WIRING	1		XA
-60	200945069-501	98748	. PLATE, IDENTIFICATION (Reference draw- . . . . ing 200945069 and Appendix A of this TO when manufacturing plate)	1		MDD



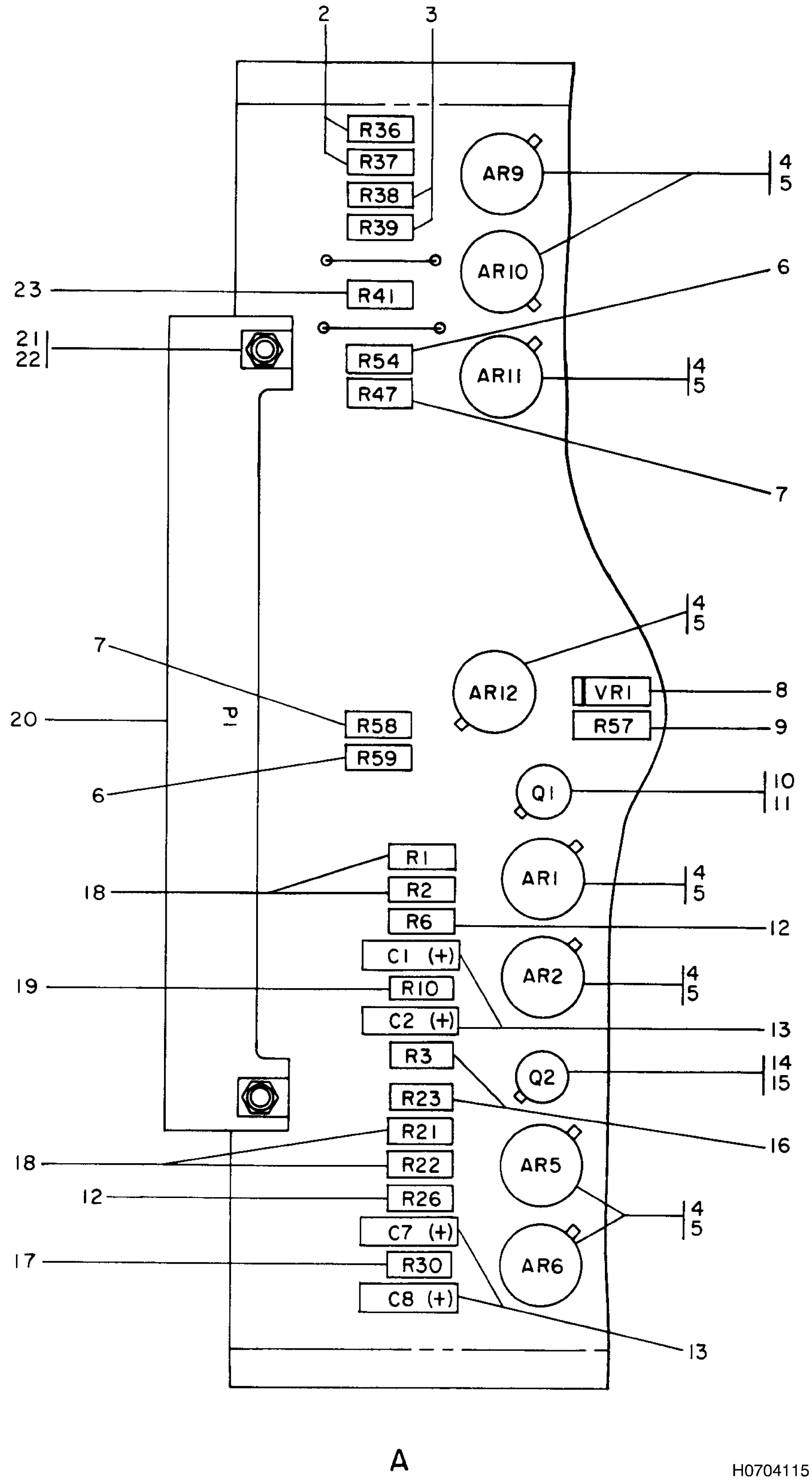


Figure 9-5. Circuit Card Assembly (A4) (Sheet 1 of 2)

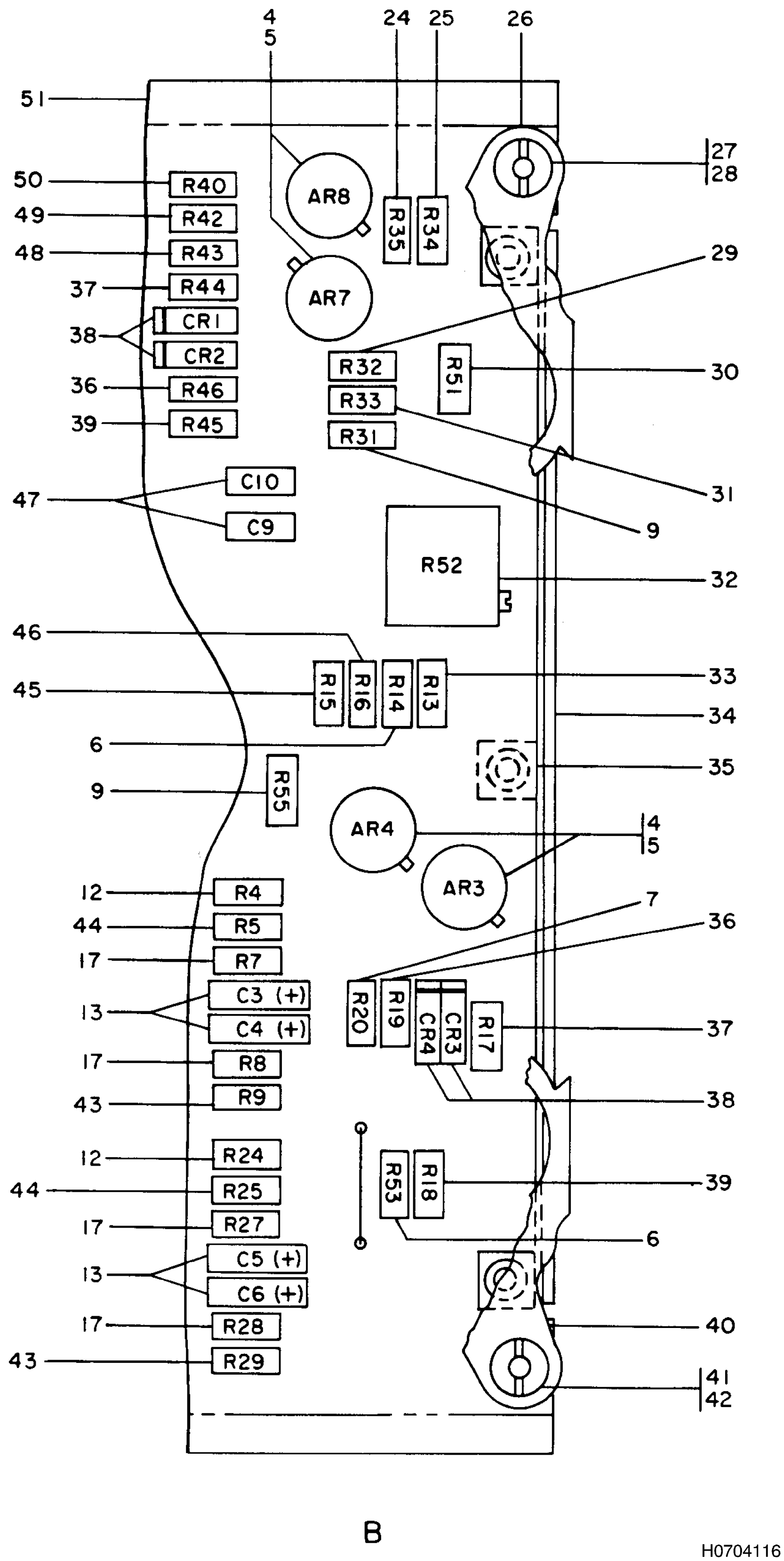


Figure 9-5. Circuit Card Assembly (A4) (Sheet 2)

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-5-			CIRCUIT CARD ASSEMBLY (A4)			
-1	123D7314G1	89954	CIRCUIT CARD ASSEMBLY (A4) (FOR NHA . . . . Figure 9-1)	REF		PADLD
2/1	RNC55H1072BM	81349	. RESISTOR	2		PADZZ
3/1	RNC55H9311BM	81349	. RESISTOR	2		PADZZ
4/1/2	JM38510-10101BGX	81349	. MICROCIRCUIT	12		PADZZ
	144A9511P1 =	89954				PADZZ
5/1/2	10336DAP	07047	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 144A8597P1) (AP)	1		PADZZ
6/1/2	RNC55H3011FM	81349	. RESISTOR	4		PADZZ
7/1/2	RCR07G510JS	81349	. RESISTOR	3		PADZZ
8/1	JANTX1N4626	81349	. SEMICONDUCTOR	1		PADZZ
9/1/2	RNC55H2001FM	81349	. RESISTOR	3		PADZZ
10/1	JANTX2N2945A	81349	. TRANSISTOR	1		PADZZ
	144A9512P1 =	89954				PADZZ
11/1	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
12/1/2	RNC55H1502FM	81349	. RESISTOR	4		PADZZ
13/1/2	M39003-01-2596	81349	. CAPACITOR	8		PADZZ
14/1	JAN2N2432A	81350	. TRANSISTOR	1		PADZZ
15/1	7717-93DAP	13103	. PAD, MOUNTING (GE SPEC CONT DWG . . . . . 928C161P1) (AP)	1		PADZZ
16/1	RCR07G432JS	81349	. RESISTOR	2		PADZZ
17/1/2	RNC55H2002FM	81349	. RESISTOR	5		PADZZ
18/1	RNC55H3012FM	81349	. RESISTOR	4		PADZZ
19/1	RCR07G203JS	81349	. RESISTOR	1		PADZZ
20/1	10-285418-5P	77820	. CONNECTOR, PLUG (GE SPEC CONT . . . . DWG 854D896P22)	1		PADZZ
21/1	68-1660-26	72962	. NUT (GE SPEC CONT DWG 107B8220P3) . . . . (AP)	2		PADZZ
22/1	MS35206-205	96906	. SCREW (AP)	2		PADZZ
23/1	RNC55H5110FM	81349	. RESISTOR	1		PADZZ
24/2	RCR07G624JS	81349	. RESISTOR	1		PADZZ
25/2	RNC55H3010FM	81349	. RESISTOR	1		PADZZ
26/2	135C4148P2	89954	. EXTRACTOR	1		PADZZ
27/2	144A5514P1	89954	. NUT (AP)	1		PADZZ
28/2	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
29/2	RNC55H5901FM	81349	. RESISTOR	1		PADZZ
30/2	RNC55H1432FM	81349	. RESISTOR	1		PADZZ
31/2	RCR07G515JS	81349	. RESISTOR	1		PADZZ
32/2	RJ22CP502	81350	. RESISTOR	1		PADZZ
33/2	RNC55H4991FM	81349	. RESISTOR	1		PADZZ
34/2	153C5392P2	89954	. STIFFENER	1		XB
35/2	MS20470B3-3		. RIVET	3		PADZZ
36/2	RNC55H5111FM	81349	. RESISTOR	2		PADZZ
37/2	RNC55H3922FM	81349	. RESISTOR	2		PADZZ
38/2	JANTX1N4148-1	81350	. SEMICONDUCTOR	4		PADZZ
39/2	RNC55H2102FM	81349	. RESISTOR	2		PADZZ
40/2	135C4148P1	89954	. EXTRACTOR	1		PADZZ
41/2	144A5514P1	89954	. NUT	1		PADZZ
42/2	MS21090-0202	96906	. SCREW (AP)	1		PADZZ
43/2	RNC55H4022FM	81349	. RESISTOR	2		PADZZ
44/2	RNC55H1002FM	81349	. RESISTOR	2		PADZZ
45/2	RCR07G150JS	81349	. RESISTOR	1		PADZZ
46/2	RCR07G513JS	81349	. RESISTOR	1		PADZZ
47/2	M39014-02-0218	81349	. CAPACITOR	2		PADZZ
48/2	RNC55H2001BM	81349	. RESISTOR	1		PADZZ
49/2	RNC55H8661BM	81349	. RESISTOR	1		PADZZ
50/2	RCR07G162JS	81349	. RESISTOR	1		PADZZ
51/2	144A9514P1	89954	. BOARD, PRINTED WIRING	1		XA
-52	200945069-501	98748	. PLATE, IDENTIFICATION (Reference draw- . . . . ing 200945069 and Appendix A of this TO when manufacturing plate)	1		MDD

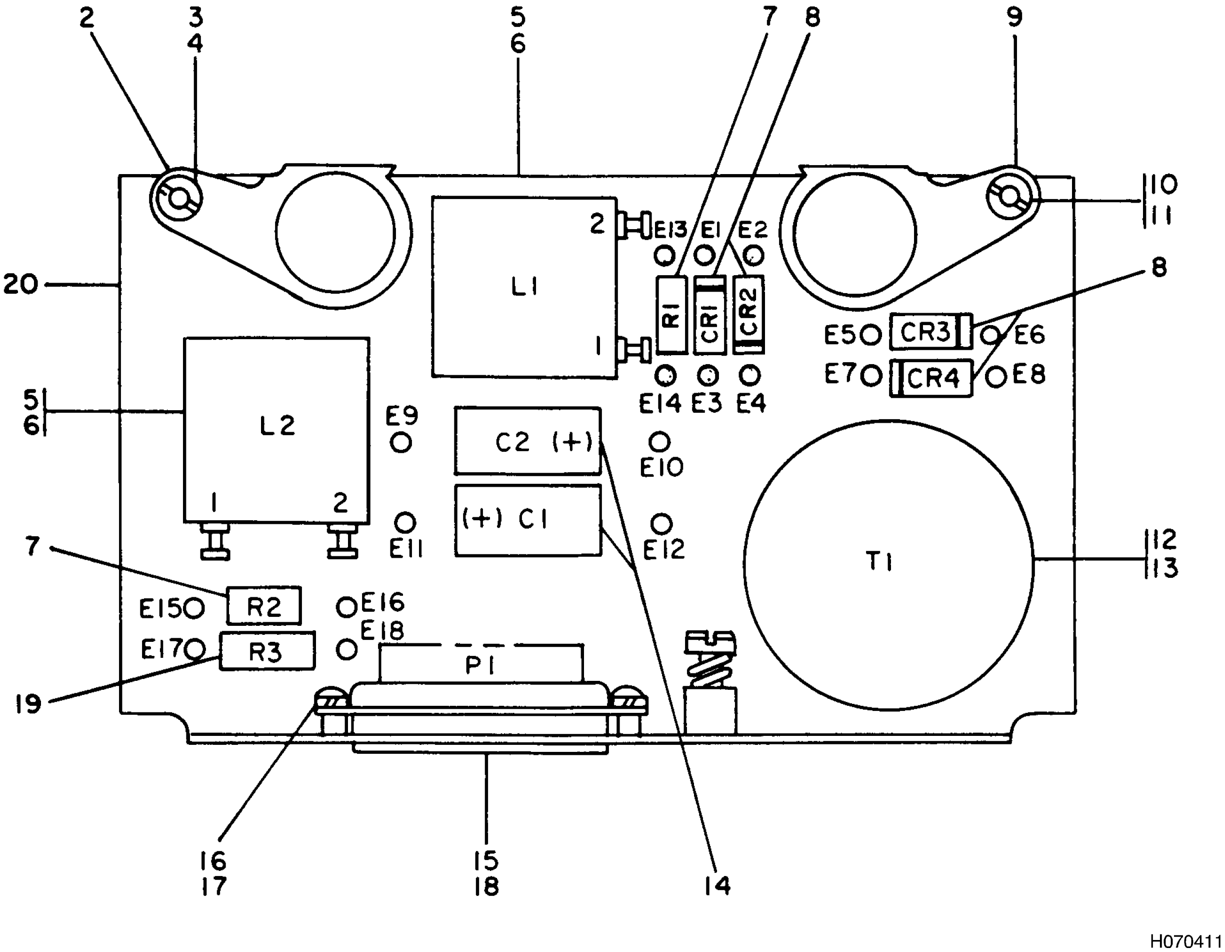


Figure 9-6. Module, Power Supply (PS1)

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-6-			MODULE, POWER SUPPLY (PS1)			
-1	123D5213G1	89954	MODULE, POWER SUPPLY (PS1) (FOR NHA . . . Figure 9-1)	REF		PADLD
2	135C4148P2	89954	. EXTRACTOR	1		PADZZ
3	144A5514P1	89954	. NUT (AP).	1		PADZZ
4	MS21090-0203	96906	. SCREW (AP)	1		PADZZ
5	TW63884	02640	. INDUCTOR, POWER (GE SPEC CONT . . . . DWG 144A8716P1)	2		PADZZ
	K4811 =	51181				PADZZ
6	MS35206-213	96906	. SCREW (AP)	2		PADZZ
7	RCR07G102JS	81349	. RESISTOR	2		PADZZ
8	JAN1N4944	81350	. SEMICONDUCTOR	4		PADZZ
9	135C4148P1	89954	. EXTRACTOR	1		PADZZ
10	144A5514P1	89954	. NUT (AP).	1		PADZZ
11	MS21090-0203	96906	. SCREW (AP)	1		PADZZ
12	TW63882	02640	. TRANSFORMER, POWER (GE SPEC CONT . . . DWG 144A8719P1)	1		PADZZ
	K4814 =	51181				PADZZ
13	MS35206-226	96906	. SCREW (AP)	1		PADZZ
14	40EW277C050WIC	26769	. CAPACITOR (GE SPEC CONT DWG . . . . 144A8721P1)	2		PADZZ

FIGURE & INDEX/ SHEET NO.	PART NUMBER	CAGE	DESCRIPTION 1 2 3 4 5 6 7	UNITS PER ASSY	USABLE ON CODE	SMR CODE
9-6- 15	DBMM17W2P	71468	. CONNECTOR (GE SPEC CONT DWG . . . . 144A8720P4)	1		PADZZ
16	MS35206-213	96906	. SCREW (AP)	2		PADZZ
17	MS35338-40	96906	. WASHER, LOCK (AP)	2		PADZZ
18	DM51158-3	71468	. PIN, RED (GE SPEC CONT DWG . . . . 144A8720P4) (USED ON P1)	2		PADZZ
19	RCR20G202JR	81349	. RESISTOR	1		PADZZ
20	153C5375G1	89954	. BRACKET ASSEMBLY, POWER SUPPLY	1		XB
-21	200945069-501	98748	. PLATE, IDENTIFICATION (Reference draw- . . . . ing 200945069 and Appendix A of this TO when manufacturing plate)	1		MDD

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CHAPTER 10
DIFFERENCE DATA SHEETS

10.1 INTRODUCTION.

Overhaul instructions for the models included in this chapter are the same as the instructions for Thrust Control Unit, 282E261G5, except for the specific differences noted by the applicable Difference Data Sheet.

10.2 INDEX OF DIFFERENCE DATA SHEETS.

Models covered by Difference Data Sheets are as follows:

DIFFERENCE DATA SHEET

THRUST CONTROL UNIT PART NO. PART NUMBER 282E261G4

THE INSTRUCTIONS CONTAINED IN PRECEDING CHAPTERS OF THIS TECHNICAL MANUAL ARE APPLICABLE TO THIS MODEL EXCEPT FOR THE DIFFERENCES CITED IN THIS DIFFERENCE DATA SHEET.

1 DESCRIPTION.

Same as for Part Number 282E261G5.

2 SUPPORT EQUIPMENT AND LIST OF CONSUMABLES.

Same as for Part Number 282E261G5.

3 THEORY OF OPERATION.

Same as for Part Number 282E261G5.

4 DESCRIPTION OF SYSTEM, TIE-IN OF EQUIPMENT AND ACCESSORIES.

Same as for Part Number 282E261G5.

5 PREPARATION FOR MAINTENANCE.

Same as for Part Number 282E261G5.

6 CHECKOUT AND TROUBLESHOOTING.

Same as for Part Number 282E261G5.

7 MAINTENANCE/REPAIR INSTRUCTIONS.

Same as for Part Number 282E261G5, except that Figure 7-3 and CAUTION following Paragraph 7.6.2.2 do not apply.

8 OVERHAUL INSTRUCTIONS.

Same as for Part Number 282E261G5.

9 ILLUSTRATED PARTS BREAKDOWN.

Refer to Chapter 9 for differences.

APPENDIX A
ITEM UNIQUE IDENTIFICATION (IUID) SUPPORTING INFORMATION

A.1 IUID SUPPLEMENTAL INFORMATION.

- Inspect Unique Identifier (UID) Label by using a 2D hand-held electronic reader. Reference TO 00-25-260.
- If UID Label is not present or damaged, manufacture new label and install as shown in Table A-1.
- Do not cover any existing markings. Reference TO 00-25-260.
- Cage code is that of the activity responsible for guaranteeing uniqueness of the sequencing code.
- For additional information, reference MIL-STD-130( ) and TO 00-25-260.

Table A-1. IUID Required Information

Part Number	Noun	UID Label Figure/Index	UID Label Part Number	Surface Preparation	Application/ Verification
123D5213G1	Power Supply Module	Figure A-1, Sheet 1	200945069-501	REF: TO 00-25-260 Premarking Activities	REF: TO 00-25-260
282E261G5	Thrust Control Unit	Figure A-1, Sheet 2	200945069-501	REF: TO 00-25-260 Premarking Activities	REF: TO 00-25-260
123D7314G1	Circuit Card Assembly	Figure A-1, Sheet 3	200945069-501	REF: TO 00-25-260 Premarking Activities	REF: TO 00-25-260
123D7316G1	Circuit Card Assembly	Figure A-1, Sheet 4	200945069-501	REF: TO 00-25-260 Premarking Activities	REF: TO 00-25-260
282E265G5	Module Assembly Component	Figure A-1, Sheet 5	200945069-501	REF: TO 00-25-260 Premarking Activities	REF: TO 00-25-260

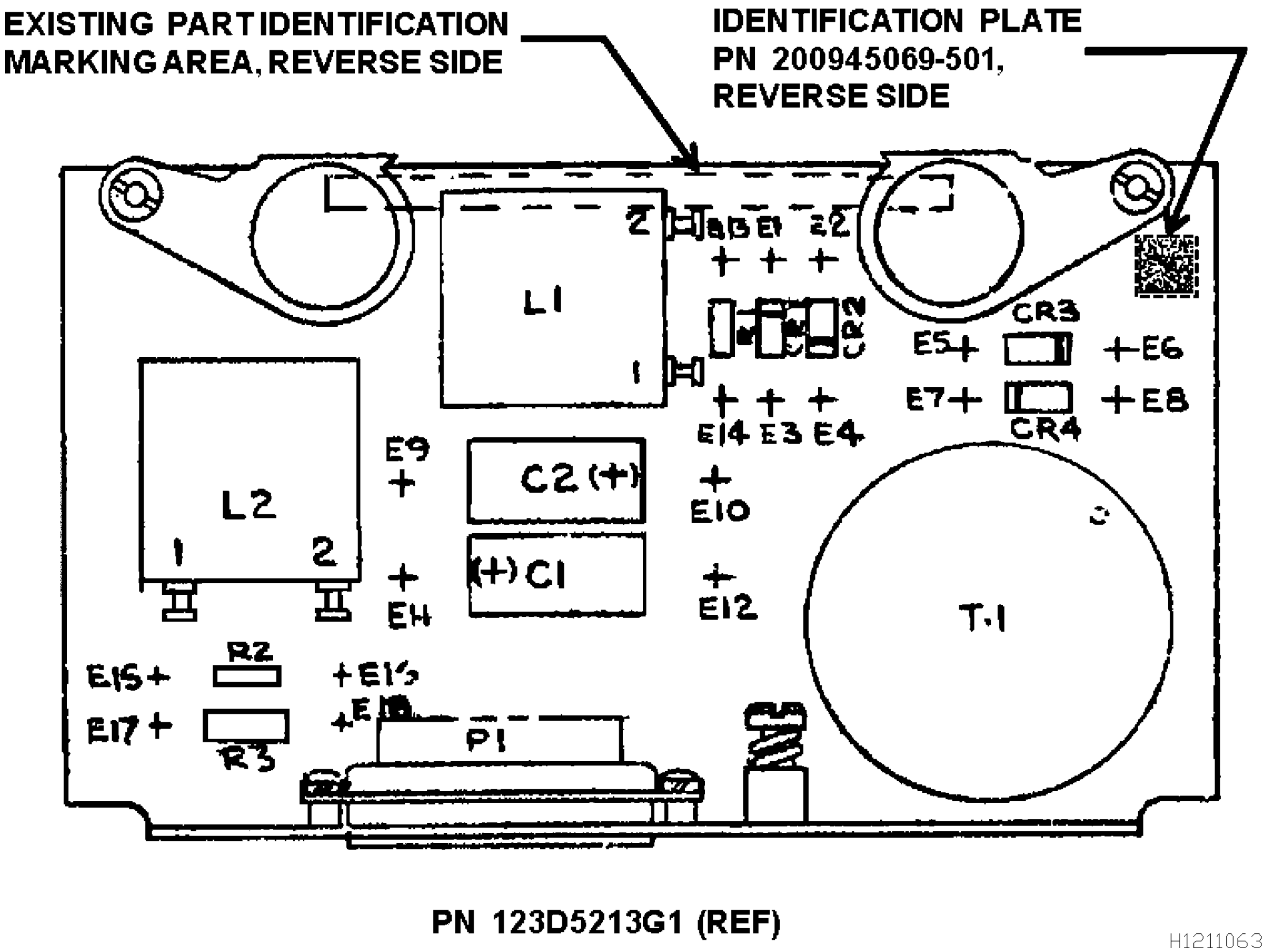

Figure A-1. IUID Identification Plate Location (Sheet 1 of 5)

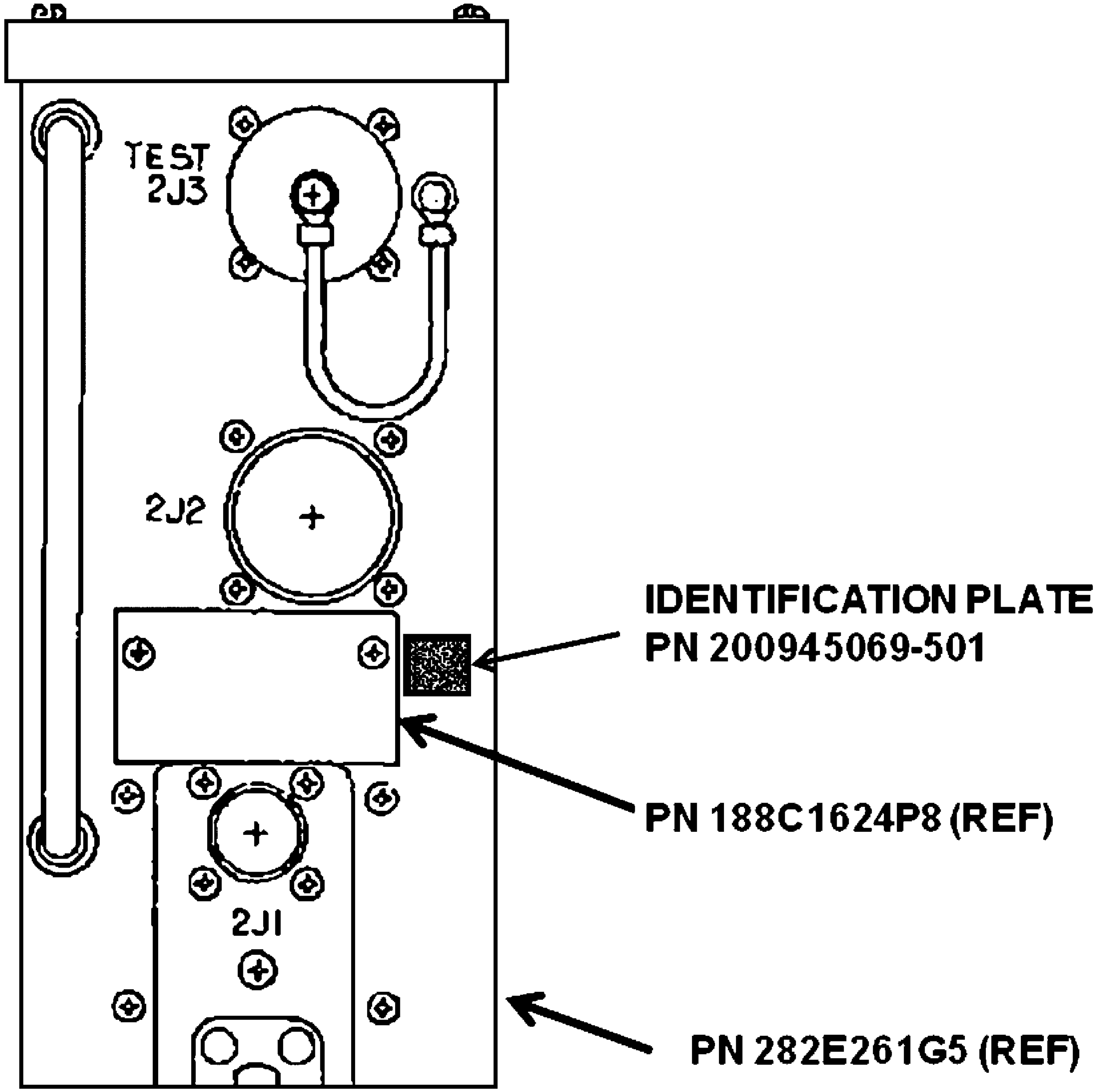

Figure A-1. IUID Identification Plate Location (Sheet 2)

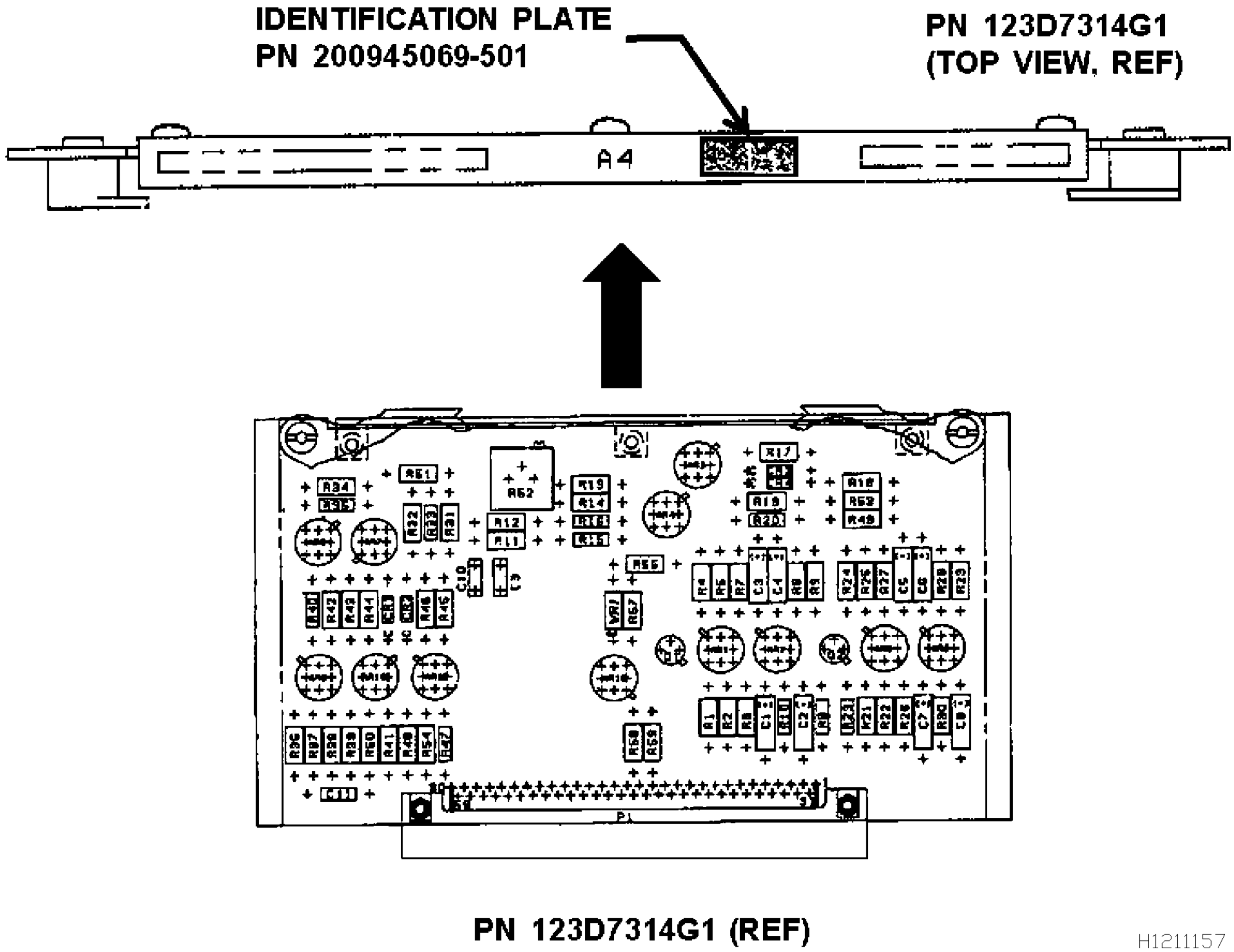

Figure A-1. IUID Identification Plate Location (Sheet 3)

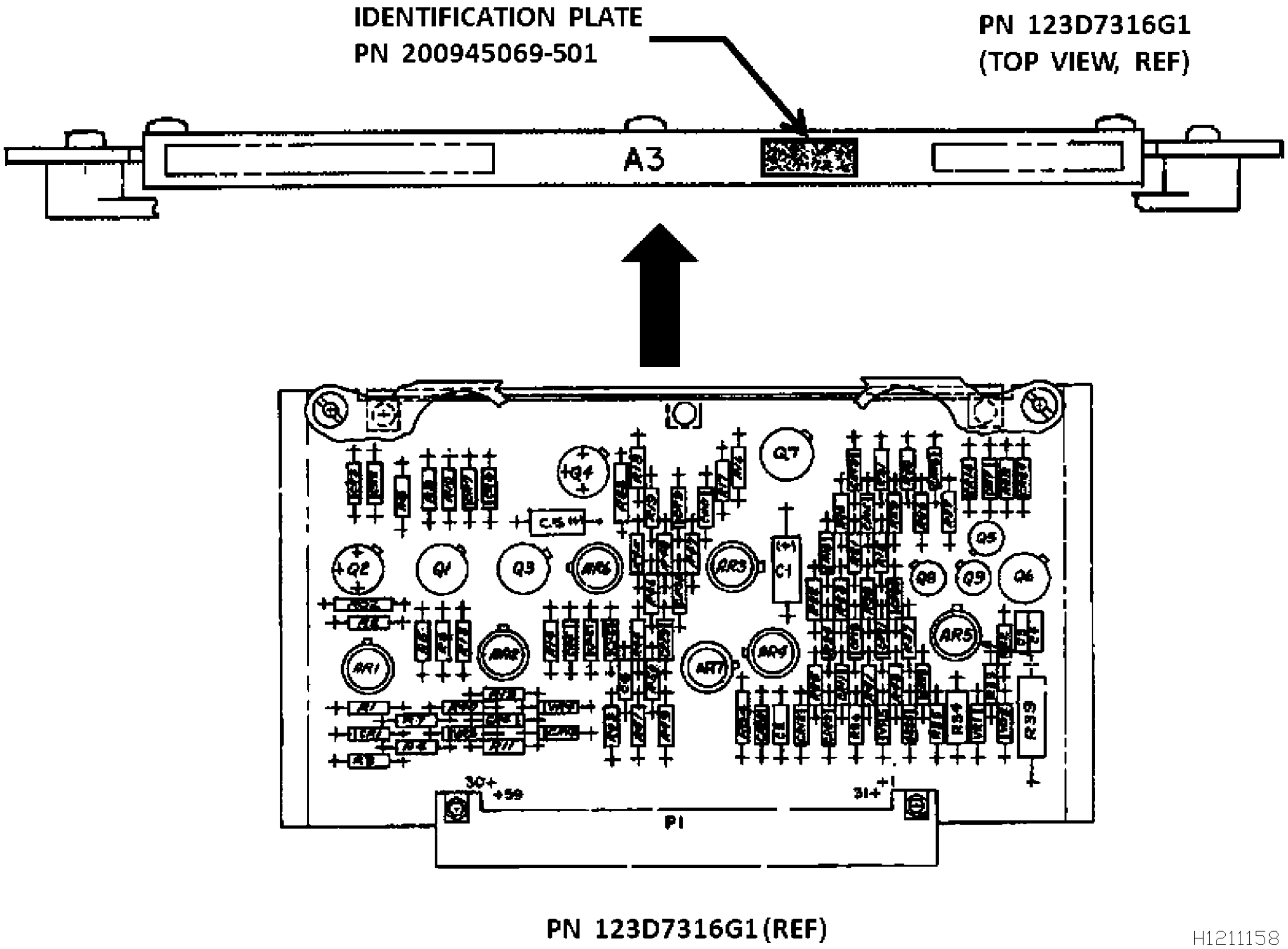

Figure A-1. IUID Identification Plate Location (Sheet 4)

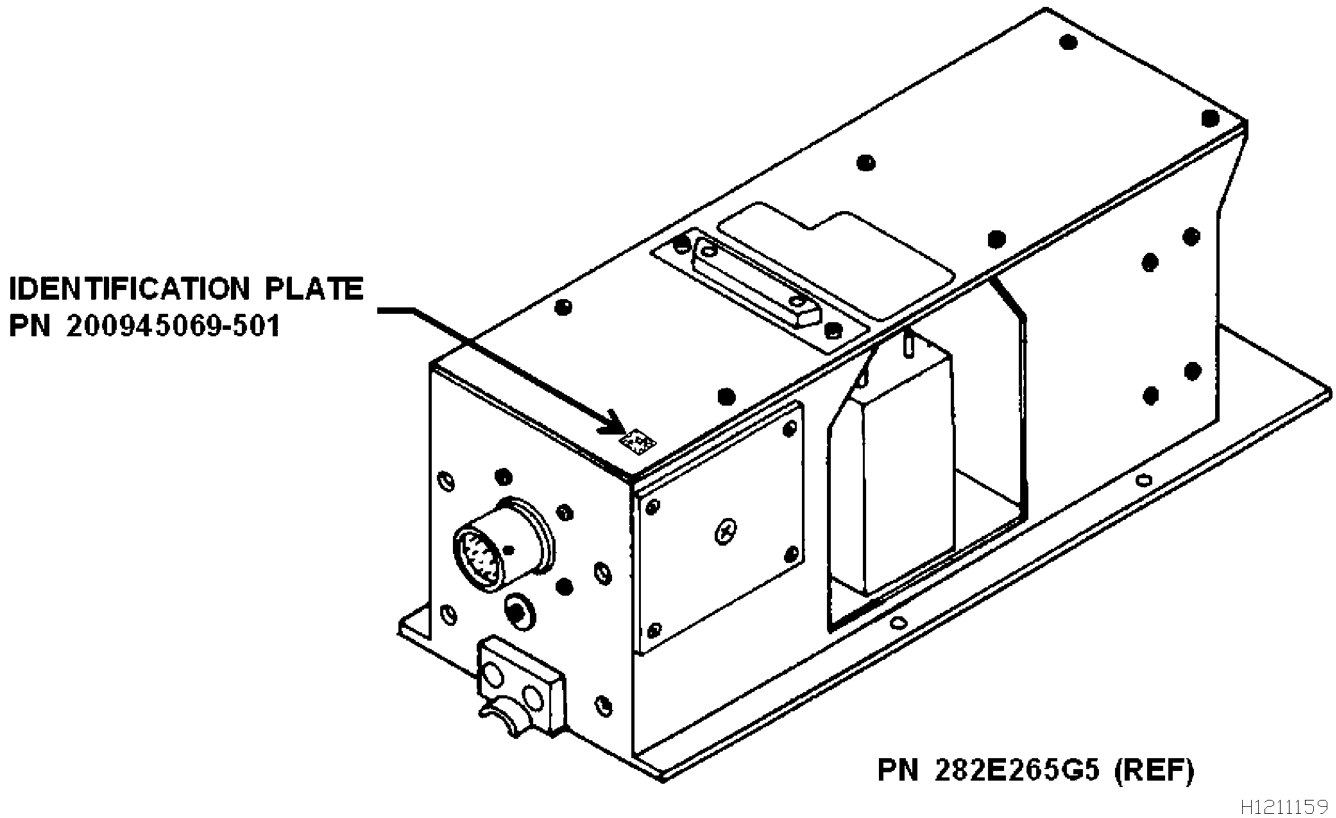

Figure A-1. IUID Identification Plate Location (Sheet 5)

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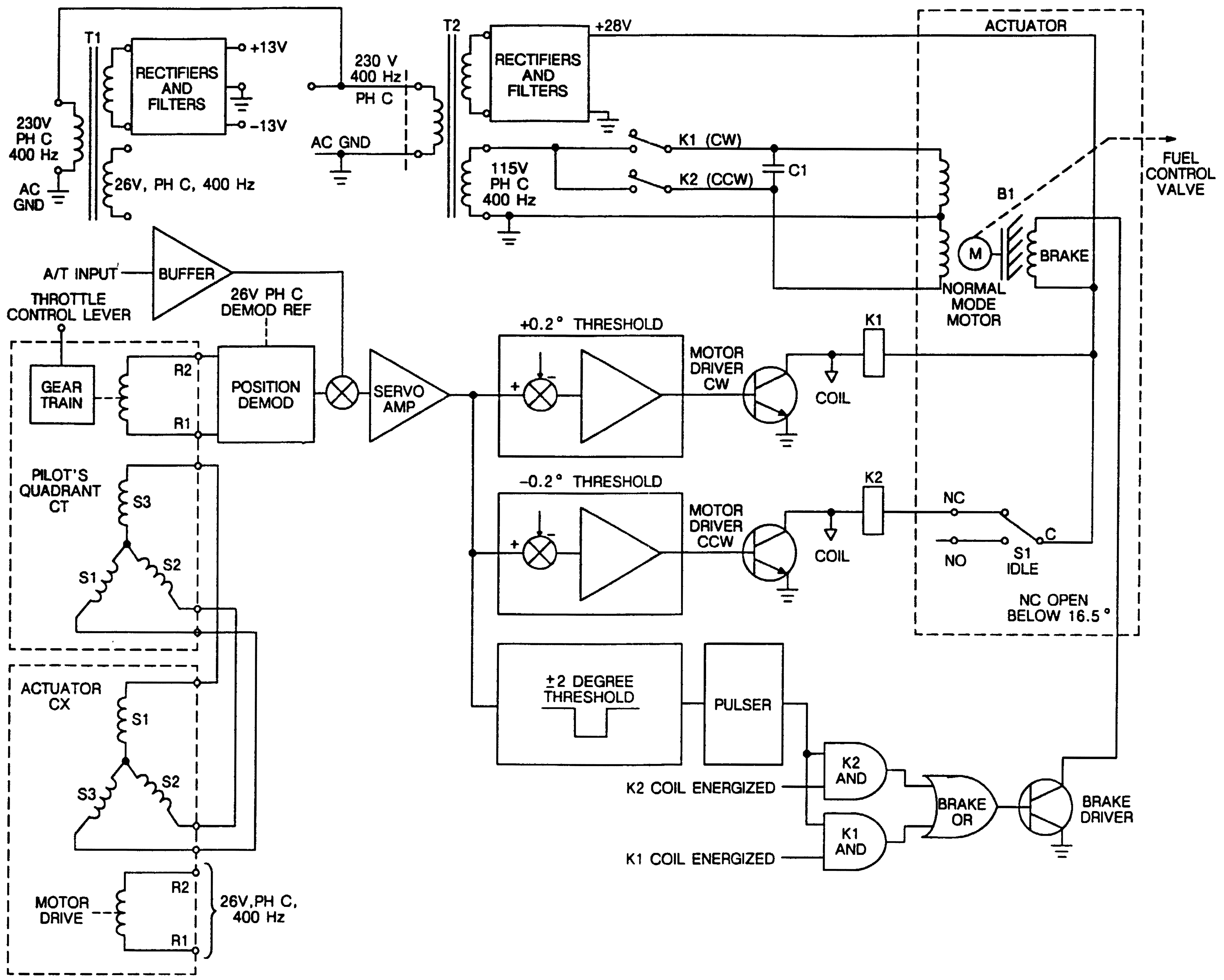
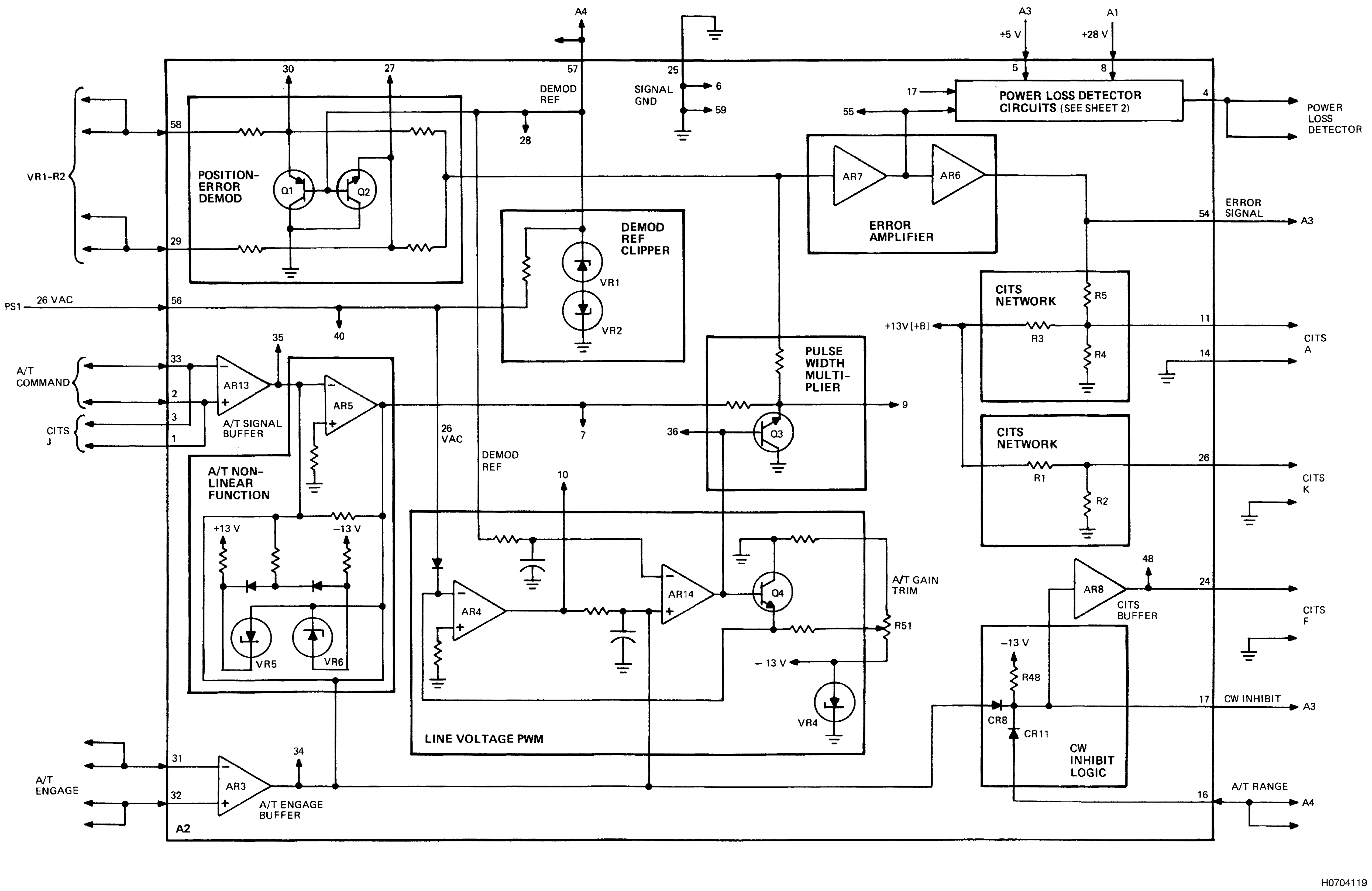
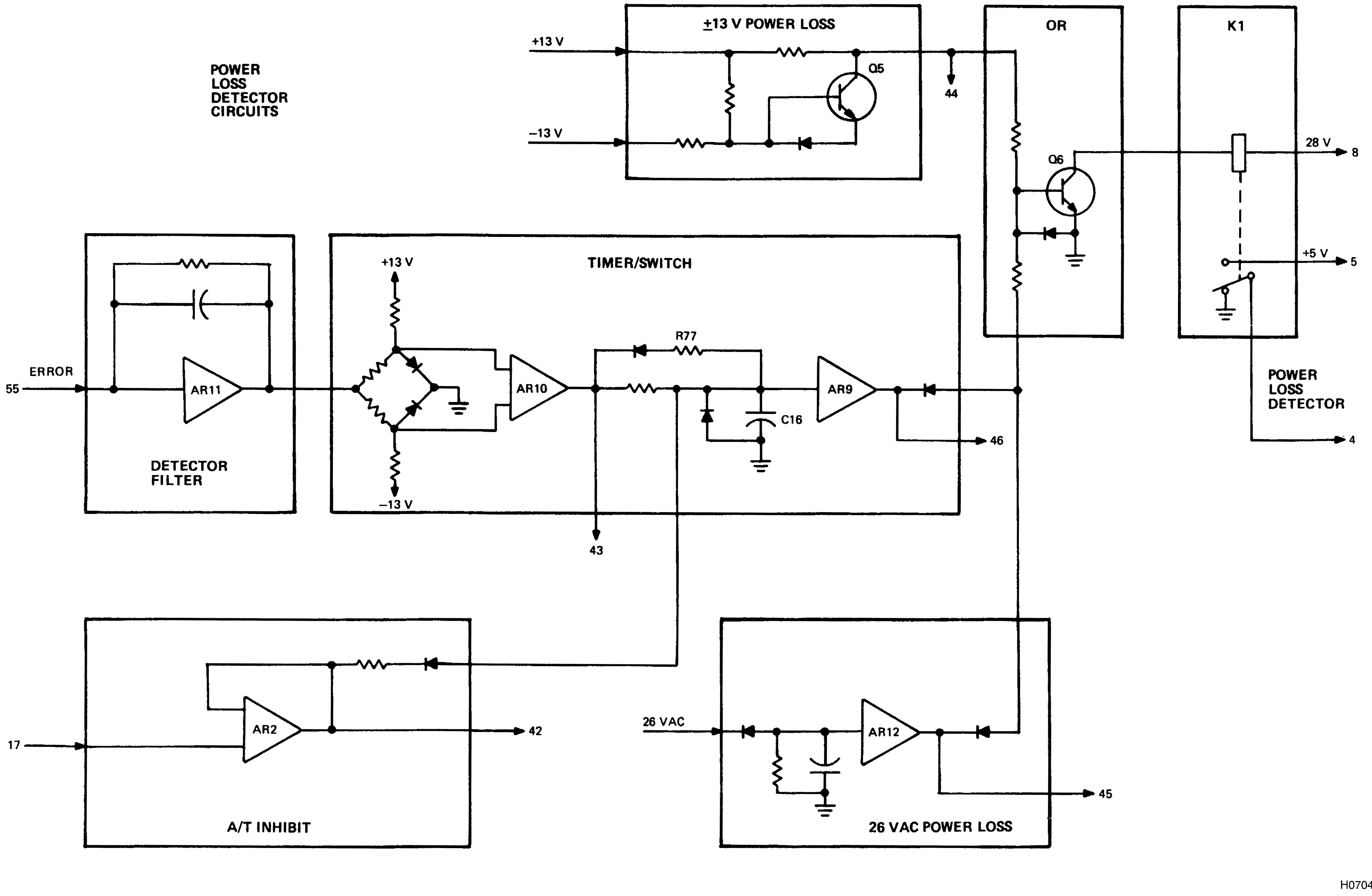


FO-1. Closed Loop Block Diagram



FO-2. CCA A2 - Simplified Block Diagram (Sheet 1 of 2)



FO-2. CCA A2 - Simplified Block Diagram (Sheet 2)